Leyla Sarfaraz

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Role Models of Growth-Oriented Iranian Women Entrepreneurs

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Leyla Sarfaraz
Department of Management
Shiraz University
Shiraz, Iran

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In Loving Memory
Of
My Beloved Parents
Mohammad Reza and Zivar Sarfaraz
Whose Infinite Love I Deeply Cherish in
Every Heartbeat

Foreword

Women entrepreneurs contribute substantially to the global economy. They create jobs for themselves and benefit other stakeholders like employees, suppliers, customers, and investors. They bring new ideas into society, advance economic development, and provide financial health and stability for their families.

Across 83 economies my coauthors and I studied in the "2014 Global Entrepreneurship Monitor (GEM) Special Report: Women's Entrepreneurship," an estimated 329 million women were in the process of starting a business or running a new or established one. In many countries, however, women start businesses at lower rates than men, and this includes Iran. To the extent women are constrained from starting businesses, their societies fail to benefit from the ambition and creativity of half their population. For this reason, researchers, policy makers, and practitioners all over the world have increasingly turned their attention toward understanding and expanding women's entrepreneurship.

More women are starting businesses in Iran: 10 % of working age Iranian women were in the process of starting or running a new business in 2014. Their efforts are narrowing the gender gap; while women started at one-fourth the rate of men in 2010, they started at nearly half the rate of men in 2014. In addition, they were less likely to start out of necessity in 2014, meaning that they were more likely to become entrepreneurs because they were attracted by the opportunity to do so. In addition, their businesses are increasingly being started in high value-added sectors. Compared to two previous GEM Women's Entrepreneurship reports I coauthored in 2010 and 2012, we can see improvements in many aspects of women's entrepreneurship in Iran.

While these results show great achievements in Iran, there is still more work to be done. A gender gap remains, and continued efforts are needed to encourage more women to start and maintain growth-oriented, innovation-based, and global businesses. This book provides a valuable resource for understanding the unique context in which women start businesses in Iran, the progress made to date, and the actions needed to further stimulate and support this activity.

The book starts out in Chap. 1 with the historical, cultural, and economic context underlying the influences on women's entrepreneurship in Iran. Entrepreneurs are highly dependent on an ecosystem that includes societal views and expectations that impact one's willingness and ease of starting a business, stakeholders who directly participate and benefit from their efforts, and conditions like infrastructure, legal and regulatory systems, financial markets, and technology transfer that facilitate (or constrain) their efforts. While these factors impact all economic activity, they often affect entrepreneurs differently than, for example, large established or government-owned businesses. More importantly, women entrepreneurs may face different conditions than men, even when these conditions don't explicitly state a gender distinction.

Chapter 2 provides insights into entrepreneurship in Iran from the Global Entrepreneurship Monitor (GEM), drawing from the University of Tehran's participation in this global research program from 2008 to 2014. The tremendous effort this team has undertaken to conduct GEM surveys every year affords an in-depth look at patterns in the rate and nature of entrepreneurship in Iran, as well as societal attitudes, over a 7-year period. Special attention in this chapter is devoted to the industry mix and international focus of female entrepreneurs in Iran.

Social, financial, and human capital can have marked effects on the resources women bring to a new business venture. Not only do these resources differ among women, including across different regions of the world, but there are some commonalities within a community on the presence and access to these resources for women versus men. Chapter 3 addresses this important influence on women's entrepreneurship.

Chapter 4 ties the challenges unveiled in the earlier chapters to interventions and decisions targeted toward improving the rate and quality of women's entrepreneurship. Professor Sarfaraz also discusses particular issues faced by rural female entrepreneurs in Iran. Additionally, she reviews the emergence of the IT sector in Iran, documenting the investments made by both private and public sources to build this foundation, and the effect it has had on female startup efforts in this sector.

The book then turns to case studies, bringing women's entrepreneurship in Iran to life and demonstrating the uniqueness and diversity of these women and the businesses they start. These stories provide guidance, inspiration, and understanding about their ambitions, the challenges they face, and their achievements.

The efforts of Leyla Sarfaraz and her GEM Iran team colleagues have brought greater awareness about the importance of entrepreneurship to Iran's economy. Prof. Sarfaraz has been a valued expert and tireless advocate for women's entrepreneurship in her country. With this book, she presents a valuable guide for the future advancement of women entrepreneurs in Iran.

Babson College
Babson Park MA, USA
November 3, 2015

Donna Kelley

Preface

In 2009, when Dr. Kristie Seawright, the then Executive Director of the Global Entrepreneurship Monitor (GEM), asked me to write a mini case study on a successful Iranian woman entrepreneur for publication in the GEM Women's Report, I started researching the subject. Surprisingly, I found that there are quite a few outstanding Iranian women entrepreneurs with a wide range of ages, educational backgrounds, social status, and incomes, setting up a diverse array of firms even in culturally male-dominated fields in various parts of the country. I thought studying the lives of these influential and inspirational entrepreneurs, who have overcome numerous barriers, would prove to others how an ordinary woman could turn into an extraordinary entrepreneur when she dares to dream big and believe in her competency. Such a study would also provide readers a better understanding of the concept of entrepreneurship, highlighting in particular women's issues within the field.

Since parts of the book present the stories of successful Iranian women entrepreneurs, I decided to also include a few stories of remarkable Iranian women entrepreneurs who left Iran as teenagers and now live abroad. I was curious to learn whether the cultural background and growing up in Iran could somehow shape the career success of those outstanding women. This would allow the reader to make comparisons and evaluate how exposure to Western culture could affect the path that these women took toward entrepreneurial success. Aside from this, I found the stories of these women to be quite inspiring. For instance, I was highly impressed by the story of an Iranian-American outstanding entrepreneur, who could reach her childhood dream of going into the space after successfully launching her own business.

When we talk about a successful woman entrepreneur, it is not just a matter of establishing and running a business. We see how they contribute to the well-being of their families, their communities, even their global environment, and future generations by changing society's gender-normative attitudes about women. Sometimes there are crucial hidden truths behind these cases that can shed light on some missing factors in women's entrepreneurial studies.

To provide insight into the ecosystem in which these women successfully started and developed their own company, I aimed to briefly explore key dimensions of the business environment in Iran. However, in my session with Springer's editor, he suggested that the subject of women entrepreneurship in Iran be the main focus of the book. Hence, this book developed from a focus on case studies to a book that also provides a general view of the Iranian economy and ease of doing business in Iran, as well as Iran's labor market and entrepreneurship from a gender perspective.

The book is organized into two parts. The first part, comprising of Chaps. 1–4, provides an overview of the business environment in Iran with a focus on different dimensions of women entrepreneurship in Iran. The second part, Chaps. 5 and 6, presents the cases of the Iranian women entrepreneurs living in Iran and the Iranian women entrepreneurs living abroad, respectively.

Chapter 1 provides a brief historical overview of the status of women in Iran, their right to own property in ancient times, and how this differs from their female counterparts in most of other Middle Eastern countries. A general overview of Iran's economic performance is also studied to shed some light on how its economy works. The position of Iranian women in the labor market and the gender gap in unemployment, as well as labor force participation at different age groups in urban and rural areas, are all studied. In a comparative perspective, the relative position of Iranian women in the labor market is investigated and compared to their female peers in some selected countries. Later sections of the chapter discuss "Ease of Doing Business" and the entrepreneurship ecosystem in Iran. Moreover, experts' perspectives on the female entrepreneurial environment in Iran are examined to obtain a general perspective on the normative support for Iranian female entrepreneurs.

Chapter 2 mainly follows the GEM framework and uses its data to study different phases of the entrepreneurial journey for both women and men in Iran. Moreover, the rate of business discontinuance in Iran is compared to different regions, and the reasons for business discontinuation in the country are explained. Then, in a gender comparison, the dynamics of women's entrepreneurial activity and entrepreneurial motivation in the country for the period of 2008–2014 will be examined. To explore the feasibility of internationalization for Iranian entrepreneurs, the performance of Iran in international trade is studied using data from the Heritage Foundation (Index of Economic Freedom), the World Bank (Trading Across Borders), and the GEM (International Entrepreneurship). The position of Iranian female entrepreneurs in international commerce is also explored and compared to other GEM participants. At the end of the chapter, the sector distribution of entrepreneurial activity in Iran is discussed.

Chapter 3 discusses the importance and availability of social capital, financial capital, and human capital for women in Iran. First, different dimensions of social capital are reviewed and the effect of social capital on women's entrepreneurship in Iran in terms of structural, relational, and cognitive dimensions is investigated. Then the demand for financial capital and also the supply side of business financing in Iran is studied in detail. Venture capital financing as a new source of financing in the country is considered and discussed as well. To develop a better understanding

of human capital in Iran, the World Economic Forum's human capital index is used to measure the performance of Iran in social capital and also to show where Iran stands in human capital compared to some selected countries.

Chapter 4 focuses on the challenges faced by women entrepreneurs and the public policy for supporting women entrepreneurs in Iran. In particular, institutional and cultural barriers to women's entrepreneurship are investigated and discussed. Balancing family and work, as one of the biggest challenges faced by a female entrepreneur, is also considered, and the approach women entrepreneurs in Iran use to overcome this issue is studied. Then the Global Gender Gap Index is used to show the difference between the performance of women and men in terms of economic participation and opportunities and educational attainment. The chapter also draws attention to the challenges of female entrepreneurship in rural areas. The emergence of women entrepreneurs in Information Technology, as a substantial change in the entrepreneurial environment of Iran, is studied, and the internal and external efforts undertaken by the public and private sectors to promote IT entrepreneurship in the country are explained.

Chapter 5 introduces and examines the cases of seven award-winning, role-model Iranian women entrepreneurs who are successfully running their businesses in Iran. The women featured in the book show how they have challenged barriers, learned from their failures, and taken courageous steps to change the stereotypes about women entrepreneurs even in a traditional society. These women are mainly involved in culturally male-dominated fields, including environmental protection technology, international transportation, mining, recycling and composting, auto-parts and gas valves manufacturing, modern livestock husbandry, and anthemia herb production. The case study of these women describes their background, dreams, motivations, challenges, and success in their entrepreneurial journey. Each story provides an insight into a different dimension of entrepreneurship and incorporates entrepreneurship and poverty reduction, philanthropy, high technology, technology transfer, and environmental protection.

The last chapter of the book presents the cases of four distinguished Iranian women entrepreneurs who were born and raised in Iran and moved from Iran to England and the USA as teenagers. The case studies describe these women's backgrounds, dreams, and motivations, as well as the challenges they faced in their adopted country. This chapter explores how these women have benefited from cultural diversity and successfully combined positive elements from two different nations to create innovative, extraordinary designs, products, and services. Their stories indicate the importance of childhood experiences in shaping the imagination, dreams, and creativity of successful entrepreneurs. These growth-oriented women entrepreneurs are involved in the following enterprises: digital technologies, residential and commercial architecture, and a chain daycare center.

The women featured in this book have been selected among award-winning entrepreneurs and have been interviewed by the author of this book via a questionnaire. The questions were designed such that these entrepreneurs could share with us the story of their entrepreneurial journey in their own words.

Very late, in the process of writing the book, I learned about the growing number of influential women entrepreneurs in rural areas and also those involved in the Information Technology industry that I could not ignore. My longtime efforts to find successful entrepreneurs and their contact information, gain access to them, and obtain their consent to be featured in the book deterred me from writing an individual case for each of them. However, I decided to introduce them in the mini case studies in the boxes using their media interviews.

Writing this book was a fantastic learning process for me. As a result of this experience, I now believe in the power of entrepreneurship more strongly than ever and deeply appreciate its incredible impact on the well-being of individuals and communities. It is interesting that since starting the book, women entrepreneurship in Iran has been fostering very rapidly.

Shiraz, Iran Leyla Sarfaraz
Spring 2016

Acknowledgment

I am very grateful to the women entrepreneurs featured in this book as without their participation, this book could not be completed. I would like to express my gratitude to Mohammad Reza Zali who generously provided me with data (including unpublished data) vital to writing this book. I am also deeply grateful to Donna Kelley, Hamid Etemad, Jonathan Levie, and Maria Minniti for their wonderful support. I would like to extend my sincere thanks to Lorraine Oak, Emeric Solymossy, Julia Tischler, and Sarra Tlili for their encouragement, feedback, and thoughtful comments on some cases in Part II of the book.

Endorsements

Professor Sarfaraz's book is authoritative, well researched, and highly informative. It is an enjoyable reading and deserves a careful examination. While reaching the last few pages, I began to crave for more and even deeper understanding that the book had methodically cultivated. I was impressed with the volume's broad scope, depth of materials, and rich perspectives in all of its six chapters. This book has, indeed, sown the seeds of scholarly exploration of differential-gender attributes in search of deeper entrepreneurial characteristics and possibly those in entrepreneurial internationalization. In a few words, therefore, this book speaks for itself and needs no endorsement or recommendation.

Hamid Etemad
Editor-in Chief, Journal of International Entrepreneurship
McGill University, Montreal, Canada
January 2016

This book provides a detailed and fascinating window on the world of women entrepreneurs in Iran. Drawing on a wide range of secondary data, Leyla Sarfaraz documents the extent to which the potential of women to contribute to the Iranian economy through entrepreneurship remains untapped. In the second half of the book, she provides inspiring examples of how women, as entrepreneurs, can succeed in difficult circumstances and gives us a rare insight into what the women of Iran can achieve when they persist and believe in themselves: as one of them said: "will is power."

Professor Jonathan Levie
Director of Teaching and Knowledge Exchange
Hunter Centre for Entrepreneurship
Strathclyde Business School
University of Strathclyde
Glasgow, United Kingdom

"There is still much we don't know about what causes differences in entrepreneurship rates between men and women. In particular, the role of geography and culture remains understudied. This interesting and important book contributes to filling this gap. The author gives us a vivid portrait of female entrepreneurship in Iran and compelling stories of Iranian business women at home and abroad. Everyone interested in entrepreneurship, gender issues, and comparative studies will find this book fascinating!"

Maria Minniti, PhD
Professor and L. Bantle Chair of Entrepreneurship and Public Policy
Syracuse University, NY, USA

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Part I
Women's Entrepreneurship in Iran: A Comparative Perspective

Chapter 1
Business Environment in Iran

1.1 Historical Background of Women in Iran

"Iran is home to one of the world's oldest continuous major civilizations" (Araee 2008). Iran, with its inception as Persia in 3200 BC, appears to be one of the most ancient countries in the world where women had the right to own property. "Judging from the Greek sources the wealth of Achaemenid royal women was well known. Royal women of the Achaemenid court (559–331 BC) such as Atossa, Amestris and Parysatis are described by Greek historians as powerful queens who exercised considerable influence on the king (Hdt. 7. 3. 4; Ctesias, FGrH 688 F15 (49)). . . and owned vast amount of property" (Brosius 1998). This was true when extensive land holding by Greek women was uncommon.

Historically, the status of women in the Iranian culture has been distinct from the status of women in many other Middle Eastern cultures. Traditionally, Iranian women could be appointed as the Chief or Sheriff in towns and villages. There were also some women who were born in educated families and had the opportunity to make their voice heard as writers, poets, and political activists. For instance, as early as in the nineteenth century, Bibi Khatoon Astarabadi (1858–1921) was the author of Men's Defects (Ma'ayeb al-Rejal), a book published in 1895 on criticizing men. Bibi Khatoon was also the founder of the first school for girls in Iran and played an influential role in the Iranian Constitutional Revolution (1905–1911). Throughout the twentieth century and the beginning of the twenty-first century, a number of other women have followed in her footsteps and established themselves as prominent and influential figures in the Iranian society and even worldwide.

Mahin Afshar, the granddaughter of Malek Altojar (owner of the traders), was the first international businesswoman in Iran. She sold some of her dowry to finance her transportation brokerage business and extended her commerce to the world's major ports such as Amsterdam, Notre Dame, London, New York, and Anvers (Saber 2013). Mahin Afshar obtained the first trade license granted to a woman issued by the Iranian government in 1958. Two recent examples of brilliant,

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internationally recognized Iranian women who have contributed to academic breakthroughs and technological advancements are Maryam Mirzakhani and Anousheh Ansari, respectively. In 2014, Maryam Mirzakhani became the first woman to ever win the Fields Medal. In 2006, Anousheh Ansari, who is also featured in this book, became the first female private space explorer; she is also the cofounder of Prodea Systems, a knowledge-based business.

Technically, there is no historical or legal/official barrier to women's employment, economic participation, and doing business in Iran (Khorshidi 2009). For instance, when Mirza Taghi Khan Amir Kabir, the prime minister of the king Nasser al-din Shah Qajar, prohibited the import of some foreign commodities to protect domestic production in 1851, Khorshid Khanom, a woman from Tehran, produced a sample of decorated military uniform and took it to the Prime Minister. When Mirza Taghi Khan noticed the high quality of the uniform, he made a 5-year contract with Khorshid Khanom and asked her to hire more employees in her small workshop to produce all the required military uniforms to substitute the import of uniforms from Austria and the Ottoman Empire. This contract was reported as important economic news in "Vaghaye-e Etefaghiyeh" newspaper and Khorshid Khanom became the first woman entrepreneur in the contemporary history of Iran.

It appears that academic research on identifying Iranian female entrepreneurs in Iran started in 2001 by Firouzeh Saber. She, for the first time, introduced 20 women entrepreneurs in her master's thesis entitled "Ways to Develop Women's Entrepreneurship in Iran" at the University of Tehran. In her work, she studied the obstacles and challenges faced by these women entrepreneurs, and paved the way for more studies in this field in the country. It seems that attention to women's entrepreneurial studies in Iran lags 20 years behind the cutting edge of worldwide women's entrepreneurial research and development. According to Buttner and Moore (1997), until the beginning of the 1980s almost nothing was known about female entrepreneurs. However, even in developed nations, lack of attention to women's entrepreneurship was still quite noticeable well into the 1990s. As pointed out by Reed (1996), gendered attitudes to entrepreneurs make women invisible. Nevertheless, this issue has been way more significant in less developed countries where the lack of studies on women entrepreneurs is considerably more propounded.

1.2 Economic Overview

Iran, with its 80 million population is the world's 17th most populous country and the second most populated country in the Middle East and North Africa (MENA) after Egypt. Comprising a land area of 1,648,195 sq. km (636,372 sq. miles), Iran is the 18th largest country in the world. The country, as an upper-middle income nation, holds the world's fourth-largest proven oil reserves and the world's second-largest natural gas reserves with an estimated Gross Domestic Product (GDP) of US $406.3 billion in 2014. Iran's proven natural gas reserves constitute 18.19 % of the

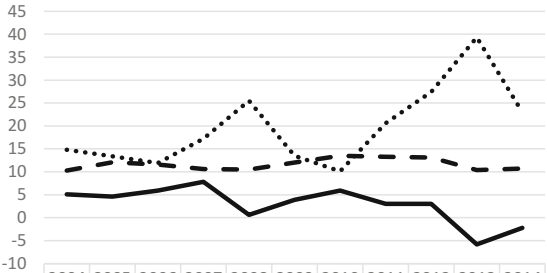
world's reserves. It is also considered the 15th richest country in the world in terms of mineral resources. The country produced 4.94 % of the world's production and consumed 4.84 % of the world's consumption in 2013 (International Monetary Fund, World Economic Outlook 2014a).

The Iranian government plays an important role in the allocation of resources in the economy by controlling and distributing oil revenue, and by owning large public and quasi-public enterprises. Despite the initiation of the privatization program in 1988, which has moved very slowly ever since, state-owned enterprises have been the major manufacturer and source of production in Iran. The scope and extent of government intervention has provided more opportunities for rent-seeking activities rather than entrepreneurial activities. Extensive government control over the economy impedes the emergence of a dynamic private sector and results in a low level of competition and causes inefficiency in the economy.

All people can have greater opportunities if the government policies support the private sector in making investments, creating jobs, and increasing productivity through a dynamic business environment (World Bank, Doing Business 2013). In particular, Estrin and Mickiewicz (2011) found that while the rule of law does not generally have gender-specific effects on women's entrepreneurship, in countries with larger state sector, women are less likely to be involved in entrepreneurial activity. Moreover, Iran is characterized by a high rate of youth population along with a high rate of youth unemployment and a significant gender gap in employment rate (especially in youth) in favor of men. The unemployment rate was 26 % for youth (15–24 years) and 42.7 % for female youth in 2013 (Devarajan and Mottaghi 2014). Women's very low economic participation rate in Iran deprives the economy from their potential contribution to economic growth; women could potentially serve as valuable resources if they had the same opportunities as their male counterparts. While the female unemployment rate in Tehran (the Capital) reached 21.6 %, the official data shows that the female unemployment rate reached 50 % in some cities outside Tehran (Devarajan and Mottaghi 2014). The inability of the country to provide a friendly environment for women's participation in the economy can be regarded as a part of the institutional obstacles to entrepreneurship in Iran and it may also be considered as one of the main factors that keep the economy from operating at full capacity.

Iran is suffering from stagflation, with negative growth rates in 2013 and 2014. As Fig. 1.1 shows, the country has witnessed double-digit inflation and double-digit unemployment since 2004. While the goal of policy makers in Iran has been to diversify the economy from dependence on oil earnings, the economy is heavily dependent on oil industry and revenue; oil constituted about 30 % of GDP, 80 % of exports, and 70 % of fiscal revenues in 2012 (Devarajan and Mottaghi 2014).

The inability of the country to have access, use, or repatriate about 44 % of its oil revenue due to US sanctions (Oyster 2013) has destabilized the economy. New tighter international trade and financial sanctions since early 2012 resulted in the reduction of oil production to a 20-year low, the decline of oil-export by more than half (by about 15 % of GDP), and the 80 % devaluation of Iranian currency (Rials)



	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014
Growth rate	5.1	4.6	5.9	7.8	0.6	3.9	5.9	3	3	-5.8	-2.2
Unemployment rate	10.3	12.1	11.6	10.6	10.5	12	13.5	13.3	13.1	10.4	10.7
Inflation rate	14.8	13.4	11.9	17.2	25.5	13.5	10.1	20.6	27.4	39.3	23

Fig. 1.1 Annual percentage growth rate of GDP, unemployment and inflation rates in Iran, 2004–2014 (World Bank 2014)

against the dollar in March 2012 (International Monetary Fund, Country Report 2014b). While the new Iranian administration's positive approach towards internal and international affairs has improved market expectations and the Iran rials could somehow recover against the US dollar, external shocks together with weak macroeconomic management contributed to the decline of the real GDP from 3 % in 2012 to the negative growth rate of about 5.8 % in 2013 (Fig. 1.1).

Iran is among the countries with the highest level of development in telecommunication (growth rate of over 20 %). In addition, the country has extended telecommunication services to rural areas (Abbasi et al. 2008). The most recent Internet World Usage Statistics shows that 57.2 % of the population in Iran use the internet (Internet World Stats 2015), which can provide entrepreneurial opportunities in Information Technology and electronic commerce. Moreover, Iran is a country with high human capital potential. The number of boys and girls studying at universities increased from 344,000 in 1991 to 4,405,000 in 2011 (a 13-fold increase). This is while many job-seeking graduates cannot find a suitable job. Statistics reflect a 1.7–2.3 million unemployed people in the age group of 15–29, in a total unemployed population of 2.6–3.5 million (Abbasi-Shavazi and Sadeghi 2013).

The relative position of Iran in the view of international organizations shows that the country is rich in terms of resources but is unable to allocate and utilize its resources efficiently. Despite being ranked 29 by GDP out of 194 countries across the world (World Development Indicators 2014), Iran ranked 83 among 144 countries in the Global Competitiveness Index (GCI) 2014–2015, and ranked 130 out of 189 economies in the World Bank's Doing Business Index (DBI), 2014 (World Bank 2014, Doing Business).

1.3 Labor Force Review

Iran is going through rapid socio-economic and demographic changes. According to the Statistical Center of Iran, One-third of the Iranian population are between 15 and 29 years old, and 50 % is under the age of 30 (Abbasi-Shavazi and Sadeghi 2013). Iran's young population is both an opportunity and a threat; while the youth economic participation can speed up the country's socio-economic development, the high youth unemployment rate can cause numerous socio-economic problems. Youth account for around 70 % of unemployment in Iran. Unemployment among educated youth is particularly substantial (37 %), including 29 % for men and 48 % for women (ibid). United Nation's Report on MENA region shows that "one in five people living in the Middle East and North Africa (MENA) region is between the ages of 15 and 24, a demographic group called 'youth'" (Roudi 2011).

Figure 1.2 demonstrates Iran's youth population with 10 year intervals between 1976 and 2011. Figure 1.3 shows a rapid growth of youth population (aged 15–29) from 8.5 million in 1976 to 25 million in 2006. While the population aged 15–29 comprised 25 % of Iran's population in 1976, the percentage of this age group significantly increased to 35.4 % in 2006 and started to decline afterwards (Abbasi-Shavazi and Sadeghi 2013). The sharp increase in the youth population during the period of 1996–2006 can be attributed to Iran's government encouragement programs to increase the population in the 1980s.

According to Iran's 1979 Constitution, the government has the duty to provide all citizens with equal and appropriate opportunities and to provide them with jobs. However, the employment rate and the economic participation rate by gender in the country show that, in practice, women (who constitute 49.6 % of Iran's population) and men do not have equal opportunities to enter the labor market. Young men's economic participation in Iran is almost five times that of young women, indicating

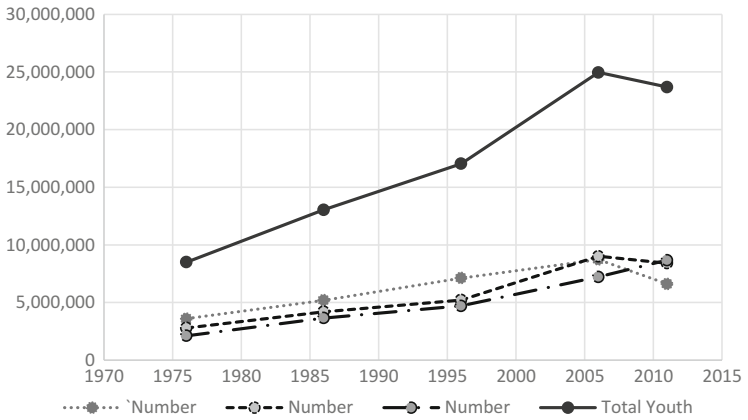


Fig. 1.2 Number of youth population by age groups, 1976–2011 (Abbasi-Shavazi and Sadeghi 2013)

a significant gender gap in economic participation (Abbasi-Shavazi and Sadeghi 2013). According to Norouzi (2008), unemployment rates for women have increased rapidly after 1997 with the rise in the number of college-educated women and the supply of labor. As a result, according to her, the unemployment rate among women with college degrees increased more than fourfolds from 1997 to 2001. Norouzi mentions that the government incentive programs to increase population in 1980s as well as promoting women's education in Iran are some of the reasons behind the increase in female unemployment rates in recent years.

Unemployment rate in Iran varies greatly by gender, age groups, region, and also with a considerable disparity between rural and urban areas. Chronic unemployment is a serious problem in the country. The latest official data on Iran's labor market shows a significant gap between female and male unemployment rates. According to the Statistical Center of Iran (SCI), 2014, the unemployment rate was 43.4 % among women of age 15–24, compared to men's unemployment rate of 21 % in the same age group. Moreover, the unemployment rate among women with a college degree or higher is extremely high across both urban (88 %) and rural (59 %) areas. Unemployment rates also vary remarkably among different regions of the country, ranging from 9.8 % in Kerman to 48.8 % in Lorestan for women versus the lowest rate of 5.4 % in Yazd to the highest rate of 22.2 % in Lorestan for men (ibid).

Figure 1.3 shows that in 2014, the overall unemployment rate for women (19.4) was more than double of that rate for men (9) and the female unemployment rate in urban areas (23.5) was 2.25 times that of rural areas.

Labor force indicators in recent years demonstrate the following points: high gender gap in the overall unemployment rates, significant differences between adult and youth unemployment rates regardless of gender, deep unemployment rate among youth in the age group of 15–24, and deeper unemployment rate among young women compared to young men (Fig. 1.4).

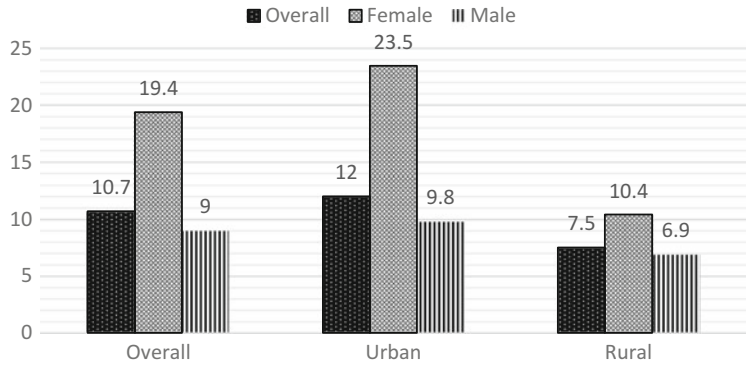


Fig. 1.3 Unemployment rates by gender and region in Iran, 2014–2015 (SCI releases the annual data based on the Iranian New Year which starts in March) (Statistical Center of Iran (SCI), Labor Force Survey Results 2014)

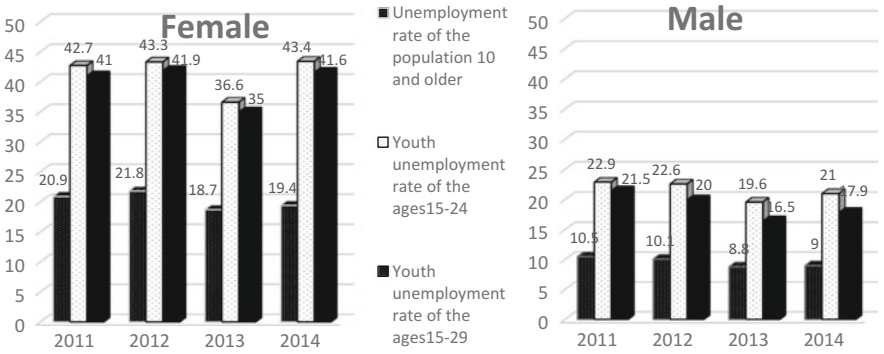


Fig. 1.4 Labor force status by gender in Iran, 2011–2014 (Statistical Center of Iran 2014)

In general, the gender gap in employment has exhibited an increasing trend in recent years. In other words, women's position in the labor market is not improving but deteriorating. Figure 1.4 shows that women unemployment rate (20.9) increased from about 2 times than that of men (10.5) in 2011 to 2.15 times in 2012, and about 2.125 times in 2013 and 2.15 times in 2014. While youth unemployment rate is significantly higher than the overall unemployment rate for both women and men, gender gap in youth unemployment rate is also very notable. Figure 1.4 shows that the ratio of female to male unemployment for youth, in the age group of 15–24, increased from 1.8 in 2011 to 1.91, 1.87, and 2.06 in 2012, 2013, and 2014, respectively. Female to male unemployment ratio of youth, in the age group of 15–29, had also an increasing trend from 1.9 in 2011 to 2.05, 2.12, and 2.32 in 2012, 2013, and 2014, respectively.

Devarajan and Mottaghi (2014) argue that the high rate of growth in labor supply (3.5 % on average), together with the increase in women's participation rate (15 %) in the labor force are the main reasons for chronic unemployment in Iran. In addition, they consider the changing skill mix and the skills mismatch in the labor market as the core issues leading to the development of unemployment in the country. Moreover, government domination over the economy, the country's rent-seeking environment, and the inability of the private sector to flourish and absorb the increasing number of job seekers in the labor market, have deteriorated the employment condition.

Gender gap in employment is even more significant in urban compared to rural areas. In urban areas, the male unemployment rate (11.4 %) is much lower than women's unemployment rate (27.4 %), while in rural areas the male unemployment rate (7.6 %) is only slightly lower than the female unemployment rate (7.8 %) (SCI 2014). This is also true with regard to the youth unemployment rate. In Fig. 1.5 a long-term comparison of labor market in the three decades shows that the gender gap in unemployment rate has persistently been more significant in the urban areas for women aged 15–24 (almost twice that of men in the same age group). It is interesting to note that the urban unemployment rate for both genders decreased from 1986 to 1996 and then increased from 1996 to 2006 while the rural

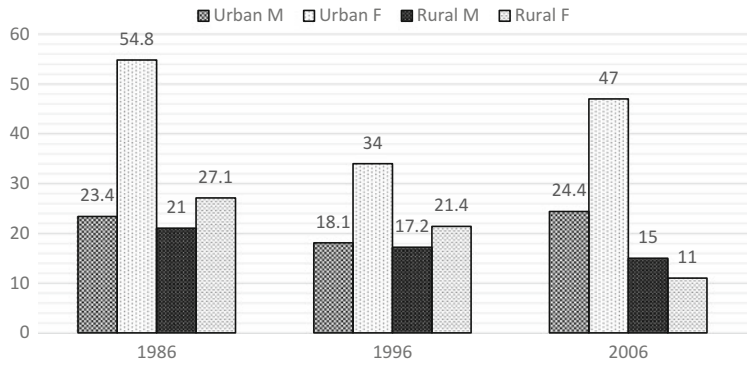


Fig. 1.5 Unemployment rates by gender in urban and rural areas in Iran, for ages 15–24 (United Nations Population Fund, Iran's profile 2014–2015)

unemployment for both men and women declined from 1986 to 2006 indicating flows of immigration from villages to cities. Figure 1.5 shows that, as a whole, the unemployment rates in the urban areas for both genders show a kind of U-shaped trend over the period 1986–2006, while in the rural areas, regardless of the gender, unemployment rate displays a decreasing trend during the period of 1986–2006.

Figure 1.5 indicates that the female unemployment rate was higher than the male unemployment rate in 1986 and 1996 in rural areas, while, due to higher rates of rural-urban migration flows among men than women, the female unemployment rate in rural areas became even less than their male counterparts in 2006. For young women and men in both urban and rural areas in 1996, the unemployment rates were below their corresponding levels 10 years earlier. While there was a marked increase in the unemployment rates in the urban areas 10 years later (2006), the rate of unemployment decreased among both sexes in the rural areas over this period.

In general, the structure of women's employment in Iran shows that highly educated young women, who live in urban areas, bear the highest unemployment rate and come across more problems in finding jobs. This means depriving the society from empowered and educated women who can be the engine of economic growth and elements of diversification in the country once they find a job or run their own startups. The high unemployment rate among young educated women has recently become an important issue in the Iranian society.

Traditionally, public sector jobs have been considered to be attractive in Iran, particularly for women, since they usually offer good salaries and bonuses, and also job security. Hence, women have always been more willing to be employed by government organizations; 27.1 % of women versus 15.2 % of men are employed in the public sector in Iran (SCI 2014). However, due to the privatization policy enforced in the last two decades, the number of such government jobs has been very limited. Moreover, Iran has been faced with both structural problems inside the country and the external pressure of sanctions from the international economies, which have undermined Iran's economy. In addition, due to the improper implementation of the privatization policy and the inefficiency of the private sector, the

Table 1.1 Adult unemployment rates by gender (% of labor force) in Iran and selected regional economies, WEF, Gender Gap Report 2014b

	Iran	Bahrain	Egypt	Pakistan	Qatar	Saudi Arabia	Turkey
Female adult unemployment rate	16.8	3.7	24.1	9	2.8	18.6	10.8
Male adult unemployment rate	9.1	0.4	9.3	5.4	0.1	3.2	8.5

rate of economic growth has not been sufficient to provide jobs for educated youth in the labor market. Furthermore, men are more likely to find a job than women are. One reason of this may be a tendency in the Iranian government and society to consider men as the main breadwinners in a family and therefore be given priority for employment. Under this situation, promoting the culture of entrepreneurship and facilitating women's entrepreneurship in particular, can be a solution to unemployment in Iran. Fostering women's entrepreneurship in the country may increase women's participation in economic and social activities leading to economic growth. There is consensus among most policymakers and academics that entrepreneurship is a catalyst for economic growth and national competitiveness (Kelley et al. 2011).

Gender gap in employment is a universal issue. Women experience higher unemployment rates globally with the female unemployment rate estimated at 6.4 % in 2014, in contrast to a 5.7 % rate for men in the same year (World Economic Forum (WEF): The Global Gender Gap Report 2014b). Women unemployment rate in Iran is dramatically higher than the corresponding world average and the gender gap in the country's labor force status (7.7 points) is also deeper than the world average (0.7 points). This disparity is even true at the regional level. Table 1.1 shows that the female unemployment rate among Iranian women is higher than their counterparts in Bahrain, Qatar, Pakistan, and Turkey.

According to the 2014 Global Gender Gap Report, Iran is among the five lowest performing countries in terms of labor participation but is also part of the countries with a higher percentage of women in tertiary education due to vast investment in women's education. Hence, in Iran "there is a real potential at integrating women into the economy in order to reap the benefits of this investment." (WEF: The Global Gender Gap Report 2014b).

1.4 Women Economic Participation

The percentage of women labor force that is economically active in Iran is extremely low compared to that of men. Figure 1.6 shows the increasing gender gap in male and female labor force status in recent years. Men's economic participation rate was 4.3 times (63.2/14.7) that of women in 2011, followed by 4.43

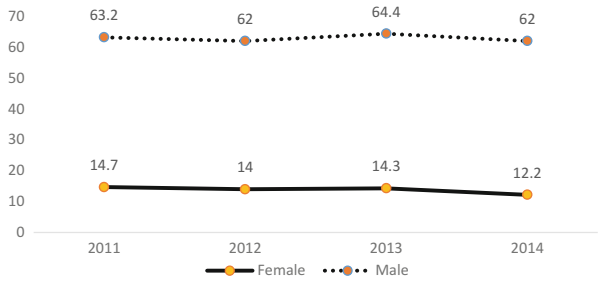


Fig. 1.6 Economic participation rates by gender in Iran, 2011–2014 (Statistical Center of Iran, Labor Force Survey Results 2014–2015)

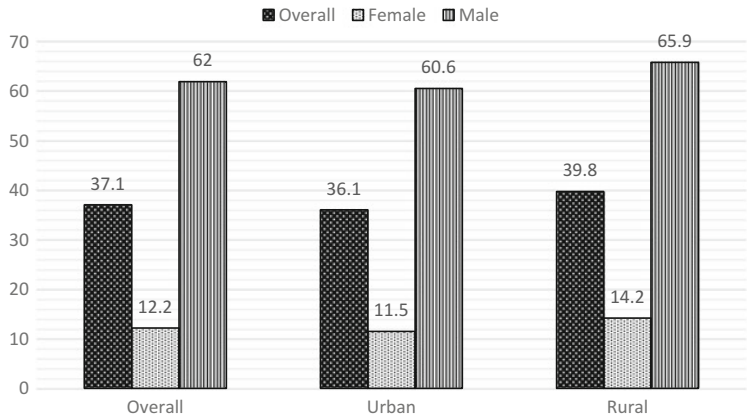


Fig. 1.7 Economic participation rates by gender in Iran, 2014–2015 (Statistical Center of Iran, Labor Force Survey results 2014–2015)

(62/14) times in 2012, 4.5 (64.4/14.3) times in 2013, and more than 5 times (62/12.2) in 2014.

In spite of high public spending on education in Iran (4.05 % of GDP in 2011) and the increasing number of young women with university degrees, the gender gap in economic participation rate in the country is still very significant in favor of men. Gender disparity in economic participation is more noticeable in urban areas compared to rural areas. Figure 1.7 shows that while on average, men's economic participation rate is 5.1 times (62/12.2) that of women in the country, the disparity is more intensified in urban areas (men 5.27 times women) compared to rural areas (men 4.64 times women).

The rate of female economic activity has been persistently lower compared to their male counterparts in the urban as well as rural areas. Figure 1.8 shows the economic participation rates by gender within a 10-year interval starting from 1986 to 2006. Men's economic participation rates follow a U-shaped curve in both urban and rural areas indicating a decrease in the rate from 1986 to 1996 and then

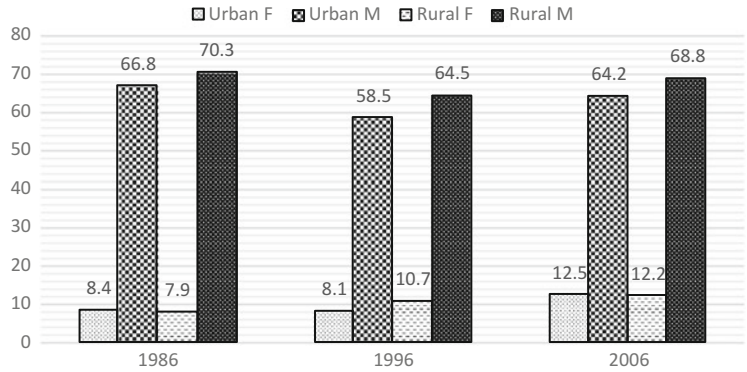


Fig. 1.8 Economic participation rates by gender in Iran in 1986, 1996, and 2006 (United Nations Population Fund, Iran's profile 2014)

increasing from 1996 to 2006 in the mentioned areas. The rate of economic participation for women declines slightly from 8.4 % in 1986 to 8.1 % in 1996 and then increases to 12.5 in 2006.

The female economic activity rate in Iran is not only very low compared to their male counterparts, but it also appears to be lower than the female labor participation rates in many other regions and across the world. Statistics show a significant progress in female labor force participation rates around the world in the last decades. For instance, the rate of female economic participation "in the prime ages of 25–54 years continued to rise in the 1990s to between 60 to 85 percent" (Lim 2002). "The Arab region has witnessed a 19 % increase in women's share of economic activity between 1990 and 2003" (Kwon 2009), and "more than half a billion women have joined the world's labor force over the past 30 years, and women now account for more than 40 percent of workers worldwide" (Revenga and Shetty 2012). The female participation rate (14.1 %) in Iran is drastically lower compared to the average of the MENA region (26 %), which is in turn well below the 39 % rate of female participation in Low and Middle Income countries as well as the world average rate (World Bank 2010). In another study, The United Nation's Human Development Report (2014) also shows low women economic activity rate in Iran (16.4 %) compared to women economic participation rate in Bangladesh (57.2 %) and the corresponding rate for women in Pakistan (22.7 %).

To develop a comparative understanding of women in the labor market of Iran, the relative position of Iranian women in the labor market will be compared to their female counterparts in some selected countries. These countries, mentioned as comparator economies throughout the chapter include Turkey and Pakistan (that share borders with Iran), Saudi Arabia (which is ranked right above Iran as the largest economy in MENA) and Egypt (that is ranked right above Iran as the most populated country in MENA). However, it should be noted that in some parts of the book, due to the unavailability of data in some years for some countries under study, the comparator economies may change.

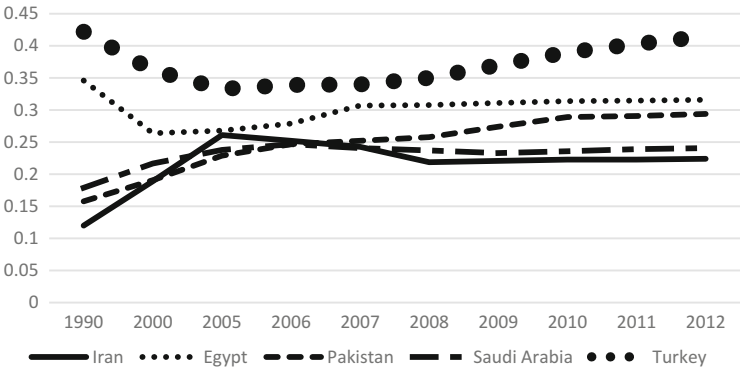


Fig. 1.9 The ratio of female to male labor force participation rate (% of population ages 15+) in Iran and the comparator economies (United Nations Development Program, Human Development Reports 2014)

Figure 1.9 displays the ratio of female to male labor force participation rate in Iran's and comparator countries' working-age population (ages 15 and older) that engages in the labor market (UNDP 2014). As Fig. 1.9 shows, the female/male ratio in the labor force participation in Iran was less than all compared sample economies in 2011. Moreover, the gap between Turkey and Iran in terms of female/male ratio almost doubled in 2011 compared to 2005. While the female labor force participation rate in Iran decreased significantly from 2005 to 2008, this rate increased in almost all selected countries with which Iran is compared.

Fig. 10.1 shows that some advances in economic participation among women in Iran occurred during 1990–2005, and started to decline during 2005–2008. It is worth noting that Mohammad Khatami's presidency was during 1997–2005 and Mahmoud Ahmadinejad's presidency was during 2005-2013. The fact that the change in the trend of Iranian women's economic participation rate coincides with a major change in the government (namely, election of a new president) could highlight the key role and the crucial impact of the government's attitudes and policy towards women issues in Iran. This observation may suggest that the gender perspective of President Khatami's administration, as a part of his government policy, might have had a role in improving the conditions for women's economic participation.

Does unemployment lead to entrepreneurship and increase female economic participation rate? Empirical results confirm the existence of a relationship between unemployment and entrepreneurial activities. However, there is no consensus on the relationship between unemployment and entrepreneurship. Entrepreneurship may be considered as a way to avoid unemployment and poverty in some countries (Allen et al. 2008). Even though unemployment could inspire women to start their own business and become self-employed, an entrepreneurial ecosystem with equal access to resources and opportunities for women and men is required to motivate women towards entrepreneurship. The feasibility of doing business, and the

existence of entrepreneurial support programs and networks can also build awareness and incentivize women's entrepreneurship. The willingness of unemployed women to become self-employed may lead to entrepreneurial activity in a favorable business environment.

1.5 Business Environment in Iran

In each economy, women entrepreneurs are closely affected by the general framework condition for entrepreneurship (Delmar 2003); notwithstanding, gender biases affect women in more profound ways. Many of the same factors influence women and men when they decide to start a business (Allen et al. 2007). In order to shed light on the general business environment in Iran, the World Bank Doing Business (DB) Indicators will be used to show the feasibility/complexity of running a business in the country, an external factor that affects both female and male business owners. In 2014, Iran's ranking in the ease of Doing Business Indicators remained unchanged from 2013 in 152th place out of 189 economies. This ranking is one place below Iran's ranking in 2012 and five places below its ranking in 2011. According to the World Bank, the index measures the regulations affecting a business and is calculated as the ranking on the simple average of its percentile rankings on each of the 10 topics (applied to domestic small, medium size businesses) included in the index in Doing Business.

Figure 1.10 shows the rankings of Iran in terms of the following topics that are included in the index of Doing Business:

(a) Starting a Business, in which Iran's rank was **62** in 2015 versus 107 in 2014 and 87 in 2013
(b) Dealing with Construction Permits, in which Iran's rank was *172* in 2015 versus 169 in 2014 and 166 in 2013
(c) Getting Electricity, in which Iran's rank was **107** in 2015 versus 169 in 2014 and 163 in 2013
(d) Registering Property, in which Iran's rank was **161** in 2015 versus, 168 in 2014 and 165 in 2013
(e) Getting Credit, in which Iran's rank was *89* in 2015 versus 86 in 2014 and 83 in 2013
(f) Protecting Minority Investors, in which Iran's rank was *154* in 2015 versus 147 in 2014 and 150 in 2013
(g) Paying Taxes, in which Iran's rank was **124** in 2015 versus 139 in 2014 and 129 in 2013
(h) Trading Across Borders, in which Iran's rank was 148 in 2015 versus 153 in 2014 and 143 in 2013
(i) Enforcing Contracts, in which Iran's rank was *66* in 2015 versus 51 in 2014 and 53 in 2013
(j) Resolving Insolvency, in which Iran's rank was *138* in 2015 versus 129 in 2014 and 126 in 2013

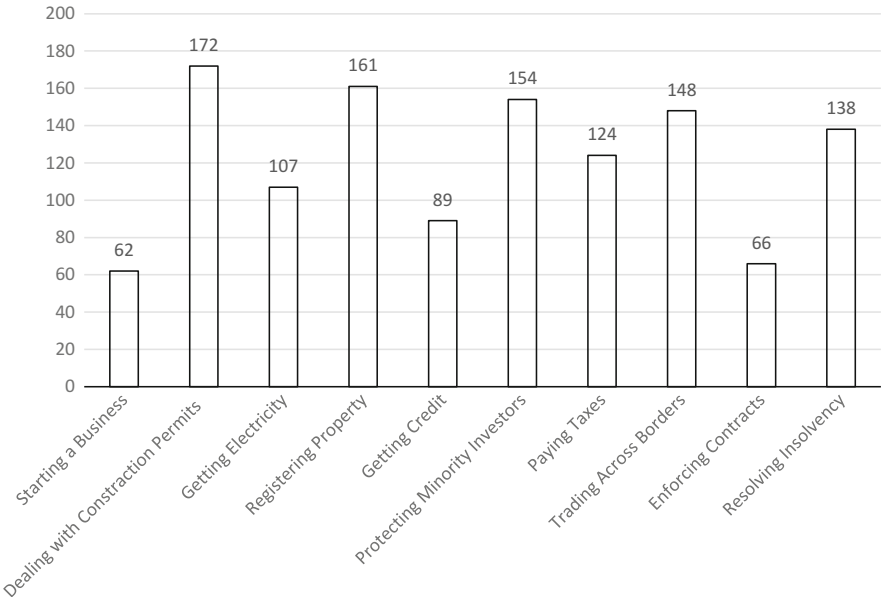


Fig. 1.10 Rankings on Doing Business topics in Iran (out of 189 economies) (World Bank 2014. Doing Business 2015: Going Beyond Efficiency. Economy Profile 2015)

The aforementioned data demonstrate improvement in the rankings of Iran with respect to some indicators of "Doing Business" (marked in **bold**) such as starting a business, dealing with construction permits, getting electricity, and paying taxes in recent years. It also shows dropping in the rankings of the country (marked in *italic*) in the topics of dealing with construction permits, getting credit, protecting minority investors, enforcing contracts, and resolving insolvency in 2015 compared to the previous 2 years.

While Doing Business ranking does not measure all aspects of business climate that affect the business owners and entrepreneurs, it displays the conductivity of the regulatory environment a government has created for the private sector (World Bank 2013). Doing Business is a benchmark to compare and track the complexity of doing business in different countries.

To identify and compare some sources of business obstacles in the sample group countries, six indicators (the ease of starting a business, dealing with construction permits, registering property, getting credit, protecting investors, and paying taxes) of doing business in these countries will briefly be reviewed. More information on all indicators on Doing Business is available in the full report of the World Bank's Doing Business 2015 (World Bank 2014).

Table 1.2 shows that Iran stands in a higher position than all sample countries with respect to starting a business. This means that the regulatory environment for opening a business is more conducive in Iran compared to other comparator countries. To start a business, companies need to be formally registered to be

Table 1.2 The rankings of Iran, Egypt, Pakistan, Saudi Arabia and Turkey (among 189 countries) on some indicators of Doing Business, World Bank (2014). Doing Business 2015

	Iran	Egypt	Pakistan	Saudi Arabia	Turkey
Starting a Business	62	73	116	109	79
Dealing with construction permits	172	142	125	21	136
Registering property	161	84	114	20	54
Getting credit	89	71	131	71	89
Protecting investors	154	135	21	62	13
Paying taxes	124	149	172	3	56

protected by law and have access to services and institutions, i.e. getting credit from banks and entering new markets. If formal regulations to enter a business is too costly or complicated, the tendency to go to informal options increases among those who want to start a business. In dealing with construction permits (DWCP), Iran (with the ranking of 172) has the weakest performance among the comparator economies. Saudi Arabia, with significant differences compared to other comparator countries, stands in a high ranking of 21 in DWCP indicator. A business is required to obtain all necessary approvals in order to build a warehouse and connect it to basic utilities. DWCP is essential for public safety. While to avoid hazardous construction, builders are required to have a permit, it needs, however, to be efficient in terms of cost and time. Builders may be tempted to pay bribes or build illegally in a bureaucratic environment where regulations are costly in time and money. When enterprises attain all the necessary approvals and build a commercial warehouse, they can register the property and use it as a collateral. Table 1.2 shows that for an entrepreneur in Iran it is excessively difficult to obtain construction permit and register the property compared to entrepreneurs in comparator economies.

Iran performs better in getting credit than many other areas of the Doing Business indicators, as is shown in Table 1.2. While Saudi Arabia and Egypt stand in a better position (both countries rank 71) in the topic of getting credit, Iran and Turkey rank 89, and Pakistan has the lowest rank of 131 in this indicator. The index of getting credit is measured by the feasibility of credit information systems and the sound collateral laws. Credit information systems help lenders to identify potential borrowers (by viewing their financial history) and also provides the opportunity for borrowers to make easier access to credit by establishing a good credit history. With respect to getting credit, except for the Pakistan, the gap between the comparator economies is not very significant.

The topic of investor's protection measures "the strength of minority shareholder protections against directors' use of corporate assets for personal gain—or self-dealing" (World Bank Doing Business 2013). Table 1.2 shows that Pakistan stands in a better position than all comparator economies in showing more transparency of related-party transactions and stronger investor protections against self-dealing in the areas measured. Table 1.2 displays the very weak performance of Iran (rank 154) and Egypt (rank 135) in protecting the investors compared to the comparator

countries. It is worthy to mention that, in 2013, Iran was the only economy ranked above 100 in the investor's protection topic among comparator countries (at that time Egypt ranked 82). According to Doing Business 2013, "Iran strengthened investor protections by requiring greater immediate disclosure of related-party transactions in 2012".

With respect to paying taxes, Iran is ranked 139; standing well below Saudi Arabia (3) and Turkey (56), and above Egypt (149) and Pakistan (172). This indicator covers the tax rate and the complexity in tax rules. The complexity of paying taxes in Iran, Egypt and Pakistan can be a motivation for business owners to end up in the informal sector which, in turn expands informal sector and decreases the revenue of the government.

1.6 Doing Business in Iran

The complicated business environment in Iran indicates how hard it is for both men and woman to run a business successfully. To gain a better understanding of the difficulties in building startups, the number of procedures, the money paid and the days required to starting a business, dealing with construction permits and registering property in Iran will briefly be studied and compared with the comparator countries. Friendly business environment is reflected in small numbers of procedures, short delays, and low costs on obtaining legal permission to start a business. Entrepreneurs in an easy business environment have cost and time advantage over their counterparts in tough business environment.

With respect to the ease of starting a business, as Table 1.3 shows, the number of procedures and the cost an entrepreneur is officially required to take to start a business in Iran is less than the comparator economies. While, in general, starting a business is more conductive in Iran than the comparator countries, fewer days is required to complete the procedures in Turkey and Egypt than other countries in the group (Table 1.3). According to the World Bank Doing Business 2013, Egypt has had the most progress in the Middle East and North Africa since 2005 which was continued in the years before 2009; since then, because of the economic, social, and political issues, the trend of reform process has been very slow.

Table 1.3 shows that the days required to deal with construction permits is considerably higher in Iran than other comparator economies. For instance, the official time of dealing with construction permit in Iran is more than three times that of Saudi Arabia, which gives Saudi Arabian entrepreneurs comparative advantage over entrepreneurs from Iran.

Regulatory processes and transparency can motivate more entrepreneurs to start businesses in the formal sector. The days required for entrepreneurs in Iran to register property is about 6 times that of their counterparts in Turkey and 4.5 times that of the start-up entrepreneurs in Saudi Arabia. While it took 22 days for new entrepreneurs in Egypt to register their properties in 2013 (Doing Business 2013), in

Table 1.3 The procedures, days and costs required for starting a business, dealing with construction permit, and registering property in Iran, Egypt, Pakistan, Saudi Arabia and Turkey, World Bank (2014). Doing Business 2015

	Starting a business			Dealing with construction permits			Registering property		
	Procedures (number)	Time (day)	Cost (% of income per capita)	Procedures (number)	Time (day)	Cost (% of warehouse value	Procedures (number)	Time (day)	Cost (% of property value)
Iran	6	12	3.1	15	318.5	5.3	9	35	10.6
Egypt	7	8	9.2	20	179	1.9	8	63	0.7
Pakistan	10	19	9.6	10	249.4	3.5	6	50	7.6
Saudi Arabia	9	20.5	4.0	11	102	0.4	5	8	0.0
Turkey	7	6.5	16.4	18	169	3.5	6	6	4

2014 the enterprises were required to wait 63 days for property registration, higher than Iran and Pakistan.

Time and costs are the main bases for competition and the crucial elements of business performance. "As a strategic weapon, time is the equivalent of money, productivity, quality, even innovation" (Stalk 1988). Whereas business-owners do not pay anything to register their properties in Saudi Arabia, entrepreneurs in Iran are required to pay 10.6 % of their property value for property registration, much higher than the cost of registering property in Egypt (0.7 %), Turkey (4.0 %) and Pakistan (7.6 %).

In the world of globalization, the role of competition in the economy is felt more than ever. Competition is the key to success in the aggressive business world. No matter how smart, capable, and skillful business-owners are, they have limited or no control on their external environment. Hence, providing a conductive business environment needs to be a strategic priority of a government for fostering entrepreneurship. Regulatory processes and transparency benefits business-owners and governments. Lower number of procedures, fewer days of processing legal documents and lower costs may encourage private sector investment in job-creating startups, create more revenue for government, and prevent bribes and corruptions in economy.

1.7 Changes in the Regulatory Environment

Doing business allows policy makers to analyze and compare "doing business indicators" in their country to those in other countries to evaluate how these indicators can be changed to improve the business environment. Here, the changes affecting starting a business, as a sample of 1 of the 11 areas of indicators in the Ease of Doing Business in Iran will be studied.

A time-period review of starting a business in Iran in Table 1.4 displays how the economy's regulatory environment has changed with respect to starting a business indicator since 2004. This would clarify the steps undertaken by the government for making business easier (or sometimes more difficult). It can also help to identify the potentials for improvement.

Table 1.4 shows that in 2012 compared to 2004 the number of procedures needed to legally start and operate a company decreased from 9 to 6, the days required to complete the procedures decreased from 29 to 8, the cost of completing the procedures decreased from 6.5 % of income per capita to 3.8 % and the paid-in Minimum Capital decreased from 2.5 to 0.7 % of the income per capita. Paid-in Minimum Capital is the amount required to be deposited in a bank or with a notary before registration. In 2013, despite lowering the cost and the paid-in-minimum capital of starting a business in Iran, the rank of the country in starting a business dropped from 50 in 2012 to 87 in 2013. This could be due to an improvement in the indicators of starting a business in other countries and/or as a result of the increase in the procedure numbers and the delay in the time of processing legal registration.

Table 1.4 How the regulatory environment changed over time with respect to the Starting a Business topic in Iran, World Bank Doing Business 2013 and 2015

Year	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2015
Rank	–	–	–	–	–	–	–	–	50	87	62
Procedures (numbers)	9	9	8	8	8	8	7	6	6	7	6
Time (days)	29	29	28	28	28	28	9	8	8	13	12
Cost (% of income per capita)	6.5	7.3	6.3	5.4	5.3	4.6	3.9	4.0	3.8	3.3	3.1
Paid-in min. capital (% of income per capita)	2.5	2.1	1.7	1.3	1.3	1.0	0.8	0.8	0.7	0.5	0.0

The rankings are measured relatively and it seems that the majority of countries attempt to move toward efficient practices and strong regulations to make doing business easier and to protect enterprises in their economies. No reform was taken in Iran in 2014 in starting a business. In 2015, starting a business became easier in Iran by streamlining the name reservation and company registration procedures (World Bank Doing Business 2015).

The speed and the extent of changes are the keys to improving the business climate in this fast-paced, competitive environment. Governments increasingly recognize the importance of providing a regulatory environment for the private sector to meet their economic development goals. In a fast changing global business environment, governments are required to create a friendly entrepreneurial environment to increase the competitive power of their economy.

Iran has taken some reforms in doing business topics in the recent years. Table 1.5 shows how doing business has changed over time in terms of starting a business, dealing with construction permits, registering property, getting credit, protecting investors, and paying taxes in 2010, 2011, 2013 and 2015; no significant reform was taken in 2008 and 2009, 2012 and 2014 in Doing Business indicators.

The Iranian government made progress on investor protections "by requiring greater immediate disclosure of related-party transactions" in 2013 (Doing Business 2013). In the same year, Iran "made starting a business more difficult by requiring company founders to obtain a criminal record clearance to register a new company" (ibid, 2013). In 2015, starting a business was simplified by streamlining the name reservation and company registration procedures (Doing Business 2015).

1.8 Entrepreneurial Framework Conditions

The dynamics of entrepreneurship is greatly affected by the business environment and the entrepreneurship ecosystem. The creation of new firms and the sustainability of the enterprises are dependent on the entrepreneurial socio-economic environment. Global Entrepreneurship Monitor (GEM) provides a model known as the "National Expert Survey" (NES) to measure the Entrepreneurial Framework Conditions (EFC) across the globe. NES provides a general understanding of how experts evaluate the EFC of a country. In NES, at least 36 experts including entrepreneurs, consultants, politicians, and academics in GEM participating countries are selected (four experts for each of the nine indicators of EFC) to complete a questionnaire relating to their domestic and/or regional entrepreneurial environment. Experts express their perspectives and judgments on a 1-to-5 scale, where 1 means complete disagreement with the statement and 5 means complete agreement with the statement. EFCs are recognized by GEM as one of the key indicators of any entrepreneurship ecosystem. The results shed light on the structural characteristics of a given country/region and "provides insights into the ways in which these EFCs either foster or constrain an entrepreneurial climate, activity and

Table 1.5 The reforms taken by Iran in Doing Business from 2010 to 2015, World Bank (2013). Doing Business 2013. Economy Profile and World Bank (2014). Doing Business 2015

Indicators	2010	2011	2013	2015
Starting a business	Electronic registration system simplified business start-up.	Installation of a web portal allowed entrepreneurs to search for and reserve a unique company name.	Starting a business became more difficult by requiring company founders to obtain a criminal record clearance to register a new company.	Streamlining the name reservation and company registration procedures.
Dealing with construction permits	This indicator improved with the introduction of e-service offices in Tehran, streamlining the process for obtaining location approvals, construction permits and building completion certificates, and the time to obtain water and electricity connections fell substantially.	No reform	No reform	No reform
Registering property	No reform	No reform	No reform	No reform
Getting credit	No reform	Access to credit information improved with the establishment of a new private credit bureau.	No reform	No reform
Protecting investors	No reform	No reform	Iran strengthened investor protections by requiring greater immediate disclosure of related-party transactions.	No reform
Paying taxes	The tax burden on business was eased by converting the sales tax into value added tax.	No reform	No reform	No reform

Table 1.6 GEM's key entrepreneurial framework conditions,[a] Singer et al. (2015)

(1) Entrepreneurial Finance The availability of financial resources, equity, and debt, for new and growing firms, including grants and subsidies.	**(2) Government Policy** Government Policy. The extent to which public policies give support to entrepreneurship. This EFC has two components: (2a) Entrepreneurship as a relevant economic issue (2b) Taxes or regulations are either size-neutral or encourage new and SMEs.	**(3) Government Entrepreneurship Programs** The presence and quality of programs directly assisting SMEs at all levels of government (national, regional, municipal).
(4) Entrepreneurial Education The extent to which training in creating/managing new, small or growing business entities is incorporated within the education and training system at all levels. This EFC has two components: (4a) Entrepreneurship Education at basic school (primary and secondary) (4b) Entrepreneurship Education at post-secondary levels (higher education such as vocational, college, business schools, etc.).	**(5) R&D Transfer** The extent to which national research and development will lead to new commercial opportunities, and is available to SMEs.	**(6) Commercial and Legal Infrastructure** The presence of property rights, commercial, accounting and other legal and assessment services and institutions that support or promote SMEs.
(7) Entry Regulations Contains two components: (7a) Market Dynamics: the level of change in markets from year to year. (7b) Market Openness: the extent to which new firms are free to enter existing markets.	**(8) Physical Infrastructure** Ease of access to physical resources-communication, utilities, transportation, land or space—at a price that does not discriminate against SMEs.	**(9) Cultural and Social Norms** The extent to which social and cultural norms encourage or allow actions leading to new business methods or activities that can potentially increase personal wealth and income.

[a]Reproduced with permission of the Global Entrepreneurship Research Association (GERA). Table 3.1 from the Global Entrepreneurship Monitor 2014 Global Report by Singer, Amoros and Moska

development" (Xavier et al. 2013). Experts also evaluate women's entrepreneurship as an additional topic related to business environment in the NES survey.

Table 1.6 shows the GEM's key EFCs. While each distinct factor in Table 1.6 is crucial for providing an efficient and dynamic business environment, they also need to interact efficiently and reinforce each other to create a highly productive entrepreneurial ecosystem. According to Zali et al. (2011), 41 % of the experts in Iran found financial problems as one of the three important constraints in entrepreneurial

activity while 9 % considered financial availability as a sign of entrepreneurial protection in the country.

The NES results indicate the poor performance of the financial system in Iran and the need for restructuring financial institutions. Over 40 % of the required financial capital in Iran is borrowed from close family members, and another 15.3 % from other relatives (Razavi and Zali 2008). The lack of financial availability in the country may motivate startups to lend from informal sector. However, while historically in Iran, the parallel financial markets have been a traditional source of financing, after the Islamic Revolution in 1979, borrowing from informal sector was made illegal. Despite illegality, when entrepreneurs cannot obtain credit from the formal sector and find no other option, they turn to informal financing which usually comes at considerably higher interest rates than formal financing.

According to the GEM Iran, 36 % of experts asserted that the government policy on entrepreneurship development was one of the three important factors in the protection of entrepreneurship in the country and only 8 % counted the government programs as barriers to entrepreneurship development (Zali et al. 2011). In order to build entrepreneurial culture, governments need to develop entrepreneurial minds through entrepreneurship education and training systems at all levels of primary, secondary, and post-school. In 46 economies that participated in GEM 2010 NES, education and training in primary and secondary school was one of the three worst performing EFCs (Kelley et al. 2011).

NES provides policy makers and other stakeholders with a better understanding of the strengths and weaknesses of a nation's ecosystem in entrepreneurship from the perspective of national experts. In the NES 2012, the Iranian experts recognized commercial infrastructure, internal market dynamics, and physical infrastructure as the most positively valued EFC, and national policy and regulations, government entrepreneurship programs, and primary and secondary education, as the most negative valued EFC (Zali et al. 2012).

1.9 Entrepreneurship Ecosystem in a Comparative View

Iran and Turkey are the only economies among the comparator countries of Egypt, Pakistan, and Saudi Arabia that participated in the GEM NES 2014. Figure 1.11 compares the expert's perspective on EFCs in Iran and Turkey.

Figure 1.11 shows that in 2014, experts in Turkey had higher evaluation of their domestic EFCs compared to the Iranian experts' evaluation of Iran's EFCs in all indicators, except for one that is "physical infrastructure". While experts in Iran gave the highest score to the Physical Infrastructure, experts in Turkey considered the dynamics of internal market as the most positive valued EFC in Turkey. Experts in Iran have a relatively positive view on the dynamics of internal market versus market openness, an indicator that exhibits the extent to which new firms are free to enter existing markets. Inconsistencies in the open market and market dynamics is observed in the developing economies (Zali et al. 2011) and may be attributed to the

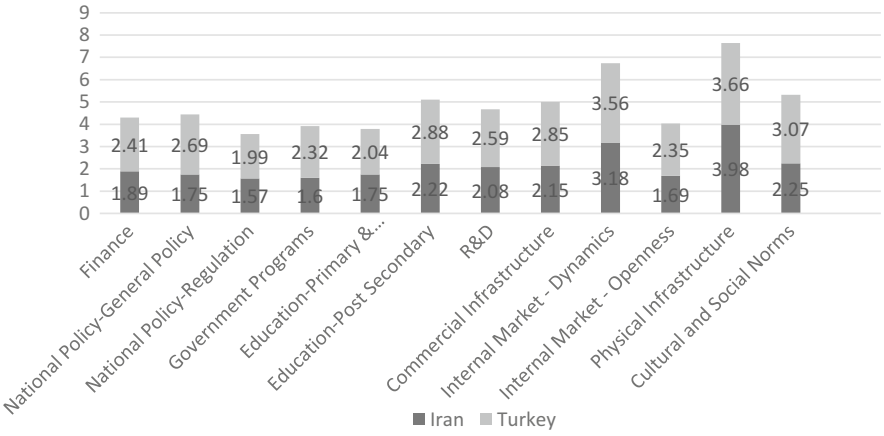


Fig. 1.11 Indicators on entrepreneurship framework conditions, Iran vs Turkey, 2014 (Zali et al. 2015)

inefficiency of institutions, laws and regulations that prevents new entries to the market in spite of the high demand for new products. Experts in both countries of Iran and Turkey gave the lowest ratings to the national policy-regulations.

Among the GEM NES Surveys in the past years, NES 2010 is the most recent one that includes all the comparator countries in this study (Egypt, Iran, Pakistan, Saudi Arabia and Turkey) in the EFCs review. Hence, NES 2010 will shortly be studied here to enable us to compare the perceptions of the experts in Iran with their peers in comparator countries. To do this, the following data from GEM Iran NES 2010 will be used (Zali et al. 2011).

Table 1.7 shows that, in their evaluation of all the indicators of EFC, experts in Iran gave the lowest score to entrepreneurship education at primary and secondary school levels (marked in *italic* in Table 1.7) and the highest score to the physical infrastructure indicator (marked in **bold** in Table 1.7). In general, experts in Iran seem to be less optimistic than their peers in other comparator countries in all indicators of EFC. Experts in Saudi Arabia were more optimistic than their peer experts in other comparator economies with respect to the indicators (marked by * in Table 1.7) of finance, regulations in national policy, government programs, R&D transfer, commercial infrastructure, physical infrastructure, and cultural and social norms.

Experts in Turkey gave greater scores to the indicator (marked by * in Table 1.7) of entrepreneurship education at primary and secondary school levels than their peers in comparator countries. Experts in Egypt had higher perception on general national policy and also on dynamics of internal market (marked by * in Table 1.7) in Egypt than their peers in the comparator countries. In comparison with their peers in the comparator countries, the evaluation of experts in Pakistan was stronger on post-secondary entrepreneurial education and also on the openness of internal market (marked by * in Table 1.7). In general, among 53 GEM participating

Table 1.7 Entrepreneurship framework conditions main indicators in Iran, Egypt, Pakistan, Saudi Arabia and Turkey

	1	2a	2b	3	4a	4b	5	6	7a	7b	8	9
Iran	1.82	2.02	1.96	1.8	*1.48*	2.47	2.02	2.52	3.24	1.74	**3.36**	2.53
Egypt	2.4	2.69*	2.2	2.12	1.27	2.11	1.83	2.64	3.36*	2.21	3.62	2.1
Pakistan	2.46	2.01	2.25	1.98	1.52	2.79*	2.09	3.3	3.03	2.79*	3.55	2.88
Saudi Arabia	3.1*	2.46	3.04*	2.28*	1.8	2.67	2.58*	3.35*	2.95	2.54	4.18*	3.14*
Turkey	2.06	2.57	2.22	2.21	2.21*	2.52	2.37	2.77	2.63	2.18	3.33	2.06

1: Finance; 2a: National Policy—General Policy; 2b: National Policy—Regulation; 3: Government Programs; 4a: Education—Primary & Secondary; 4b: Education—Post-Secondary; 5: R&D Transfer; 6: Commercial Infrastructure; 7a: Internal Market—Dynamics; 7b: Internal Market—Openness; 8: Physical Infrastructure; 9: Cultural and Social Norms; Zali et al. (2011)

countries in 2010, the best rank of Iran in EFC was on market dynamics (15) and the worst rank was on the market openness (53).

1.10 Expert's Perspective on Female Entrepreneurial Environment

While startup owners (regardless of gender) are affected by the business environment, it seems that female entrepreneurs are more likely to face barriers to start a venture than their male counterparts. In the GEM National Expert Survey, experts are asked, through a harmonized internationally comparable questionnaire, to evaluate the level of support for women's entrepreneurship in their countries. There are five questions in the NES questionnaire that are directly related to the female business. Experts are asked to complete and score the following NES questions:

1. In my country, men and women have the same level of knowledge and skills to start a new business.
2. In my country, men and women get equally exposed to good opportunities to start a new business.
3. In my country, women are encouraged to become self-employed or start a new business.
4. In my country, starting a new business is a socially acceptable career option for women.
5. In my country, there are sufficient social services available so that women can continue to work even after they start a family.

The experts' responses are based on a 1–5 scale, where 1 shows complete disagreement with the statement and 5 expresses complete agreement with the statement. Figure 1.12 summarizes the results of the evaluation of the experts in Iran, Pakistan, and Turkey.

Figure 1.12 shows that in general, experts in Iran have a more negative perception of the conductibility of entrepreneurial climate for women compared to their counterparts in Turkey and Pakistan. Experts in Turkey, in comparison with experts in Pakistan and Iran, have a stronger belief that to start a business, men and women have the same level of knowledge and skills and get equally exposed to good opportunities. They also give higher ratings to the availability of sufficient social services for women to continue to work even after they start a family than the experts in Pakistan and Iran. On the other hand, experts in Pakistan have a stronger evaluation on the social acceptability of women's startups and the encouragement for women to become self-employed or start a new business than the experts in Turkey and Iran.

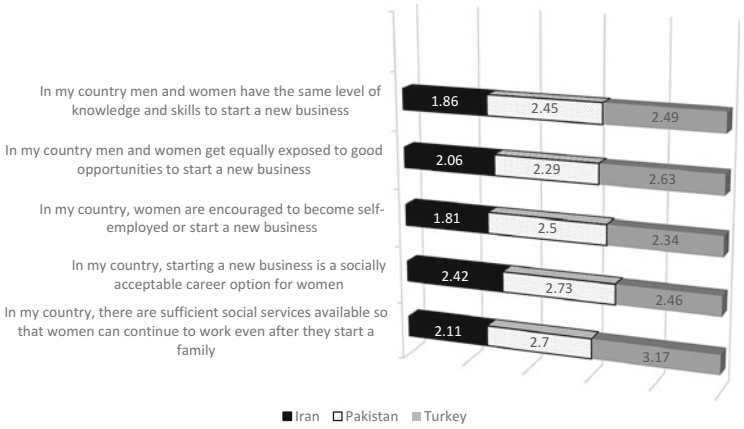


Fig. 1.12 NES questions and score results related to female entrepreneurship in Iran, Pakistan, and Turkey, 2010 (Zali et al. 2011)

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Chapter 2
Dynamics of Women's Entrepreneurship Development in Iran

2.1 Launching the Entrepreneurship Program in Iran

The issue of entrepreneurship has been an important subject in Iran's national strategic objectives since the Third Economic Development Plan[1] (2000–2004). In spite of prioritizing women issues in Iran's Third and Fourth (2005–2009) Economic Development Plans and allocating one chapter to "Women and the Economy", there have been no programs devoted specifically to women's entrepreneurship development. Different sources show limited opportunities for women in Iran to be involved in business activities.

As part of the aforementioned national plans, the Entrepreneurship Development Program in Universities was launched in 12 universities (and later extended to 44 universities) under the supervision of the Management and Planning Organization and the Ministry of Science, Research, and Technology. In the pursuit of promoting entrepreneurship in the country, the Faculty of Entrepreneurship was founded at the University of Tehran, the university that also represents Iran in the Global Entrepreneurship Monitor (GEM).

GEM, which started in 1999, is a major research project designed to describe and analyze entrepreneurial processes within a wide range of countries (Allen et al. 2007). This international consortium measures and systematically studies the attitudes, activities and characteristics of individuals participating in various stages of entrepreneurship through the GEM Adult Population Survey (APS). The GEM APS database has grown to nearly two million observations in 104 economies that have participated in GEM between 1999 and 2013 (Amorós and Bosma 2014). Complementing the APS is the National Expert Survey (NES); an instrument that develops insight into the entrepreneurial environment of the participating economies based on the judgement of national experts. Iran has participated in GEM since

[1]Iran's strategic objectives are classified in the country's Economic, Social, and Cultural Development Plans for a 5-year period.

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2008. Joining GEM has provided the Iranian policymakers and academics a more comprehensive perspective on entrepreneurial activity, and has given them insight into the country's relative position in terms of entrepreneurship compared to other GEM members.

The low rate of female entrepreneurship—accompanied by a high rate of female unemployment—in Iran may indicate environmental constraints and the inability of the society to benefit from women as untapped potential entrepreneurial human resources. While the number of young women with university degrees is increasing very rapidly in Iran, the unemployment rate among Iranian women is still very high compared to their male counterparts in the country, region and across the world. Promoting gender equitable socio-economic opportunities in a conductive environment in Iran can provide avenues for women's involvement in business activity. Women may be more motivated to choose entrepreneurship as a career in a friendly, dynamic business environment. It is also logical to assume that entrepreneurship could be an option for those who are looking for a job.

To provide a comparative perspective on women's entrepreneurial activity in Iran, entrepreneurial behavior of women in Iran will be compared to that of selected countries including Egypt, Pakistan, Saudi Arabia and Turkey. These four countries comprise our benchmark group, which shall be referred to as the "comparator economies" in the remainder of this chapter. According to the World Economic Forum's (WEF) Global Gender Gap Report 2013, the ability of women in Iran and the MENA region to rise to positions of enterprise leadership is 4 based on a 1–7 scale. This rate is the same as the score that female business-owners earned in Egypt, Pakistan and Turkey, and is higher than the female related score in Saudi Arabia (3).

2.2 GEM Conceptual Framework

To view the process of the entrepreneurial journey, the attitudes and planning involved in starting a venture—from just starting to running a new business and to developing it into an established enterprise in Iran—will be studied for both women and men. Every GEM participant across the world is required to interview a random representative sample of at least 2000 adults (aged 18–65) in the APS, and 36 experts in the NES, at the same time each year, to be part of the GEM survey.

In 2014, more than 206,000 individuals and 3936 national experts on entrepreneurship from 73 countries were interviewed by GEM across the globe; these participants constituted 72.4 % of the world's population and 90 % of the world's GDP (Singer et al. 2015). The number of interviewees in Iran were 3178 individuals and 36 national experts on entrepreneurship in 2014. The data related to entrepreneurship in Iran is mainly based on the face to face interviews collected by GEM Iran in different years since 2008.

2.3 Entrepreneurial Attitudes

Attitudes toward starting a business may be both the beginning of a new life for an individual and also a step toward economic growth. It is important to realize the potential entrepreneurs in an economy, those who believe that they are capable and are seeing opportunities to start a business and are undeterred by fear of failure when they find opportunities. While potential entrepreneurs may or may not actually become entrepreneurs, they are considered as the starting point to an entrepreneurial journey. According to the GEM 2012 Women's Report, overall, women across the world have lower opportunity perceptions than men, with the greatest difference visible in the MENA/Mid-Asia region. The report states that the Sub-Saharan Africa region has the highest average level (73 %) of perceived capabilities. Numerous factors such as economic framework conditions, culture, history and psychological phenomena have influence on individual perceptions (Koellinger et al. 2007).

GEM introduces several variables to measure entrepreneurial perception in participating countries. In Table 2.1, these perception variables are used to demonstrate the extent of positive and negative attitudes towards entrepreneurship in Iran and comparator economies by gender. Individuals who see good opportunities are expected to have a positive effect on entrepreneurial propensity (Koellinger et al. 2007). Table 2.1 shows that in all comparator economies, men see more opportunities and also have higher perceived capabilities to start a business than women. Iranian women's self-perception of their ability to start a business is higher than their counterparts in Pakistan, Turkey and Egypt.

Female self-perception of the ability to start a new business varies from 5 % in Japan to 87 % in Nigeria (Kelley et al. 2013). Perceived capability to start a new business is a subjective perception and does not necessarily indicate the actual ability of potential entrepreneurs. There is a possibility that the individuals have unrealistically positive images of themselves and overestimate their actual abilities.

Different experimental studies argue about the possibility of creating a new business because of over confident self-perceptions (Camerer and Lovallo 1999;

Table 2.1 Entrepreneurial perceptions in Iran and comparator economies[a] by gender, Kelley et al. (2012)

	See good opportunities		Has capabilities to start a business		Fear of failure for those seeing opportunities		Intend to start in the next 3 years (non-entrepreneurs)		Personally knows an entrepreneur	
	F	M	F	M	F	M	F	M	F	M
Iran	35	43	49	59	44	39	19	28	35	45
Egypt	53	55	45	72	38	28	35	51	23	38
Pakistan	35	57	34	61	25	35	12	40	20	55
Turkey	35	44	37	62	36	26	12	18	24	44

[a]Saudi Arabia was not included in the GEM Women's Report 2012

Koellinger et al. 2007). Koellinger et al. (2007) found that, while the rate of start-up activity in the countries with high rate of entrepreneurial confidence was significantly high, the average chance of a business staying in the market for more than 42 months was low in these countries.

Fear of failure for those seeing opportunities is considered as a deterrent factor in the entrepreneurial journey with a negative effect on the propensity to start a business. Table 2.1 shows that while the fear of failure is lower among men compared to women in Iran, Egypt and Turkey, in Pakistan, the fear of failure is significantly higher among Pakistani men than women.

Overall in GEM member countries, women have lower levels of entrepreneurial intentions than men, with the exception of Thailand where women have 45 % higher intentions than men (Kelley et al. 2013, Women's Report). Table 2.1 also shows that men are more likely to intend to start a business in future than women in all comparator economies; this discrepancy is lower in Turkey and Iran, and is more significant in Pakistan and Egypt respectively. The intentions to start a business does not necessarily imply taking the steps to start a business. The ratio of intentions to actual starts is more than one in all GEM member countries, the only exception is the U.S. where this ratio is one. According to the GEM Women's Report, the ratio of intentions to actual starts is alike for both genders in most of the regions. The largest discrepancy is observed by the GEM in the MENA/Mid-Asia region, where one out of six women who intend to start a business actually takes the step to do that and for every male entrepreneur, two and a half men intend to start a business.

Table 2.1 shows that women in Iran are more likely to know entrepreneurs than women in Turkey, Egypt, and Pakistan respectively. Moreover, the discrepancy between men and women is more significant in Pakistan, Egypt and Turkey compared to Iran. Knowing entrepreneurs can motivate and inspire individuals to start a business by exposing them to entrepreneurial processes and ecosystems. It can also provide advice, information and connections to other sources. Mian et al. (2013) observed that amongst individuals in Iran, Pakistan and Turkey who believe they have the knowledge, skill, and experience required to start a business, knowing other entrepreneurs has a positive effect on them in their being involved in Total Early-Stage Entrepreneurial Activity (TEA) for each country. Individuals who are in the process of starting a business and those launching a new business are classified by GEM under the category of TEA. Those entrepreneurs who are able to run their business for more than 42 months are categorized by GEM as Established Business Owners.

GEM survey shows that a higher average percentage of individuals know an entrepreneur in the regions where there are lots of early-stage entrepreneurs like Zambia, Nigeria, Malawi, Uganda and Namibia and a lower number of individuals know an entrepreneur in the regions like Developed Asia with low rates of TEA (Amorós and Bosma 2014). Men are, on average, 50 % more likely than women to know entrepreneurs across the world; only in Belgium men and women are equally likely to know entrepreneurs. This may indicate that men are more likely than women to participate and get involved in social activities.

It is important to follow up the entrepreneurial ventures to realize if positive perceptions can lead only to early stage entrepreneurship or it could continue to established entrepreneurial activities. The motivation to start a business can be driven by necessity or opportunity.

2.4 Entrepreneurial Perceptions in Iran: 2008–2014

While in general, Iranian women are less likely to see opportunities for starting a venture than their male counterparts, the gender gap in perceived opportunities varies across the years of study. Figure 2.1 displays the fluctuations in the perceived opportunities of both Iranian males and females during the period of 2008–2014. Figure 2.1 shows that the gender gap in perceived opportunities was lower in 2008 and 2010, increased from 2010 to 2013 and decreased in 2014.

The perception on starting a business in Iran did not follow the same trend during the period of 2008–2014. While women see less opportunities than men, they follow the same trend direction as men do (although not necessarily at the same rate), probably reflecting their similar views on the environmental situation.

Men in Iran are more likely to believe that they have the required skills, knowledge and experience to start a business compared to women. Figure 2.2 displays that the gender gap in perceived capabilities has decreased from 2009 to 2012 but has widened in the year of 2012–2013. Comparing Figs. 2.1 and 2.2 reveals a higher gender gap in perceived capabilities compared to perceived opportunities.

Different factors such as personal characteristics, social perceptions and economic environment can impact the individual's decision to convert potential entrepreneurs to actual entrepreneurs. In general, women in Iran express a higher fear of

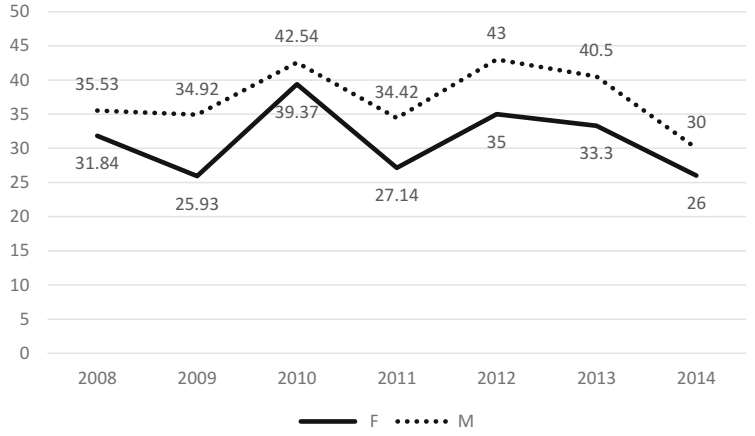


Fig. 2.1 Female and male perceived opportunities in Iran, 2008–2014 (Zali et al. 2015)

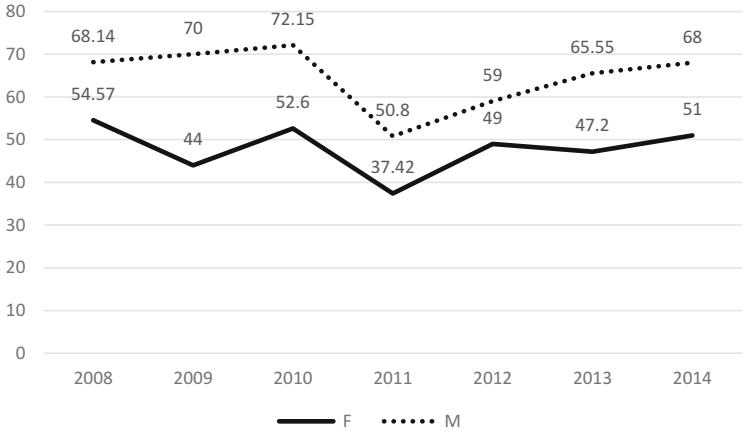


Fig. 2.2 Female and male perceived capabilities in Iran, 2008–2014 (Zali et al. 2015)

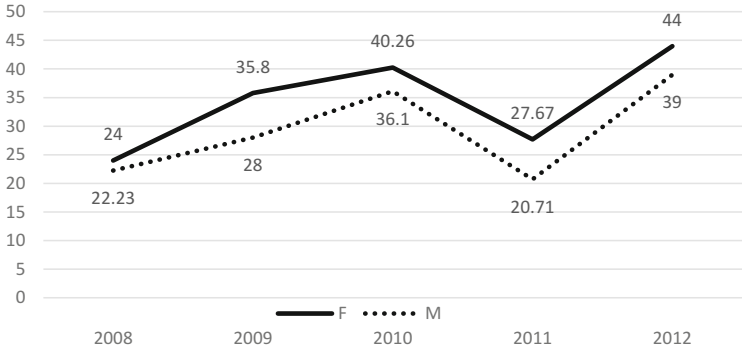


Fig. 2.3 Fear of failure among female and male in Iran, 2008–2012 (Zali et al. 2013)

failure to start a business than men. Fear of failure may refrain individuals from exploiting new opportunities.

Figure 2.3 shows high fluctuations in fear of failure and lack of confidence in starting a business in Iran for both women and men. However, it seems that both genders follow a similar trend of fluctuations in fear of failure.

2.5 Women's Entrepreneurial Activity in Iran

The contribution of entrepreneurs to the economic growth depends on the quality and quantity of entrepreneurship. Not all entrepreneurs who start a business can maintain an established business or can have noticeable impact on the economy. So it is important to study the process and consequences of starting a business and the

types of entrepreneurial ventures. The motivation and reason for starting a business is very important. Some individuals are necessity-driven entrepreneurs; those who are pushed into starting a business because they find no better option for work, in contrast with those who see the startup as an opportunity and are recognized as opportunity-driven entrepreneurs.

According to GEM, overall, regardless of country, men are more likely to be involved in entrepreneurial activity than women (Allen et al. 2007). The participation of men and women in entrepreneurship varies notably across regions and countries. While in Sub-Saharan African economies, there is almost no gender gap in TEA rates, in the MENA region, on average, only one third of the early-stage entrepreneurs are women. It is interesting to note that low female TEA levels and gender disparity in entrepreneurship exist in both less developed and some more developed nations, indicating that the level of economic development does not alone describe this gender gap.

The highest and the lowest rate of TEA is observed in factor-driven economies,[2] the countries mainly dependent on natural resources. Women's participation in TEA varies from 1 % in Pakistan to 40 % in Zambia (Kelley et al. Women's Report 2012). A high or low rate of TEA does not give us a clue as to the level of economic development. For instance, the Sub-Saharan economies, and Trinidad and Tobago reported a high rate of TEA with the first being labeled as factor-driven economy and the second as an innovation-driven country, a country where innovative and knowledge-based enterprises are the main driver of development (Xavier et al. 2013).

Men are about 2.6 more likely to start a business in Iran than women. Table 2.2 shows that the TEA female participation rate in Iran is lower than Turkey but higher than Egypt, Pakistan and the average of the MENA region. The ratio of female to

Table 2.2 Key activity and profile indicators in Iran, comparator economies, and MENA region by gender, Kelley et al. (2012)

	TEA Rate		Opportunity-motives (%TEA)		Necessity-motives (%TEA)		Established business ownership		Discontinuing a business in the past year	
	F	M	F	M	F	M	F	M	F	M
Iran	6	16	62	56	38	43	3	16	1	7
Egypt	2	13	30	41	31	34	1	8	2	7
Pakistan	1	21	22	47	72	52	2	6	0	3
Turkey	7	17	64	68	33	30	3	15	8	5
MENA avg. (unweighted)	4	14	48	54	43	39	1	8	10	10

[2]The World Economic Forum's "Global Competitiveness Report" categorizes countries based on their utilization of resources (GNP per capita and the share of exports comprising primary goods) into three economic levels: factor-driven, efficiency-driven, and innovation-driven. Iran is classified as a factor-driven economy (in transition to efficiency-driven economy).

male TEA rate in Iran is slightly lower than Turkey but higher than Egypt and Pakistan. While gender gaps in TEA rates exist in all comparator economies, the gender gap is highly significant in Pakistan where men are over 20 times more likely to start a business than women.

The GEM survey shows that in all country groups, "a significant gender gap exists between the overall activity of male versus female early stage entrepreneurs and established business owners" (Allen et al. 2007). While early stage entrepreneurs in Iran are 2.6 more likely to be men than women, the likelihood of owning an established business is 5.3 times higher for men compared to women. An even more significant gender gap in the established business ownership compared to the early-stage entrepreneurial activity is observed in all sample countries except for Pakistan. In this country, men are 3 times more likely to be engaged in established business ownership than women, while men are 21 times more likely to be involved in early stage entrepreneurial activities than women.

In general, a higher percentage of women in early-stage entrepreneurship are motivated to start a business by opportunity versus necessity. While this is also the case in Iran and Turkey, women in Egypt are slightly more motivated by necessity than opportunity, and women in Pakistan are predominantly motivated by necessity rather than opportunity. This may be justified by the fact that women with large households and low levels of education in less developed regions are more likely to start businesses out of necessity. "Opportunity motivation is assumed to be a sign of better planning, sophisticated strategy, and higher-growth expectations as compared to necessity motivation" (Bosma et al. 2008). In MENA/Mid-Asia regions, on average, 36 % of women started their businesses out of necessity, whereas this rate was 27 % in Developed Europe (Kelley et al. 2013, Report on Women's Entrepreneurship).

Table 2.2 shows that, unlike other comparable countries, Iranian women entrepreneurs are more likely than their male counterparts to be motivated by opportunity. Among women entrepreneurs in the comparable group, as Table 2.2 shows, women in Turkey (64 %TEA) and Iran (62 %TEA) are more likely to venture into entrepreneurship because of opportunity motives compared to their female counterparts in Egypt (30 %), and Pakistan (22 %). On the other hand, the percentage of women who start their business out of necessary is significantly higher in Pakistan (72 %) than Iran (38 %), Turkey (33 %), and Egypt (31 %). While in Iran and Egypt, men are more likely than women to be motivated by necessity, in Pakistan and Turkey women are more likely than men to be motivated by necessity. The TEA rate among Iranian women is 1.5 times higher than the women TEA average rate in MENA and they pursue opportunity entrepreneurship 1.3 times more than the MENA average. Hence, it seems that the results from Iranian women entrepreneurial motivation and activity are somehow inconsistent with some of the findings of the GEM about the majority of the MENA region's entrepreneurial performance. For instance, according to the GEM Report on Women's Entrepreneurship, "MENA, combines low female TEA rates with high necessity motives, indicating that few women start businesses and when they do so, it is often because they need a source of income" (Kelley et al. 2013).

Can early stage entrepreneurs turn their businesses into sustainable successes and become established business owners? Not all TEA entrepreneurs can develop a sustainable entrepreneurial path. While the rate of early stage entrepreneurs indicates the entrepreneurial dynamism in a country, the established business ownership rate shows the ability of the entrepreneurs to survive in the market. As in the early-stage entrepreneurial activity rate, in all sample countries, men are more likely to survive in the market than women. Mian et al. (2013) found that in Iran, Pakistan and Turkey, being younger, male, having a higher income, having social capital, having education and believing that one has knowledge/skills are the factors that are positively correlated with being an early stage entrepreneur.

TEA Entrepreneurs are more likely to have a substantial degree of stability in higher levels of economic development compared to the entrepreneurs in lower developed economies where individuals usually have a short term perspective on starting a business. Table 2.2 shows that TEA rates for women in Iran, Egypt, and Turkey, as they are in many countries, tend to be higher than the rates for established female business owners. In Iran and Egypt, half of the women who start a new venture can stabilize their business, while in Turkey, less than half of the women in early stage entrepreneurship can become established entrepreneurs. In contrast, in Pakistan, the rate of women with established business ownership is higher than the rate of women who have started a business (TEA). This can be due to the very low TEA rate for women in Pakistan.

While high rates of established business may signal appropriate conditions for enterprises to survive in the market, the low rate of TEA may indicate a low level of dynamism, particularly when it is combined with a high rate of established business ownership (Kelley et al. 2011a). The higher rate of established business owners to TEA rates for men compared to women in Iran, Egypt and Turkey, indicates that men are more likely to survive in business than women in these economies. In Pakistan, on the other hand, the rate of business survival for women is higher than men, indicating that, in spite of low TEA rates among women, once they start a business they are more likely to be sustainable.

In its survey, GEM tracks the business discontinuance of participating economies in the last 12 months. Table 2.2 shows that business discontinuation is lower for women than men in all comparator countries with the exception of Turkey. Women in Turkey are also more likely to close their business compared to their female peers in Egypt, Iran and Pakistan. It should be noted that more women are entrepreneurs (the sum of the TEA rate and established business ownership) in Turkey compared to other comparator economies, indicating that, as the GEM has found, countries that see more businesses being started also witness high percentages of business owners leaving their business.

The TEA rate in Iran is higher than the average TEA rate of innovation-driven economies. The low level of entrepreneurship in the innovation-driven economies may indicate that both men and women have relatively more secure jobs and are less motivated to start up their own businesses; whereas this does not seem to be the case in Iran. With the high rate of unemployment, the tendency toward entrepreneurial activity may be an alternative employment option for numerous

unemployed individuals in Iran. However, this can happen in a friendly business environment with the practical support of the government for promoting entrepreneurship with special emphasis on women's entrepreneurship.

2.6 Total Entrepreneurial Activity in Iran: 2008–2014

Like almost all other countries across the world, the TEA rate for men is higher than women in Iran. Figure 2.4 shows a chart of the average annual TEA rates in Iran for both women and men from 2008 to 2014. As Fig. 2.4 displays, the TEA rate for men has been well above that rate for women since 2008. The gender gap in TEA rates widened enormously in 2011 but narrowed significantly in 2014. While the TEA rate for women has increased since 2010, this rate for men has been fluctuating since 2008. In 2011–2012, the sharp decrease in men's TEA rate was accompanied by a smooth increase in women's TEA rate. Except for the year 2009, the TEA rate for women has steadily been increasing in recent years.

In GEM survey individuals are asked about their age groups and the level of education they have received. According to the GEM Report 2014, the most dynamic individuals in early-stage entrepreneurial activity across the globe are in the age group of 25–34 years. Figure 2.5 also shows that the highest prevalence rates of early-stage entrepreneurial activity in Iran have been in the 25–34 age groups for all years under study. The second highest prevalence rate of TEA varied in different years, e.g. the second highest TEA rates belonged to the age group 35–44 in 2012 and 2014.

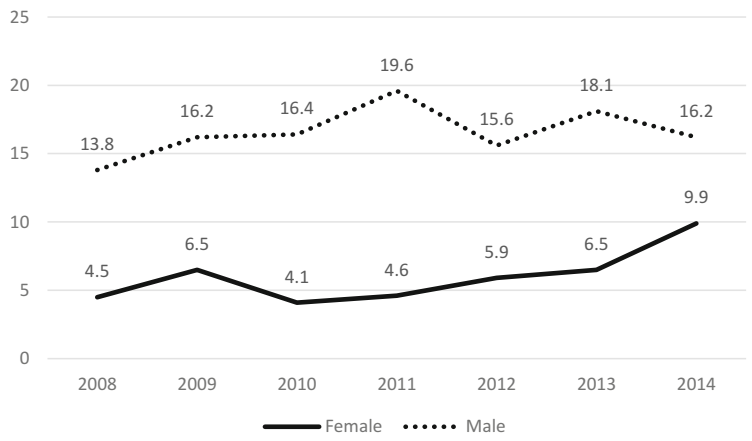


Fig. 2.4 Female and male early-stage entrepreneurial activity rates in Iran, 2008–2014 (Zali et al. 2015)

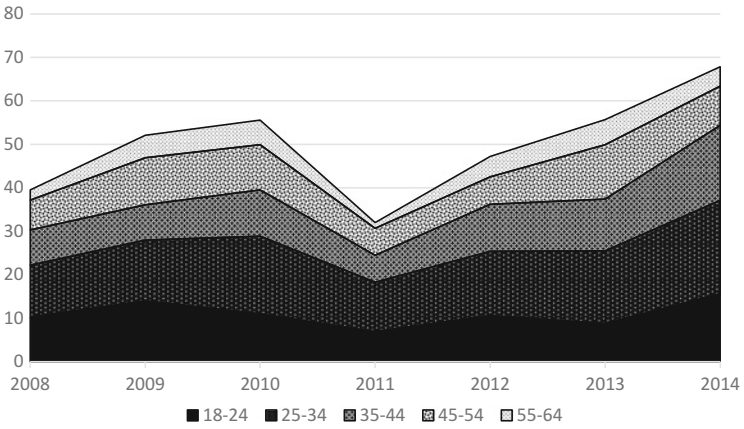


Fig. 2.5 Early-stage entrepreneurial activity rates within age groups in Iran, 2008–2014 (Zali et al. 2015)

2.7 Established Business Ownership in Iran: 2008–2014

The business sustainability is measured by the rate of established business ownership. This rate shows the survival rate of start-ups, those firms who have been paying wages and salaries (or any other payments) for more than 3 years and 6 months. Hence the higher rate of established business ownership indicates the higher sustainability of early stage entrepreneurs. The difference between the rates of the established businesses and early-stage entrepreneurs depends on the level of economic development. According to GEM 2014, the ratio of early-stage entrepreneurs to established businesses was 23.3 vs. 12.7 in factor-driven economies, 14.0 vs. 8.5 in efficiency-driven economies and 8.5 vs. 6.7 in innovation-driven economies (Singer et al. 2015).

The survival rate of start-ups varies significantly between men and women in Iran. Figure 2.6 displays the trend of established business owners by gender in the country. Figure 2.6 shows that the prevalence rates for both men and women established business owners increased in 2009 and during the period of 2011–2013.

While the established business ownership rate for men decreased sharply from 17.9 % in 2013 to 10.97 % in 2014, this rate for women has more than doubled in the same period of time. The years 2013 and 2014, respectively, show the highest and the lowest gender gap in established business ownership in Iran.

Figure 2.7 shows that the individuals aged 35–54 were the largest age group of established business owners in Iran in the years under study. Figure 2.7 also displays that the highest rate of established business owners was among the 35–44 age group in 2008 and 2010, whereas individuals from 45 to 54 years constituted the highest rate of established entrepreneurial activity in 2009 and

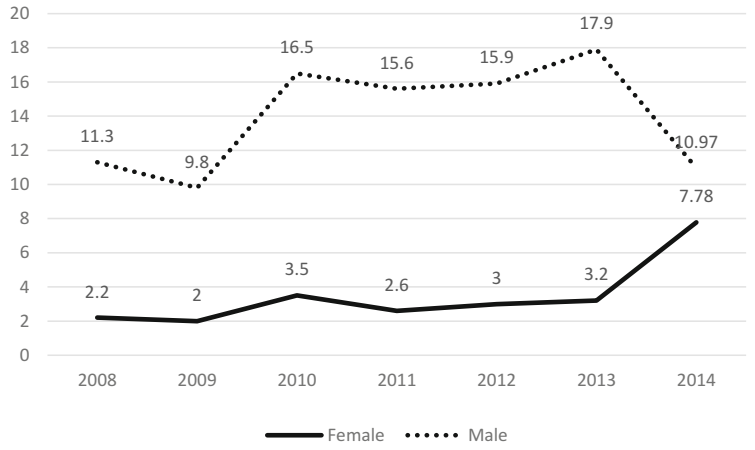


Fig. 2.6 Female and male established business ownership rates in Iran, 2008–2014 (Zali et al. 2015)

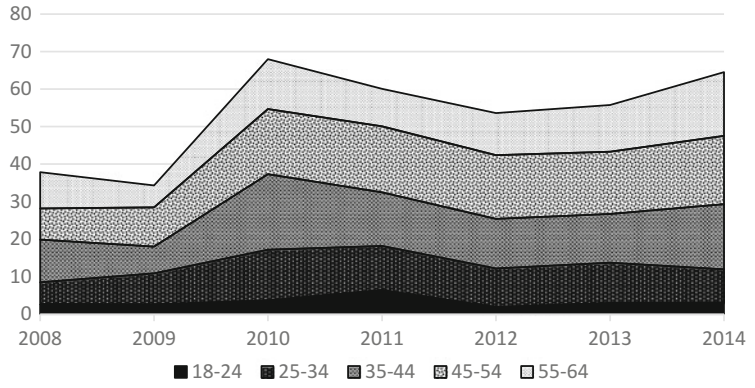


Fig. 2.7 Established business ownership rates within age groups in Iran, 2008–2014 (Zali et al. 2015)

2011–2014. Not surprisingly, the lowest established entrepreneurial activity rate belonged to the age group of 18–24 in all years under study.

2.8 Level of Education and Entrepreneurship

Figures 2.8 and 2.9 show the level of education among early-stage entrepreneurs and established business owners in Iran respectively. The average level of education of early stage entrepreneurs has increased in more recent years in Iran. Figure 2.8 shows that the significant change in the composition of early stage

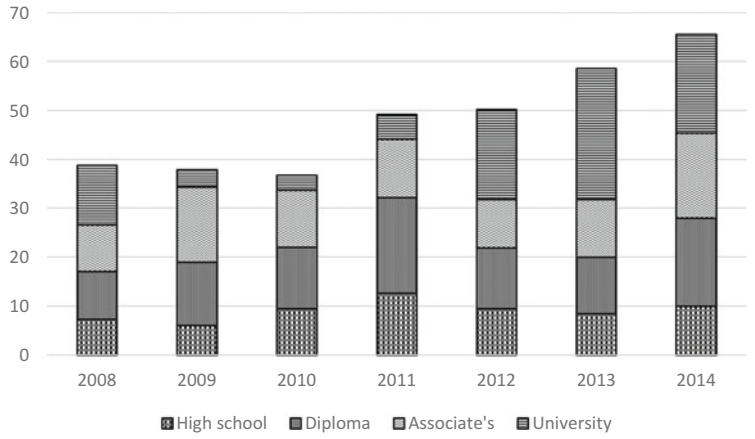


Fig. 2.8 Level of education among early-stage entrepreneurs in Iran, 2008–2014 (Zali et al. 2015)

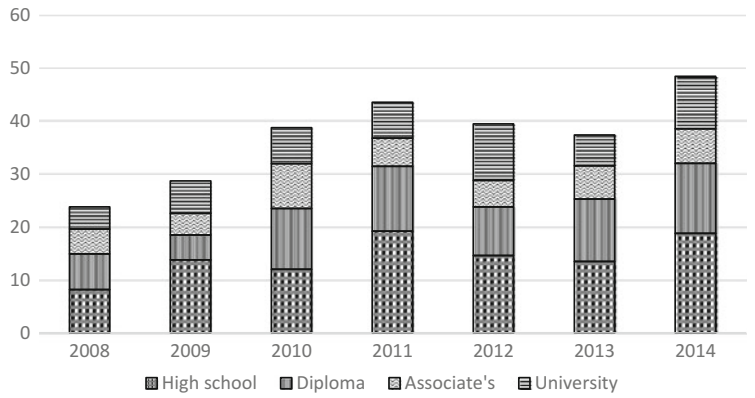


Fig. 2.9 Level of education among established business owners in Iran, 2008–2014 (Zali et al. 2015)

entrepreneurs by education has started since 2012. Establishing the School of Entrepreneurship at the University of Tehran and the Department of Entrepreneurship in the deprived province of Sistan-Baluchestan, and offering an elective course in entrepreneurship in all fields of education in Iran's universities can be mentioned among the major causes of this change.

Figure 2.9 shows that the individuals with high school education are more likely to become established business owners than the entrepreneurs in higher levels of education in all years of study. Comparing Figs. 2.8 and 2.9 shows that early stage entrepreneurs have higher levels of education compared to established business owners particularly in more recent years.

This may indicate that the growing younger educated generation in Iran has a greater tendency to start a business than their previous generation. Another reason may be that traditionally educated people were more likely to be employed by the government; hence they were not interested in entrepreneurship. The recent attention to entrepreneurship in the country may also provide motivation for young, educated individuals to start their own venture. Thus, it is more likely that the average education level among established business owners increases in the future.

2.9 Entrepreneurial Motivations in Iran: 2008–2014

For most entrepreneurs across GEM participating countries, opportunity is the dominant motivation to start a business regardless of gender (Allen et al. 2008). Figure 2.10 displays the rate of opportunity entrepreneurs by gender in Iran. Figure 2.10 shows that the rate of Iranian female motivated by opportunity decreased from 2008 to 2009 but increased smoothly during 2009 to 2012, slightly decreased in 2013 and sharply increased in 2014. The trend of male opportunity TEA rates in Iran increased from 2008 to 2009, slightly decreased in 2009, sharply declined in 2010, and increased afterwards from 2011 to 2014.

While significant gender gap in opportunity-driven rates has existed over 2008–2014 period, in the year 2011 the decline in male opportunity motives, which was accompanied by an increase in opportunity motivation for women, narrowed the gender gap in opportunity motivation to start a business. Hence, the gender gap in recognizing opportunities decreased notably in 2011 compared to the previous years in study, increased significantly in 2013 but decreased notably in 2014. It should be mentioned that a reason for low rate of opportunity-driven TEA rates for women is their low rate of TEA, both in absolute and relative terms.

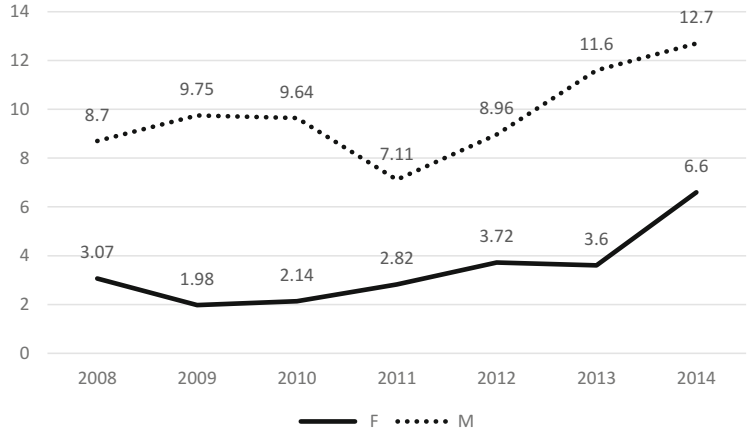


Fig. 2.10 Female and male opportunity-driven TEA rates in Iran, 2008–2014 (Zali et al. 2015)

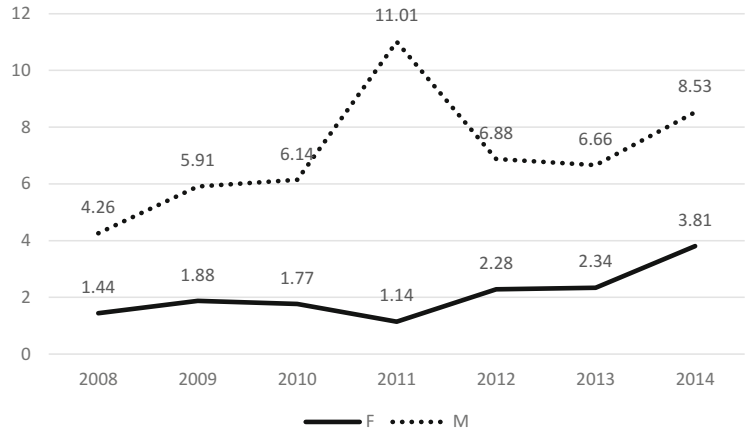


Fig. 2.11 Female and male necessity-driven TEA rates in Iran, 2008–2014 (Zali et al. 2015)

The female necessity-driven TEA rate in Iran is low in both absolute and relative terms compared to this motivation drive for men. The rates of female necessity-driven TEA in Iran ranged from the low of 1.14 in 2011 to the high of 3.81 in 2014, well below the averages of MENA region.

As Fig. 2.11 illustrates, the women's necessity motives to start a venture in Iran increased slightly in 2008, gradually decreased from 2009 on, and reached the minimum of 1.14 % in 2011 but increased afterwards. Increase in necessity-driven ventures may show lack of employment opportunities, more subsistence-oriented entrepreneurship and the need to find a source of income. Comparing the two Figs. 2.10 and 2.11 reveals that from 2009 to 2011 the female necessity-driven TEA rates declined, reversing the trend of opportunity-driven TEA rates for the same period.

The data shown in Figs. 2.10 and 2.11 also suggest that in 2011 the gender gap in recognizing opportunities was the lowest while the gender gap in the necessity-driven motivation was the highest for the same year. As it is illustrated in Fig. 2.11, men are significantly more motivated by necessity than women and the trend of necessity-driven TEA fluctuates less for women compared to men. The gender gap in necessity motivation has also broadened among all the years between 2009 and 2014 relative to 2008.

Statistics show that women entrepreneurs in Iran are more motivated by opportunity entrepreneurship rather than necessity entrepreneurship, indicating that Iranian women are more likely to explore opportunities to start a business rather than being motivated by necessity. Figure 2.12 shows that Iranian women have been more motivated by opportunity than necessity during the period of 2008–2014. The opportunity-driven motivation of women entrepreneurs in Iran moved in the opposite direction of their necessity-driven motives within the periods of 2008–2011 and changed in the same direction from 2011 to 2014.

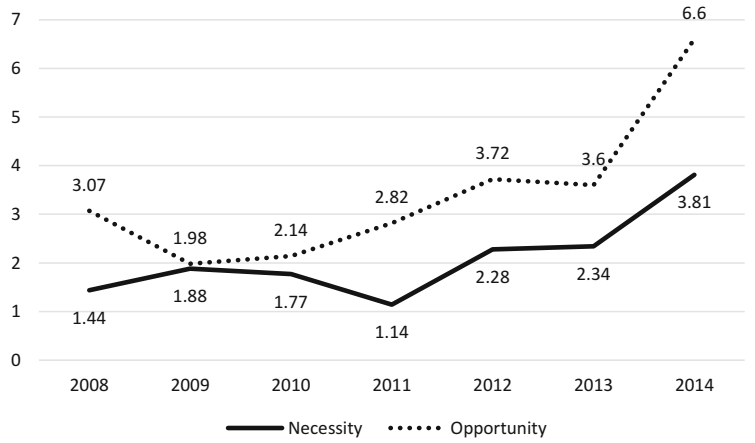


Fig. 2.12 Iranian women's opportunity-driven and necessity-driven motivations, 2008–2014 (Zali et al. 2015)

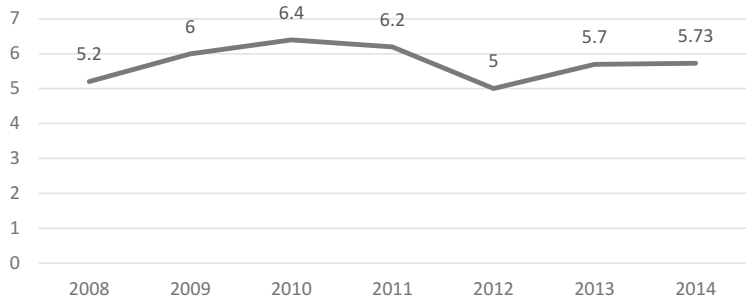


Fig. 2.13 Business discontinuance in Iran, 2008–2014 (Zali et al. 2015)

2.10 Business Discontinuance in Iran

The rates of business discontinuance are inversely related to the level of economic development. According to the GEM Report 2014, business discontinuance decreases from 11.0 % in factor-driven economies to 4.5 % in efficiency-driven economies and to 2.7 % in innovation-driven economies (Singer et al. 2015). Figure 2.13 shows the business discontinuance in Iran for the period of 2008–2014. The prevalence rate of business discontinuation in Iran is 5.73 (Zali et al. 2015), significantly below business discontinuance in factor-driven economies and slightly above that in innovation-driven economies.

Business exit is highly connected with TEA rate (Bosma and Levie 2010). Financial issues (unprofitable businesses or problems obtaining finance) remain the most important reason mentioned for business discontinuation in the majority of economies. However, business owners may discontinue their business for positive or negative reasons. According to GEM's different reports, the reasons for business

discontinuance appear to be rather positive, including opportunity to sell, exploring new business opportunity and exit planned in advance, rather than constraining factors such as unprofitability, problems getting finance, incidents, personal reasons and retirement (Kelley et al. 2011b). The cause of business discontinuation varies among entrepreneurs in different regions and countries with respect to gender, the levels of economic development and institutional environments. GEM measures business discontinuance as the ratio of discontinuance to total business ownership (including TEA and established) in the last 12 months. Compared to other levels of economic development, personal issues are more likely to be reported as the reason for business discontinuance in factor-driven economies.

Women in less-developed economies are closing businesses at a proportionately higher rate than in developed ones (Kelley et al. 2011b). Gender gap in the ratio of exits also gradually vanishes as the level of economic development increases (ibid). Generally, regardless of gender, business exit rate decreases with greater levels of economic development (Amorós and Bosma 2014).

Arasti (2011) found "management deficiency, non-financial support from banks and financial institutions, inadequate economic circumstances, and government policies" to be the four main reasons for business failure for both men and women in Iran. However, in her study, women and men did not consider the same reasons for stopping their business. Both genders "attribute their failure more to external factors (i.e., inadequate economic resources) than internal ones." It is interesting to know that Iranian women did not cite "management deficiency" as the main cause of their business failure, they consider "non-financial support from banks and financial institution and dealing with market as the most causes of their business failure." (ibid). Among the women entrepreneurs featured in this book, Ashoob Cook (Chap. 6) is the one who decided to exit from "Happy Day Care" business when she found a good opportunity to sell her company at the right time for a good price.

A GEM Survey in Iran, Fig. 2.14, demonstrates that business not profitable (31 %), problems getting finance (19 %), and personal reasons (16 %) are

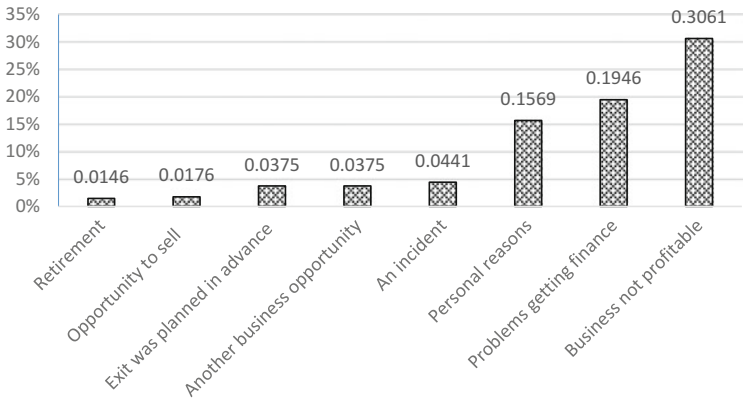


Fig. 2.14 Reasons for business discontinuance in Iran, 2010 (Zali et al. 2011)

respectively the most important reasons for business discontinuance in Iran. Figure 2.14 shows that the three reasons mentioned above are considered as the cause of more than 60 % of business failures in Iran.

Learning about the reasons behind closing an enterprise somehow reveals the bottleneck of an entrepreneurial venture and can provide valuable information on the obstacles preventing entrepreneurs from continuing their business. Recognizing and solving these problems can foster entrepreneurial activity by creating a supportive ecosystem.

2.11 Entrepreneurship and International Trade

Internationalization is extremely important for the economy of Iran. Oil has been the main source of foreign exchange earnings in the country for a long period of time. Although different national economic plans emphasize export diversity, non-oil export promotion and decreasing oil dependency, oil has accounted for 60–80 % of Iran's government revenue in recent years. Entrepreneurship and internationalization in particular can be a solution for product diversification and promoting non-oil export in Iran. Hence, the issue of internationalization will be further discussed in this section.

Internationalization can accelerate the growth of businesses. The quality of entrepreneurship and growth ambitions determines how entrepreneurs can contribute to economic growth and development. "GEM tracks ambitious entrepreneurship by observing early-stage entrepreneurs with high expectations related to job creation (20+ in the next 5 years), innovation (new products/services) and internationalization" (Singer et al. 2015). Different studies found internationalization to be a form of entrepreneurial aspiration that is positively correlated with economic development (Wong et al. 2005; Bosma 2011).

Women entrepreneurs create value for their societies through internationalization (Kelley et al. 2013). Different internal and external factors may influence and motivate entrepreneurs to operate across their national boundaries (Sarfaraz et al. 2015). The ability of an entrepreneur to extend her/his activity abroad is enormously affected by the trade policy of the countries at the origin and destination. In general, in a closed economy the residents lose the competitive advantage due to high cost of export and import as well as costs of time and bureaucratic procedures. So, people in an open economy, regardless of gender, benefit from a better competitive position relative to those in relatively closed economies. While trade openness and internationalization can bring numerous opportunities to the residents and business owners of a country, a restricted trade policy would deprive the business owners from having access to international markets (Sarfaraz et al. 2013).

Even though many countries in developing areas chose protectionism as the strategy toward economic development in the 1960s, in recent decades an increasing number of countries switched from protectionism policy (import substitution)

toward open trade policy (export promotion). The growing world-wide tendency toward regional and global trade agreements and free trade expansion can, to some extent, be attributed to the positive experience of the countries implementing a free trade policy such as Hong Kong, South Korea, Singapore, Malaysia, etc. in recent decades. Iran has traditionally chosen import substitution and protectionism as the strategy toward economic development, and is still pursuing the same policy with some amendments. While globalization, regional trade agreements, and, in particular, the World Trade Organization (WTO) have provided incredible opportunities for many capable people and business owners across the world to participate and benefit from the world market, Iranian business owners have been left behind.

Iran's limited contribution to bilateral, regional, and international trading blocs has prevented its economy from obtaining free trade agreements (Sarfaraz et al. 2013). While the Global Preferential Trade Agreements Database (World Bank 2010) shows Economic Cooperation Organization (ECO) as the only in force trade treaty for Iran, it demonstrates numerous bilateral, multilateral, regional, and international free trade agreements for the economies of Egypt, Pakistan, Saudi Arabia, and Turkey. Iran, Pakistan and Turkey have been the co-founders of ECO, an intergovernmental organization established in 1985 for promoting economic, technical and cultural cooperation that expanded its members to Afghanistan, Azerbaijan, Kazakhstan, Kyrgyz Republic, Tajikistan, Turkmenistan and Uzbekistan in 1992. Besides the trade barriers (including Tariff and non-Tariff) inside Iran and the lack of involvement in international trade treaties, the international sanctions imposed against Iran in recent years can be considered as the main obstacle in extending business across the border. Numerous sanctions against Iran started by the United States in 1979, intensified by the UN Security Council sanctions in 2006, 2007, 2008 and 2010, and pursued by the European Union in 2010 and 2012 have extremely limited Iranian enterprises' involvement in international trade.

In this chapter, three different sources of data are used to investigate the position of Iran in the international trade arena: the Heritage Foundation (Index of Economic Freedom), the World Bank (Trading Across Borders) and the GEM (International Entrepreneurship). According to The 2015 Index of Economic Freedom, Iran's score in economic freedom is 41.8, making its economy the 171st freest out of 178 countries (Miller and Kim 2015). A more detailed index of economic freedom is the Trade Freedom Index. This indicator that measures the absence of tariff and non-tariff barriers (NTBs) is calculated based on the trade-weighted average tariff rate and NTBs of each country. Trade Freedom reveals the degree of economic openness to the flow of goods and services inside and outside national borders and the ability of citizens to join the international marketplace as buyer or seller.

Figure 2.15 indicates the low performance of Iran in terms of free trade since 1996. The peak of trade freedom during 1996–2014 is observed in 2004 and 2005 with the score of 78.8 under Mohammad Khatami's administration (1997–2005). The relative positive performance of Iran's score in Trade Freedom in the early 2000s can be attributed to the authorization of private banks and foreign banks for the first time in 2001 (after the Islamic Revolution of 1979) under then President Khatami's liberalization administration policy (Sarfaraz et al. 2013).

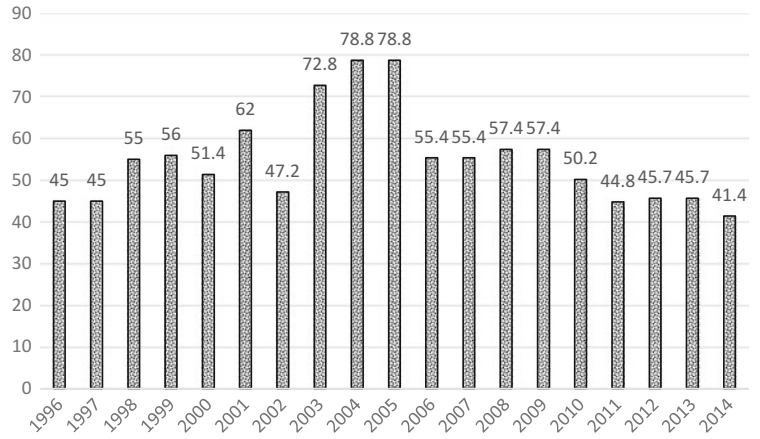


Fig. 2.15 Trade freedom scores in Iran, 1996–2014 (Miller et al. 2014)

Table 2.3 Trading across borders indicators in Iran and compared economies, World Bank Group Doing Business ranking 2014

	Iran	Egypt	Pakistan	Saudi Arabia	Turkey
Rank	148	99	108	92	90
Distance to frontier (DTF)	56.81	71.56	69.05	73.01	73.26
Documents to export (number)	7	8	8	6	7
Time to export (days)	25	12	20.7	13	13
Cost to export (US$ per container)	1350.00	625	765	1285.00	990
Documents to import (number)	11	10	8	8	8
Time to import (days)	37	15	18.4	17	14
Cost to import (US$ per container)	1555.00	790	1005.00	1309.00	1235.00

In order to have a comparative view on the Iran's international trade position in the region, "Doing Business: Trading Across Borders" indicators produced by the World Bank are used here. The ease of doing trade in Iran is compared to the feasibility of trade in the economies of Egypt, Pakistan, Saudi Arabia and Turkey. Table 2.3 shows that Iran ranked 148 out of 189 countries in Trading Across Border indicators in 2014, well below the compared economies. The score of Iran in Distance to Frontier (DTF) is lower than all comparator economies; and is 12.24 below Pakistan, the country that has the lowest score in the compared group after Iran.

DTF is based on a scale of 0 (for the lowest) to 100 (the frontier) and demonstrates "the distance of an economy to the "frontier", which represents the best performance observed on each Doing Business topic across all economies and years included since 2005"(World Bank 2014. Doing Business 2015). Hong Kong has the highest score of 95.36 in DTF.

As observed in Table 2.3, the time and cost associated with importing and exporting in Iran is well above all comparator economies. This creates less ability

for Iranian business owners to compete in the world market and causes more restriction in their international business involvement. No matter how efficient and successful a business may be, it is affected by external environment and the trade policy in its economy. If we agree with the quote stated by Benjamin Franklin (1706–1790) in the eighteenth century that "Time is money," now at the age of globalization, it would not be unreasonable to say, "Moments are money." Losing time is losing money and opportunities. Compared to other economies in the group, Iran is well behind on days required for exporting and importing, causing huge losses for the enterprises and the country. While it takes 25 days for an Iranian company to comply with all procedures required to fulfill the documents and export goods, it takes 12, 13, 13, and 20.7 days respectively for the citizens in Egypt, Saudi Arabia, Turkey and Pakistan to export. The cost associated with all required procedures to export goods in Iran is 2.16 times more than that of Egypt, 1.77 times more than Pakistan, 1.37 times more than Turkey and US$65 higher than the amount an exporter pays in Saudi Arabia.

Importers in Iran also need 37 days to comply with all procedures required to import goods from abroad while this time for the enterprises in Turkey, Egypt, Saudi Arabia and Pakistan is 14, 15, 17, and 18.4 days respectively. Importers in Iran are required to pay 1.97 times higher cost per container for importing than an importer in Egypt, and also pay the following extra amounts per container compared to their counterparts in the comparator economies, US$550 in Pakistan, US $320 in Turkey and US$246 in Saudi Arabia. The number of documents for importing and exporting seems to be more appropriate compared to the other indicators in Iran and also compared to the number of documents required for trade in other comparator economies. This can be attributed to some reforms and improvements during the Khatami administration including the liberalization of 600 importing items in 2000, removal of non-tariff barriers on 1150 items in 2001 (Sarfaraz 2002), and automated trade tax payment in 2004 (World Bank. Doing Business 2006).

The last source used here to show the performance of Iran in international economic competition is the GEM consortium. GEM considers internationalization as one of the measurements for quality entrepreneurship, besides growth and innovation. Iran's ranking on International Entrepreneurship among GEM members during 2008–2013 also demonstrates the weak international position of the country. Table 2.4 shows that Iran had the lowest ranking on International Entrepreneurship among GEM members in 2010 and could move to its highest position among GEM members in 2011.

Table 2.4 Iran's ranking on international entrepreneurship among GEM members, 2008–2013, Zali et al. (2013)

	2008	2009	2010	2011	2012	2013
Internationalization ranking/no. of participant	40/43	49/53	59/59	39/60	64/67	61/67

2.12 Internationalization and Female Entrepreneurs

"Overall, for women, the degree of internationalization increases with economic development level" (Kelley et al. 2011b). Entrepreneurs decide to expand their domestic market to international markets for different reasons. The small size of the internal market and intense local market competition is among the several factors recognized as motivations for entrepreneurs to find clients abroad. Iran, classified by the World Bank as an upper middle income country with a population of over 75 million, has a large internal market. Moreover, due to the dominant role of the government in the economic activity and the inefficiency of the private sector, business competition is often limited. Despite large domestic market and trade restrictions, foreign trade is assumed to be profitable in Iran. As Fatemeh Daneshvar (see Chap. 5), one of the businesswomen featured in the book, mentions, "I chose to establish an export-import company because I knew that it was profitable."

GEM provides information on the average percentage of entrepreneurs who export commodity across the border of their countries. In calculating internationalization, GEM asks entrepreneurs what percentage of their customers come from outside their economy. The answers are categorized based on the percentage of their export within the ranges of 76–100 %, 26–75 %, 11–25 %, and 1–10 %. Table 2.5 shows that 89.6 % of the early-stage male entrepreneurs and 89.2 % of the early-stage female entrepreneurs in Iran had no customers living outside their county. While only 0.8 % of early-stage male entrepreneurs in Iran had more than 75 % customers abroad in 2012, no Iranian women entrepreneurs had more than 75 % customers outside Iran. Table 2.5 shows that men in Iran are more likely to have higher than 26–100 % customers abroad (compared to women), whereas Iranian women entrepreneurs are more likely to have 1–25 % customers in foreign countries than their male counterparts.

According to the GEM 2010 Women's Report, Iran ranked the fourth lowest economy among GEM member countries in terms of women having less than 25 % foreign customers in 2010, and businesswomen in Iran did not sell more than 25 % of their products outside the country in that year (Kelley et al. 2011b).

The low rate of women's involvement in international business in Iran can be attributed to the relatively closed economy of Iran. Figure 2.16 shows trade freedom in Iran compared to Turkey, Egypt, and the world average. Figure 2.16 displays that, except for the period of 2003–2005 (under the government of President Khatami), trade freedom in Iran has been lower than that of the world average and Turkey. The level of trade freedom in Iran has also been lower than that of the world average, Turkey, and Egypt since 2006.

Table 2.5 Export intensity of early-stage entrepreneurial activity in Iran by gender, Zali et al. (2013)

	76–100 %	26–75 %	11–25 %	1–10 %	None
Male	0.8 %	1.2 %	2.8 %	5.6 %	89.6 %
Female		1.1 %	3.2 %	6.5 %	89.2 %

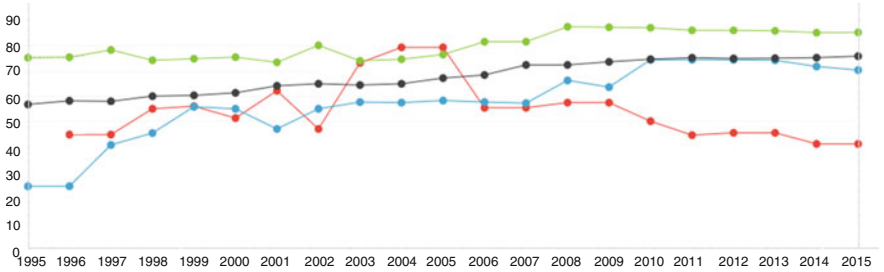


Fig. 2.16 Trade freedom in Iran, Egypt, Turkey, and the world average (*red* = Iran, *black* = World average, *green* = Turkey) (Miller and Kim 2015)

Iran's participation in the international and regional trade treaties is very limited. In spite of Iran's attempts to join WTO since 1996, its "application for membership has been rejected 21 times by the USA" (Amiri and Seidi 2012). Eventually, in 2005, "The WTO agreed to start membership negotiations with Iran after the United States dropped a long-standing veto"; since then, Iran has held observer status at the WTO (Mojaver and Fayazmanesh 2006).

Iranian entrepreneurs do not benefit from the potential gain of different bilateral/regional treaties either. For instance, Iran is not a member of the following regional trade agreements established between some countries in the MENA region: Gulf Cooperation Council (GCC, 1981), Arab Maghreb Union (AMU, 1989), Common Market for Eastern and Southern Africa (COMESA, 1994), Greater Arab Free Trade Area (GAFTA, 1997), and AGADIR Agreement including, Jordan, Egypt, Tunisia and Morocco, in 2004 (Minot et al. 2009).

Moreover, different economic and trade sanctions against Iran imposed by the USA, European Union, and the international community, particularly the sanctions on the central bank, have created a hostile environment for Iranian business owners attempting to gain access to foreign markets. Hence, in this circumstance, both female and male entrepreneurs are faced with serious obstacles. However, in spite of unfavorable environmental conditions for doing business abroad, there are some distinguished women entrepreneurs in Iran who are successfully involved in international markets.

2.13 Trend of Internationalization in Iran

Figure 2.17 shows the trend of internationalization for Iranian entrepreneurs with more than 25 % customers across Iran's borders. As Fig. 2.17 displays, except for the years 2009 and 2012, the percentage of entrepreneurs in Iran who sold more than 25 % of their products to foreign markets has an increasing trend.

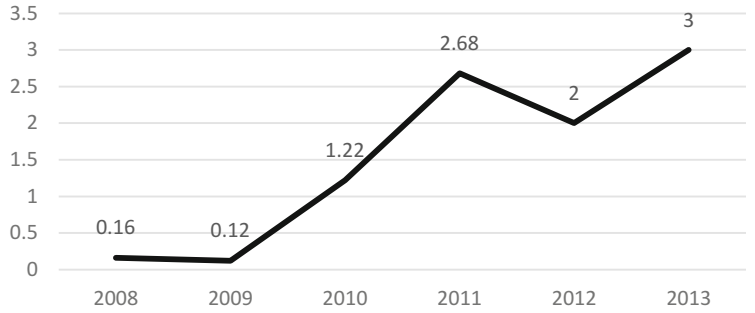


Fig. 2.17 Trend of internationalization levels (customers outside country > 25 %) in Iran, 2008–2013 (Zali et al. 2014)

Restrictive trade policy, lack of regional and multilateral trade negotiations, and international sanctions are among the reasons for the low rate of internationalization in Iran.

2.14 Industrial Sector

The proportion of individuals engaged in three main sectors of an economy—agriculture, industry, and services—can be a way to identify the level of income generation, economic growth and development. The structure of employment changes during economic development; as per capita income increases, agricultural share of the economy decreases drastically while the industry sector declines gradually and the service sector becomes the dominant contributor of the economy (Soubbotina and Sheram 2000). Thus, in more developed nations, higher proportions of people are employed in services. Accordingly, entrepreneurs are more likely to start business in services in more developed economies compared to less developed ones.

According to the Food and Agriculture Organization (FAO) of the United Nations, women constitute about 43 % of the agricultural labor force globally and in developing countries. Within a regional perspective, women's involvement in agricultural activity varies from about 20 % in Latin America to 50 % in parts of Africa and Asia, and exceeds 60 % in a few countries (SOFA Team and Doss 2011). At the global level, compared to other regions, women are more likely to be in agriculture in South Asia and Sub-Saharan Africa, and are less likely to work in farms in Latin America and the Caribbean. While in some areas, like Sub-Saharan Africa and East and Southeast Asia, the gender gap has almost disappeared in agricultural employment, in the region of South Asia the gap exists in favor of female workers in agriculture, versus in Latin America and the Caribbean where more males are involved in agriculture. SOFA Team and Doss (2011) also found that, in all regions, men are more likely to be employed than women in the industry

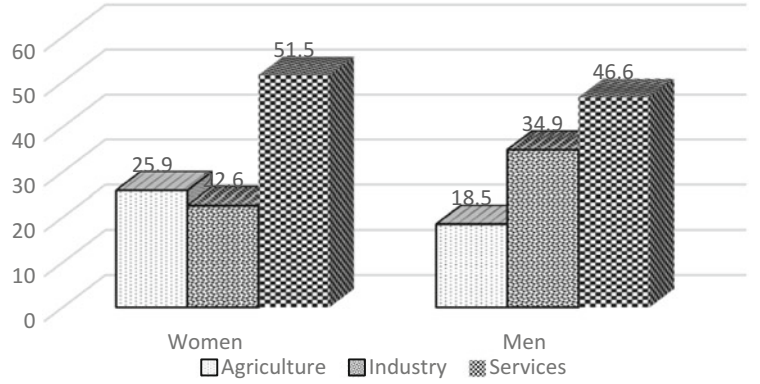


Fig. 2.18 The share of women and men employment in Iran in different sectors of the economy for ages 10+, 2014 (Statistical Center of Iran (SCI), Labor Force Survey Results 2014)

Table 2.6 The percentage distribution of employed population by economic sector in Iran and four comparable economies, 2004–2007, United Nations, the World's Women 2010

	Women			Men		
	Agriculture	Industry	Services	Agriculture	Industry	Services
Egypt	43	6	51	28	26	46
Iran	33	29	38	21	33	47
Pakistan	72	13	15	36	23	41
Saudi Arabia	–	1	99	5	72	23
Turkey	47	15	38	19	52	29

sector. Males are more likely than females to be involved in services in the regions of South Asia, and Near East and North Africa, whereas females are more likely than males to be employed in services in East and Southeast Asia, Latin America and the Caribbean, and Sub-Saharan Africa. The East and Southeast Asia regions have about the same participation rate for males and females in services.

According to the Statistical Center of Iran 2014, the proportion of the labor force engaged in services (47.3) is higher than industry (33 %) and agriculture (19.6) in Iran. Figure 2.18 shows that women in Iran are more likely to be engaged in services (51.5) than agriculture (25.9) and industry (22.6). The proportion of men in the labor force is higher in services (46.6) compared to the industrial sector (34.9) and agricultural sector (18.5). Figure 2.18 indicates that, while Iranian women, compared to men, are more likely to be engaged in the services sector (51.5 vs. 46.6) and agricultural sector (25.9 vs. 18.5), Iranian men are more likely to be employed in industry (34.9 vs. 22.6) than women.

The United Nations Center for Social Development, in the World's Women 2010, also shows that the proportion of women engaged in agriculture in Iran, Egypt, Pakistan, Saudi Arabia and Turkey is higher than their male counterparts. Table 2.6 shows that the proportion of Iranian women involved in the agricultural

sector is considerably lower than their female counterparts in Pakistan, Turkey and Egypt, whereas their involvement rate in the industry is significantly higher than all the comparable economies. It is worthy to note that the very low agricultural engagement of the population in Saudi Arabia is due to the desert climate of the country, which is not suitable for agriculture.

Men in Iran, Pakistan and Turkey are more likely to be employed in the services sector than their female counterparts. It is worthy to note that in 2010 Iranian women were less likely to be involved in services compared to their male counterparts, but in 2014 more women contributed to services than men. Iranian women pursued services at the expense of the other two sectors.

Table 2.6 indicates that in all compared economies men are more likely to be employed in the industrial sector, however, the gender gap in industry is notably low in Iran compared to all comparable economies. This can show the high human capital of the Iranian women and also may indicate that, if Iranian women have the opportunity to participate in economic activity, they are capable of working in the more complicated sector of the economy. Aside from the lack of employment opportunities for women in Iran, some companies prefer to hire men rather than women and there also exist limitations to the type of jobs for women in the country (Mahajerani 1998).

More detailed classification of economic sectors indicates that women entrepreneurs establish and run enterprises across all of the broad industrial sectors including extraction, transformation, business services and consumer-oriented products (Georgeta 2012). The extractive sector includes the extraction of goods from nature (agriculture, forestry, fishing, and mining); transforming industries include construction, manufacturing, transportation, and wholesale distribution; business services include the activities where a business is interacting and dealing with another businesses; and consumer services entail doing business directly with private customer, e.g. retail, education, health services, social services, restaurants, etc.

Figure 2.19 shows that entrepreneurs are more likely to be in the consumer sector, compared to extractive, transforming, and business services, at all levels of economic development. However, as the economies move to a higher level of economic development, entrepreneurs are less likely to be in the consumer sector. Figure 2.19 demonstrates that compared to other levels of economic development, entrepreneurs in factor- driven economies are more involved in extractive sector, while entrepreneurs in innovative economies are more likely to have enterprises in business services. The gap in the transforming sector is not significant at the three levels of economic development.

The high level of business services in innovation-driven economies demonstrates the wide availability of highly educated human capital in supportive entrepreneurial environments. The high participation level of entrepreneurs in extractive businesses in factor-driven economies shows the high availability of natural resources.

The GEM Women's Entrepreneurship Report (Kelley et al. 2013) indicates that overall, across countries participating in the GEM survey, about two thirds of women entrepreneurs are in the consumer sector while fewer than half are men in

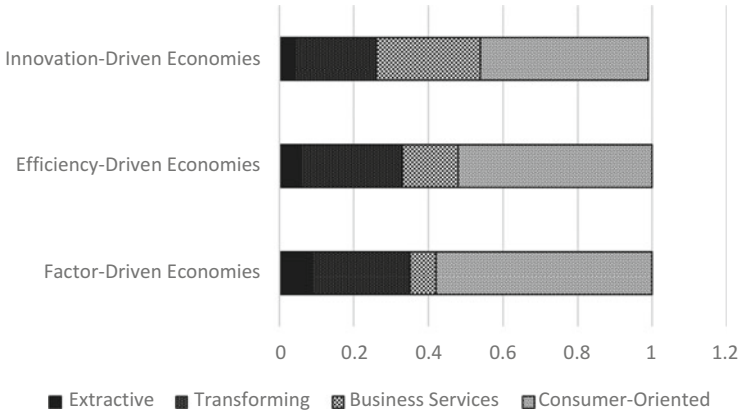


Fig. 2.19 Sector distribution of early-stage entrepreneurial activity by phase of economic development (Kelley et al. 2011b)

this sector. While in each region, on average, at least half the women entrepreneurs are running business in the consumer sector, this sector absorbs four-fifths of the women entrepreneurs in Sub-Saharan Africa. On the other hand, women and men in innovative-driven economies, compared to other development levels, are more likely to participate in business services than the consumer sector. However, women in innovation-driven economies are engaged in the consumer sector more than twice as business services. In general, entrepreneurs are more likely to operate in transforming compared to business services and the extractive sector. This is likely due to the fact that consumer businesses typically have low entry barriers and require less capital investment at startup. Men are generally more likely than women to operate in the capital-intensive transforming sector and knowledge-based business services. This is even truer for women in business services in countries with lower levels of economic development. While in some countries like Pakistan, Malawi and Ghana no women entrepreneurs in business services have been reported, above 35 % of women in France, Sweden, Poland and Slovenia are engaged in this sector.

The GEM Iran survey shows that in Iran more than half of the early stage entrepreneurs are in the consumer-oriented sector, 32.28 % in the transforming sector, 7.19 % in business services and 8.58 % in the extractive sector (Fig. 2.20). Business owners in established business activity also follow the same pattern among the four sectors, however with different proportions. Figure 2.20 displays that established business owners, compared to early stage entrepreneurs, are more likely to participate in the transforming sector (38.3 % vs. 32.28 %) and less likely to be in the extractive sector (6.96 % vs. 8.58 %), business sector (6.01 % vs. 7.19 %), and in the consumer-oriented sector (48.72 % vs. 51.95 %). This indicates that, for entrepreneurs who are starting a business, it is easier and less complicated to be in the consumer-oriented sector versus the transforming sector. Compared to early-stage entrepreneurs, established business owners are relatively

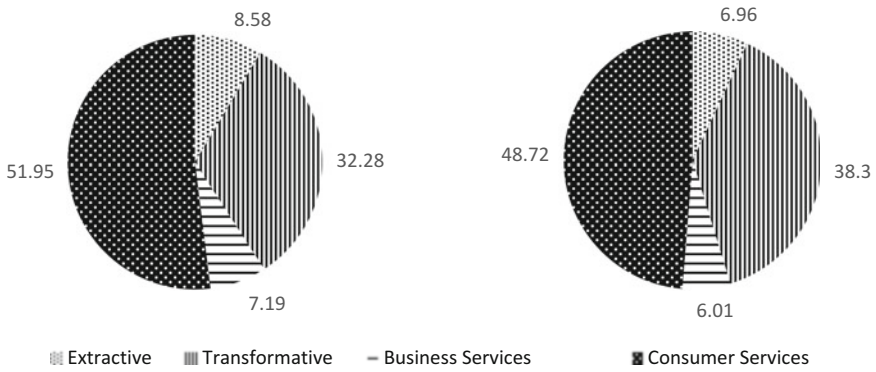


Fig. 2.20 Sector distribution of early-stage (*left*) and established entrepreneurial activity (*right*) in Iran (Zali et al. 2013)

Table 2.7 TEA by industry sector in the MENA/Mid Asia region by gender, Kelley et al. (2013)

	Extractive	Transformative	Business services	Consumer services
Women	4	36	8	51
Men	13	32	10	43

more likely to participate in the transforming sector, which requires more capital intensive investment.

Individuals in early-stage entrepreneurial activity are more likely than established owners to be in business services. This can be explained by the higher tendency of the younger generation to be in business services in Iran. As was found earlier, the highest prevalence rates of early-stage entrepreneurial activity in Iran have been in the 25–34 age group versus the largest age group of 35–54 for the established business owners. On average, men are more likely to have enterprises in business services and are over three times more likely to be engaged in the extractive industry than women in the MENA/Mid Asia Region. Table 2.7 shows that in the MENA/Mid Asia Region, for both men and women, Total Early-Stage Entrepreneurs are more likely to participate in consumer services and transformative industries.

2.15 Women's Business Enterprises in Iran

There exists a lack of data regarding the structural sector of women's business enterprises in Iran. According to Arasti and Kanani (2011), 44 % of the businesses owned by Iranian women are in manufacturing and 56 % are in services. About 39 % of these activities are in cultural and educational areas, 14 % in the food industry, 4 % in transportation, 4 % in trade, 4 % in the production and education of

carpet and 4 % in counseling services. Iranian women featured in this book are doing business in all industrial sectors of extraction, transformation, business services and consumer-oriented goods.

Among the 11 Iranian-born women entrepreneurs featured in this book, 7 who live in Iran and have established business in the country are engaged in the following sectors:

- Extractive:
  - Khiyal Abdolahzadeh (see Chap. 5), the founder of Anthemia Herb Production & Distribution, Ltd.
  - Fatemeh Daneshvar (see Chap. 5), the founder of Sepehr Asia Mineral Import-Export, Ltd.
- Transforming:
  - Mahrokh Falahi (see Chap. 5), the founder of animal husbandry wholesale producer and distributor, Fallahi Livestock, Ltd.
  - Fatemeh Moghimi (see Chap. 5), the founder of Sadid Bar International Transportation, Ltd.
  - Banafsheh Sedigh (see Chap. 5), the founder of Auto Parts and Gas Valves Shigal MFG, Ltd.
  - Haydeh Shirzadi (see Chap. 5), the co-founder of Kermanshah Recycling and Composting, Ltd.
- Business Services:
  - Parvin Baradaran Ghahfarokhi (see Chap. 5), the founder of Kuhrang Knowledge-Based Instrumentation Technology, Ltd.

The four Iranian born entrepreneurs who live in the USA and Europe are doing business in the following sectors:

- Transforming:
  - Gisu Hariri (see Chap. 6), the co-founder of Hariri & Hariri Architecture, Ltd, New York, USA.
  - Mojgan Hariri (see Chap. 6), the co-founder of Hariri & Hariri Architecture, Ltd, New York, USA.
- Business Services:
  - Anousheh Ansari (see Chap. 6), the co-founder and CEO of Prodea Systems, Ltd, Texas, USA.
- Consumer Services:
  - Ashoobgar Cook (see Chap. 6), the former founder of Happy Child Care and Education Chain Company, Happy Child, Ltd., London, England.

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Chapter 3
Social, Human and Financial Capital and Women's Entrepreneurship in Iran

3.1 Women's Entrepreneurship and Capital Accessibility

If we have consensus that the ultimate goal of entrepreneurship is to create jobs, reduce unemployment and generate wealth for individuals and nations, then how can we ignore the participation of women as one-half of the population in the entrepreneurial activity? The extent to which women can play a role in the socio-economic welfare, prosperity, happiness and development of each country depends on the women's access to resources, the roles that are defined for women by formal and informal institutions, and the women's self-perception (Sarfaraz et al. 2015). Adequate accessibility to different forms of capital provides women the possibility of utilizing their skills, talents and creativities to launch their businesses and to contribute effectively to their countries' sustainable development. "Conceptual development and empirical research in the entrepreneurship area have established the positive effect of human, social and financial capital on the survival and growth prospects of entrepreneurial ventures" (Manolova 2006). Yet access to capital is a frequently cited problem for women business owners (Coleman and Robb 2009).

3.2 Social Capital

Like entrepreneurship, social capital is a multidimensional concept. Different studies support the key role of social capital in the process of establishing, growing and developing a business (Welter 2006). Social capital, as an intangible asset, provides the possibility for entrepreneurs to have access to the external resources that they do not possess themselves (Neergaard et al. 2006). Research studies demonstrate the crucial role of social capital at both micro and macro levels of an economy. Baron and Markman (2000) define social capital in terms of social networks (formal and informal ties), social skills (interpersonal and communicative

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ability), as well as social identity (status, identity and reputation). Putnam (1995) considers the social capital dimensions to be a high research priority.

In spite of different forms of social capital, each form contains two common characteristics: (1) they demonstrate some facet of social structure; (2) they simplify the action of people within the structure (Nahapiet and Ghoshal 1998). With the above approach, the structural dimension refers to the impersonal arrangement of connections between individuals or units which expresses the overall pattern of linkages between actors. The relational dimension describes the assets created and strengthened by relationships. The cognitive dimension demonstrates the resources that provide shared "representations, interpretations, and systems of meaning among parties" (ibid).

Rabiee and Sarabi (2013) investigated the effect of social capital on women's entrepreneurship in Iran in terms of three dimensions of social capital including the structural, relational and cognitive aspects. They found that while the relational dimension of social capital has a lower impact on female entrepreneurship, the structural dimension has the most significant effect on female entrepreneurship, followed by the cognitive dimension. In their study, Rabiee and Sarabi also investigated the impact of the components of the structural, cognitive and relational dimensions of social capital on Iranian female entrepreneurs. The structural dimension includes "network ties, network configuration, and appropriable organization". The cognitive dimension includes "shared code and language, and shared narratives". The relational dimension includes "trust, norms, obligations and identification" (Nahapiet and Ghoshal 1998). According to Rabiee and Sarabi, all the aforementioned factors have a positive influence on female entrepreneurship in Iran. Moreover, they ranked the significance level of social capital's sub-dimensions on Iranian female entrepreneurship, sorted in descending order of importance as follows: network ties, appropriable organization, identification, shared narratives, trust, network configuration, shared codes and language. Hence, Iranian women entrepreneurs seem to be mostly affected by structural social capital, including network ties and appropriate organization.

Are there any differences between men and women regarding social capital? While many studies show gender differences in social capital, there is no consensus among researchers on this issue. For instance, Cliff and Jennings (2004) found no significant gender differences in social capital, but Staber (1997) observed strong differences between men and women regarding social network structures. In a comparative gender study, Mian et al. (2013) analyzed social capital in terms of networks in Iran, Pakistan and Turkey. They employed Global Entrepreneurship Monitor (GEM) data and studied the relationship between gender and social capital in the three economies. The measurement of social capital was based on the answer to this question: "Do you know someone personally who has started a business in the past 24 months?" Their result shows that men are more likely to know an entrepreneur than women. They found that there exist significant relationships between gender and social capital in the three economies. Hence, they argue that traditionally, women have different networks from men and networks are important in determining a propensity to start new business. They also found that networks are

positively related to the likelihood of becoming an entrepreneur in the three countries regardless of gender differences (Mian et al. 2013).

Women have different social preferences than men. Women are more likely to concentrate in a limited number of jobs compared to men. In 2005, about 36 % of women in the European Union were engaged in just six occupational groups (women-dominated jobs) out of the 130 ISCO occupational categories, whereas 25 % of men were occupied in the top six occupations (male-dominated jobs) of ISCO categories (Bettio and Verashchagina 2008). This shows that the scope of jobs is very limited for women, while men are working in a more diversified category. Arasti et al. (2014) compared the level of social capital in different women's and men's occupations. They measured the level of social capital for top entrepreneurs in terms of trust, participation, activity and social networks in occupational segmentation in Iran. They found that in a male-dominated environment, where the majority of the labor force are men, the level of social capital (trust, participation, activity and social networks) is lower than the female-dominated environment, where women make up the majority of the labor force. Moreover, women have the highest level of social capital in a female-dominated environment whereas they have the lowest level of social capital when they work in a male-dominated environment. This can be explained to some extent by the reluctance of women to communicate and interact with men due to social and religious constraints. The presence of women in both male and female-dominated work places increases the level of social capital. This can explain the relational dimension of social capital (including emotional, sympathy, friendship and trust) that exists among women. In general, women have more social capital when they work together and in homogeneous gender groups than in a mixed workplace.

In general, studies show that women have less social capital than men; what is the reason for that? Boden and Nucci (2000) argue that women have less social capital because they have had fewer years of work experience and they have a lack of managerial experience due to less exposure in the labor market. Putnam mentions the quality of ties as a crucial factor in gaining productive social capital. Social capital consists of both strong and weak ties that provide resources which enable individuals to access financial and technological resources, as well as offering access to information about contacts with new customers, access to distribution channels, new contacts, general advice and market information, and providing a bridging lubricant (Putnam 2000). Weak ties consist of the channels that exchange socially distant ideas, influences and information which create instrumental benefits, e.g., economic gains (Granovetter 1973), and have heterophilous ties. Strong ties, providing homophiles' links, create an interconnection between similar individuals who share the same sources of information, which are inefficient for instrumental purposes (Kinunda-Rutashobya 2014). Thus, the composition and strength of ties are related. Individuals in heterophilous networks have more tendency to connect with diverse groups and benefit from more and various sources of information than the individuals tied to a narrow group. The numerous stereotypes against women impede them from developing heterophilous ties that are essential for instrumental benefits.

Arasti et al. (2014) found that women have higher relational social capital when they work with their female counterparts and lower relational social capital when they work with male peers and in male-dominated occupations. This may explain why women prefer to be connected with homogeneous ties and less diverse networks.

3.3 Sources of Networks

Studies show that women are more connected with limited and non-professional networks compared to men. In different countries, women are more likely to have their personal network, usually including their husband, extended family and friends (Nielsen and Kjeldsen 2000).

It seems that overall, regardless of gender, the tendency to use one's personal network is more prevalent than other forms of networking in many countries. Schøtt (2011) investigated gender behavior with respect to the sources of advice which entrepreneurs and business owners receive from their personal, work, entrepreneurial, professional and market environments in 37 economies. He found that compared to men, women are more likely to have lower and less diversified networks. According to Schøtt men are more likely than women to have advice from friends, work, entrepreneurial, professional and market environments while women have more tendency to have advice from their family network.

The degree to which entrepreneurs can gain benefit from social capital depends on the composition, size, and relations of the players in the network. As has been observed, when it comes to women entrepreneurs' social capital, the family in general and the spouse in particular play an important role. Staber (1997) also confirms that women are more likely to report their spouse as their first network link in their business venture than their male counterparts. He found that 23 % of women versus 9 % of men have a tendency to affiliate with their spouses, indicating that the social network reciprocity is not the same among spouses. Kin play a more important role in networks of Iranian entrepreneurs as well. Zali et al. (2010), in a comparative study between Iranian and Danish social networks, found that while Iranians are more likely to receive advice from family, relatives and friends, entrepreneurs in Denmark have more tendency to acquire advice from formal networks. Moreover, they found that networks are larger and more diverse in Denmark compared to Iran.

Bastani and Hikuei (2007) studied the social capital differences between 320 - Iranian women and men with respect to the structural characteristics (size and composition of the network), interactive features (frequency of contact, method of contact, intimacy for relationship), and functional properties of the network (supported by interactive tools and resources). They found that in contrast to men, women often have less chance and opportunity to communicate with people outside their family and are less likely to have extra familial bonds. While more women than men receive emotional support from close family and relatives

(especially siblings, spouse and children), men are more likely to receive support (financial support, practical information) from friends, colleagues and other members of their networks.

A more recent study also shows a significant difference between gender and social capital in a sample of 300 women (54 %) and men respondents in Iran (Kiani 2012). In her investigation, Kiani divided networks into "emotional and supportive networks." Kiani studied the relationship between social capital and personal features (gender, marital status, birthplace and age), socio-economic status, and types of network. In general, Kiani found a lower rate of social capital for females compared to their male counterparts. Since men are more likely to be involved in official occupations, their networks are more formal than those of women and include mostly colleagues and less relatives. On the other hand, women's networks are less formal and include more relatives compared to men's networks. Kiani's results also show that there is a significant relationship between the level of education, socio-economic status, and social capital. In her study, Kiani adds that "family life makes men's and women's contacts with the opposite sex possible and provides an opportunity on interacting with men and women, apart from their same-sex inclinations." This is meaningful because Iranian women have less restriction to interact with men in their family and among their relatives due to tradition and religious concerns. In this study, no difference was observed between emotional networks of men and women, whereas a meaningful difference was reported between supportive networks of women and men.

Atkinson (2001) revealed that social capital is very important for providing women with access to mentors, as well as different forms of informal support and expansion of their business contacts. Social capital can be considered as an empowerment capital that opens new perspectives for women in business. It can be argued that improving social capital may increase the quality of entrepreneurial activity. Razavi et al. (2011) have found that as social capital rises, necessity-based entrepreneurship decreases while opportunity-based entrepreneurship increases.

3.4 Financial Capital

No business can be established and run without financial capital. Financial capital is flexible and can be used to acquire other resources (Brush et al. 2001). Financing as a special act is essentially required in practice as in theory (Schumpeter 1959). Studies show that men entrepreneurs are more likely to operate with relatively higher financial capital than their women counterparts. This seems to be true in more developed economies as well. For example in Canada, "firms headed by women appear to operate with less financial capital overall—in terms of both debt and equity—than firms headed by men." (Jennings and Cash 2006).

Do women and men use the same sources of financing? Women differ from men in sourcing finance for their businesses. Women in Australia are more likely than men to use their own financial resources and the sources of family and friends to

start a business (Barrett 2006). In Canada, women entrepreneurs seem less likely to utilize financing provided by formal and external sources (Jennings and Cash 2006). According to the International Finance Corporation (IFC 2011), women in the developing countries have less access to basic banking services such as saving accounts, and they are more likely to rely on internal and informal sources of funding (e.g., family, friends, community groups, mini finance institutions, NGOs, moneylenders, etc.). This may be due to different factors. For example, the difficulties reported by IFC 2011 in MENA include "high interest rates, collateral, lack of track record, and complexity of the application process" as the main obstacles to financing. Diaz and Jimenez (2006) consider women's poorer relationships with lenders, less experience, and inability to provide guarantees as the reasons for women not using external financing in Spain.

Women in Iran are also more likely than men to acquire internal financial capital. Arasti and Kanani (2011) studied and compared the reasons for Iranian women and men entrepreneurs not seeking external financing. They asked 24 women and 72 men in the industry sector about the difficulty of obtaining formal financing in terms of cost of debt, collateral, fear of being in debt, process rigidity and time consumption. Their results show that compared to men, women are more likely not to seek external capital because of 'high collateral' and 'process rigidity'. However, they observed no difference between women and men in terms of "debt time-consuming", "cost of debt financing" and "fear of being in debt". The reason can be that both genders pay the same amount for the cost of financing bank debt and spend equal time in waiting for the bank to process their applications. It appears that women cite more reasons for not requesting formal financing than men. The authors also found a significant gender gap in the rate of bank loan turndown. The loan applications of 45.8 % of women were rejected at least once while this rate for men was only 15.3 %. This can be attributed to the weaker entrepreneurial networks of Iranian women entrepreneurs.

The Iran National Plan Report 2004–2005 also points to the high loan interest rates, inability of women to provide collateral to apply for loans, short periods of repayment, women's lack of information about the availability of loans and credits in the financial system, and inadequate access by women to micro-credit funds as the reasons for Iranian women entrepreneurs' low demand for formal financing.

If for any reason women seeking formal financing cannot acquire a loan, then the other choice would apparently be borrowing from informal financial sources. What determines the ability to attain informal financial sources? Ammini Nejad et al. (2011) investigated the effect of the four elements of social capital (including social networks, trust, willingness to collaborate and social skills) on the ability to acquire informal funds on 77 early-stage entrepreneurs (30 % female and 70 % male) in Iran. Their study confirm the significant impact of social networks, trust and willingness to collaborate on acquiring informal financing and rejects the role of social skills in this issue. Moreover, the study shows that women are more likely to use their savings account and family and friends' financial support to start their business due to the difficulties in obtaining funds from banks or other financial institutions. According to GEM National Expert Survey, 2010, 41 % of the experts

in Iran consider financing as one of the three main entrepreneurial constraints in the country (Zali et al. 2011). Schøtt (2011), and Zali et al. (2011) also show that social capital has a key role in financing. Florin et al. (2003) found human capital to be an intangible asset that "is an important factor for obtaining financial resources."

3.5 Sources of Financing in Iran

Financing entrepreneurial ventures has repeatedly been mentioned as a main obstacle in starting a new business by many scholars, entrepreneurs and policymakers. Entrepreneurial development depends not only on the demand for financial capital but also on the supply side of business financing. The financial sources include: personal savings, family members, friends, banks, venture investors and some government organizations. According to the Iran Entrepreneurship Association (IEA), commercial banks, saving and interest-free funds, insurance companies, investment companies and credit institutes are the most important financial institutions in Iran (IEA 2013). While the quantity and the quality of financing are both very important, by focusing on the quantity we should not neglect the quality and mechanisms of financing. The structure and complexity of the financial system (financial markets and the availability of financial instruments) varies in developed versus developing countries. The financial system in Iran, as in many developing countries, is not as complicated and diversified as it is in developed economies. This may mean that diversified sources of financing are not available to entrepreneurs. Iran's financial system is unsophisticated, heavily bank-dependent with undeveloped stock market (Moayedi and Aminfard 2012). Domestic credit provided by the banking sector is 37 % of Iran's GDP. The most common means of business financing in Iran are the bank loans offered by public banks, due to the high cost of debt financing in private banks (Arasti and Kanani 2011). Furthermore, it is hard for the private sector to compete with the public sector to obtain a business loan. Public sectors are more likely to acquire loans from the financial institutions than private sectors; due to the low degree of autonomy delegated to the central bank in Iran. The Central Bank of Iran (CBI) is institutionally designed not to be independent from government interference. The governor and the deputy of CBI are both nominated by the President. The influence of government over the banking system has significantly increased in recent years. The government debt to the banking system increased four times during 2005–2011, and the government debt grew 48 % from spring 2011 to spring 2012 (Pakzad 2012).

One of the large and controversial government loan programs in Iran has been loans to "Quick Return Enterprises (QRE)" introduced during Ahmadi Nejad's presidency (2005–2013). According to the then minister of labor, cooperatives and welfare, Asadollah Abbasi, some 3.5 trillion rials (~$388 million in 2005) worth of credit was allocated to support QRE and to create jobs in the country) Iran Daily 2013). The program has been under the criticism of policy-makers and economists

as a non-performing one. They argue about the failure of QRE and the inability of the program to achieve its intended objectives. For example, Mous al Reza Servati, a member of the Plan and Budget Committee of the Parliament, stated that the QRE project was expected to create 3,100,000 jobs in the country from 2006 to 2009 and to reduce the unemployment rate to 7 % at the end of the Fourth Economic Plan (2009), whereas the realized unemployment rate moved to 12.5 % and the job creation was only 900,000 at the due date (Modir 2015; Nassaji Emrouz 2011). Moreover, Servati argued that the QRE program had reached about 50 % of its goals, and only 10 % of QRE have repaid their loans and the other 90 % have problems in repaying their outstanding debt. "QRE have proved delinquent in servicing their debts" (Amuzegar 2014).

The Presidential Office is one of the fund sources of entrepreneurial activities in Iran. The Directorate of Science and Technology in the Presidential Office maintains an Innovation and Flourishing Fund to support the innovative technological and promotion plans of knowledge-based companies. The Support Fund of Researchers and Technologists is another fund source that was established in 2013 under the supervision of the Presidential Office to provide funding to researchers and technologists. According to IEA, "this fund is managed by a board of trustees and its capital is supplied through the annual aid of the government, the aid of governmental and non-governmental systems, public donations, endowed properties, specific revenues, banking facilities and revenues generated by investing on the surplus balance of the Fund" (IEA 2013). Another fund provided by the Presidential Office is the National Researchers and Industrialists Support Fund (NRISF). The objective of this fund is to develop research activities in science and technology and commercialization of technology production. NRISF supports and assists Iranian universities, seminary researchers and industrialists, either legal or natural (IEA 2013).

Iran's Ministry of Cooperative, Labor and Social Security (MCLS) also offers loans to creative and innovative individuals who have operational ideas but cannot afford to borrow from other lending sources. These individuals are motivated to launch public joint stock co-operatives, under the supervision of MCLS.

3.6 Microfinancing in Iran

There are some private and public fund sources that provide microloans to needy individuals for lifecycle needs as well as self-employment, independent home jobs and other small investment opportunities. There are about 6000 private interest-free funds that provide micro credits to the necessity-based entrepreneurs at national, provincial, urban and community levels (Standing Committee for Economic and Commercial Cooperation of the Organization of Islamic Cooperation Coordination Office (COMCEC) 2015).

One of the microfinancing services in Iran is Agricultural Bank (Bank Keshavarzi, BK). Among the 5 public Banks, 13 private banks, and, non-bank

credit institutions in Iran, BK is the only bank that offers supportive services to women in rural and urban areas (Farzin Motamed 2011). BK became a member of the Asia Pacific Rural and Agricultural Credit Association in 1991 (www.agriculbank.org). Since 1994, KB has started a micro-financing project (Hazrate Zeynab) for women who are the head of a family in rural areas. Eligible women can borrow up to 5 million rials (~$1900.00 in 1994, equal to ~$200.00 in 2013 due to drastically devaluation of rials against dollar) with no collateral and a repayment period up to 5 years (Bakhtiari 2009). According to Mohammad Talebi, the Chairman and Managing Director of BK, from the beginning of the project in 1994 until the end of 2006, US$2.6 billion loans were disbursed to more than 129,000 eligible women in rural areas in Iran (Talebi 2012). The loans are interest free and Bank charges only 4 % operation fees under the contract called Gharz-al-hasaneh Fund (Interest-free loans). In another credit project (Irani's project), also implemented by BK, all women who have saving/checking accounts, and/or investment accounts with the BK, can borrow up to 3 million rials for economic activities with no collateral requirement (Bakhtiari 2009).

Government has also provided subsidized microfinance facilities to youth, women, and individuals in deprived areas to earn income to support themselves. Some of these government fund services are the Job Opportunities Fund, the Employment Fund at Imam Khomeini Relief Foundation, the Rural Micro Credits Fund at the Ministry of Jihad and Agriculture, and the Imam Reza Mehr Fund (IRMF). IRMF (currently renamed the Omid Entrepreneurship Fund) was established in 2006 to protect employment opportunities for people and provide facilities for marriage and housing of needy people. One of the prerequisites for investment loan at the IRMF is to have vocational skills or university degree. These subsidized investment loans were provided for job creation in terms of self-employment, independent home jobs, and entrepreneurship jobs. The upper limit of the loan offered by IRMF was as follows: self-employment, a maximum of US $3540 (now increased to US$7078), independent home jobs, a maximum of US $1062 (now increased to US$1770), and for entrepreneurship employment, a maximum of US$10,617 (now increased to US$35,390), (Bultannews 2014). The IRMF also provides loan to the entrepreneurs in scientific parks and incubators. According to Article 3 of the Fund's lending guidelines, female-headed households have priority in obtaining loans.

In spite of the aforementioned governmental projects, unemployment has remained a serious problem in Iran. Hence, it seems that the subsidized loans have not been successful in generating sufficient employment opportunities for an increasing workforce. Several reasons may be given for the inefficiency of the loans. Within a rent-seeking environment of Iran, borrowers may find it more profitable to use their loan for something other than its initial purpose. The business environment may prevent the private sector from investing in the production of goods and services. Due to the high inflation rate, there is also a tendency for loan users to invest in other areas such as the exchange market, gold market and real estate. Some of the borrowers may use the loan to fix their short-term financial

problems. It seems that there is no adequate monitoring system and control over the way a loan is invested.

3.7 Venture Capital Investors in Iran

Venture capital financing is not a very common or publicly known source of financing in Iran. Sarava Pars Investment Company is the first private venture capital investor in Iran. This company invests in the area of internet and e-commerce and also provides various business consulting services including marketing, design, financial, legal, etc., to entrepreneurs in Iran. Pishgaman Amin Sarmayeh Pasargad Company (Shenasa), a subsidiary of Pasargad Financial Group, is another venture capital investor that was established in 2012. Shenasa supports and invests in start-up businesses involved in information and communication technology. Simorgh Group is another venture capital firm that invests in knowledge-based businesses and support entrepreneurs with the greatest potential for high-growth.

Besides the private venture capital investors, there are few government venture capital investors that protect entrepreneurial activities in Iran. Nokhbegan (Elite) Technology Development Institute (NTDI), as the first venture capital institute in the country, is an independent legal entity that was established in 2002 to protect investments in advanced technologies. "NTDI provides a full package of feasibility studies, business plan, technology evaluation and companies' assessment and evaluation to all the governmental and private organizations, as well as natural and legal persons" (IEA 2013).

In order to provide qualitative and quantitative support for applied research and commercialization of technology, Novin Technology Development Fund (NTDF) was established in 2007. With the help of government and Pardis Technology Park, NTDF provides unique financial services to innovative and specialized entrepreneurs. NTDF was established in accordance with Article 100 of the third 5-year development plan and Article 45 of the fourth 5-year development plan in Iran (IEA 2013).

Another support for commercialization of innovative research is provided by Sharif Export Development Research and Technology Fund (SEDRTF). SEDRTF provides services and consulting or financial partnership in projects offered by inventors. However, once the company does not need to be protected, due to its sustainability or the entrance of new shareholders, SEDRTF will take its capital out of the supported company to invest in new innovative business plans.

3.8 Human Capital

Human capital is a broad concept and investing in human capital is more than an economic necessity: it is the basis for individuals to rise up to their full potential (World Economic Forum (WEF) 2015). Human capital is defined as the sum total of education and experience (general and specific) that an individual possesses (Neergaard et al. 2006). Human capital "expansively includes the meaning of 'human as creator' who frames knowledge, skills, competency, and experience originated by continuously connecting between 'self' and 'environment'" (Kwon 2009). This type of capital takes different forms, including education, experience derived from paid employment or managerial responsibilities, industry experience related to venture, and prior self-employment or start-up experience (Brush et al. 2006). Davidsson and Honig (2003) support the impact of human capital in predicting entry into nascent entrepreneurship, but they only weakly regard it as carrying the start-up process towards successful completion.

Human capital can be regarded as one of the production elements which can generate added-values when it is inputted (Kwon 2009). WEF, in its recent Human Capital Report, emphasizes the importance of human capital as a key determinant to long term economic success. According to WEF 2015, "A nation's human capital endowment—the skills and capacities that reside in people and that are put to productive use—can be a more important determinant of its long term economic success than virtually any other resource" (WEF 2015). The first and second (revised) editions of Human Capital Index (WEF 2013, 2015) are used here to measure the human capital in Iran.

The first Human Capital Report, 2013, captured the main contributing factors to the development of a healthy, educated and productive labor force. The Human Capital Index in 2013 contained four pillars: Education, Health and Wellness, The Workforce and Employment, and The Enabling Environment. These pillars represent the factors that are essential to the development of a healthy and productive labor force.

To provide a comparative perspective on women's human capital in Iran, the human capital indicators in Iran will be compared to those of Egypt, Pakistan, Saudi Arabia and Turkey; these four countries shall collectively be referred to as the "comparator economies" in the remainder of this chapter. As illustrated in Table 3.1, Turkey ranked higher than Iran in three of the four indicators of human capital. Among the four main components of social capital, Education is the only indicator in which Iran ranked higher than Turkey. The Education Index covers "the quantitative and qualitative aspects of education across primary, secondary and tertiary levels and contains information on both the present workforce as well as the future workforce" (Schwab and Sala-i-Martín 2013). With the overall ranking of 39 among 122 countries, Saudi Arabia performed better than the comparator economies in human capital as illustrated in Table 3.1. However, as will be mentioned later, in the new measurement of Human Capital applied in 2015 by WEF, Iran and Turkey ranked higher than Saudi Arabia in the overall rankings.

Table 3.1 Detailed rankings (among 122 countries) and scores of human capital in Iran and comparator economies, WEF 2013

	Overall index		Education		Health and wellness		Workforce and employment		Enabling environment	
	Rank	Score	Rank	Score	Rank	Score	Rank	Score	Rank	Score
Iran	94	−0.487	68	−0.051	87	−0.274	119	−1.059	88	−0.564
Egypt	111	−0.790	113	−1.206	97	−0.521	116	−0.878	86	−0.555
Pakistan	112	−0.837	111	−1.166	115	−0.920	104	−0.545	95	−0.718
Saudi Arabia	39	0.245	55	0.098	72	−0.041	61	−0.099	21	1.023
Turkey	60	−0.065	77	−0.0220	51	0.0117	83	−0.337	45	0.181

Table 3.1 shows that Iran's ranking in Workforce and Employment was the lowest of its rankings in the four components of human capital indicators and also the lowest ranking among comparator economies. This indicates, to some extent, the inability of the economy to absorb and benefit from the educated resources. "Workforce and Employment indicator quantify the experience, talent, knowledge and training in a country's working-age population" (Schwab and Sala-i-Martín 2013). Iran ranked higher than Pakistan in all three indicators except for Workforce and Employment. Iran performed better than Egypt with respect to Education, Health and Wellness, but ranked lower than Egypt in terms of Enabling Environment, Workforce and Employment indicators. "The Enabling Environment pillar captures the legal framework, infrastructure and other factors that enable returns on human capital" (ibid). According to the WEF 2013 Report, "strong enrollment rates push up Iran's scores for the Education pillar, whereas the bottom five positions for the economic participation gender gap and labor force participation (15–64) indicators contribute to driving down the aggregate scores for the Workforce and Employment pillar."

In 2015, WEF decided to change the aforementioned four original pillars of Education, Employment, Health and Enabling Environment to be replaced by five vertical age bands, selected to capture the major phases in an individual's human capital development lifecycle and countries' demographic structure: Under 15; 15–24; 25–54; 55–64; and 65 and Over. The new age band indicators completely changed the rankings in human capital among countries. Table 3.2 shows the ranking of Iran and some comparator economies in human capital in the WEF 2015 Report.

Table 3.2 presents the result of the new approach to measuring human capital in Iran and comparator economies. The new measurement studies the composition of human capital and allows a time-series analysis approach that provides information on demographic and labor force structure in countries. The index draws attention to population dynamics including "youth bulges, aging populations and shrinking workforces" (WEF 2015). As Table 3.2 shows, in the new measurement of human capital, Iran performs better and ranked higher than Egypt, Saudi Arabia, and Pakistan. The ranking of Iran has improved substantially (from 94 to 80) while the ranking of the comparator countries has deteriorated. The best performance of Iran in human capital index is among the under 15 age group, in which the country ranks better than comparator economies. Turkey performs better than the other comparator countries at the age group 15 and over.

3.9 Education in Iran

Iranian families have fundamental concern about their children's education as it has a very great social value in the country. Education is also one of the most important priorities in the Iranian government's policy. In 2013, education accounted for 21.7 % of government expenditure and 3.7 % of gross national income in Iran

Table 3.2 Detailed rankings (among 124 countries) and scores of human capital in Iran and comparator economies in 2015, WEF 2015 Report

	Overall index		Under 15 age group		15–24 age group		25–54 age group		55–64 age group		65 and over age group	
	Rank	Score	Rank	Score	Rank	Score	Rank	Score	Rank	Score	Rank	Score
Iran	80	63.20	52	84.82	73	65.17	96	53.73	96	56.72	90	44.79
Egypt	84	61.38	82	77.73	82	61.66	87	55.66	98	56.12	115	33.34
Pakistan	113	52.63	115	60.52	113	50.85	103	51.37	108	50.96	105	39.15
Saudi Arabia	85	61.38	66	82.38	70	66.10	97	53.69	107	52.23	115	33.34
Turkey	68	67.09	61	83.26	50	71.75	77	58.37	78	63.41	70	53.86

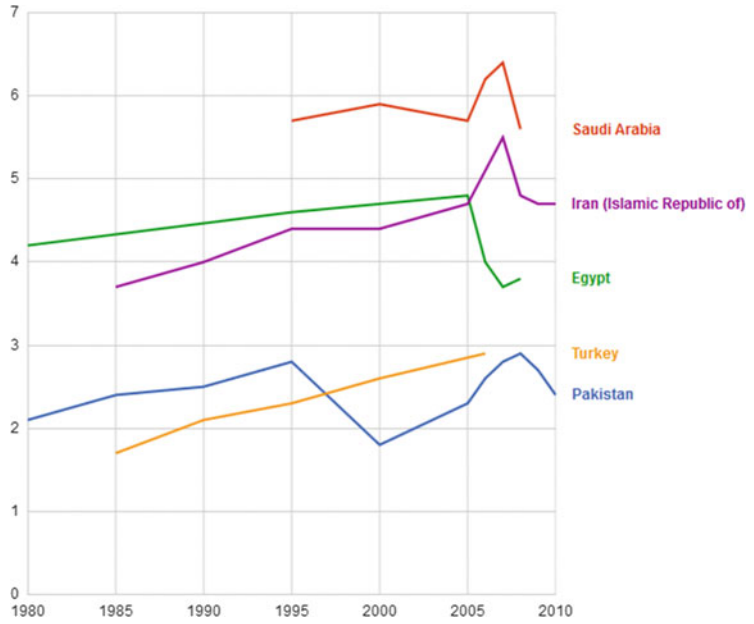


Fig. 3.1 Public expenditure on education (% GNP) in Iran and comparator economies (United Nations Development Program 2012)

(World Bank 2014).The number of students climbed rapidly in 1991–2011, rising 13 times from 344,000 in 1991 to 4,405,000 in 2011 (Abbasi-Shavazi 2013). In spite of high spending on education, the economic participation rate of the working age population in Iran is only 37.9 % (Statistical Center of Iran 2014).

Iranian government spends a lot of money on education. As Fig. 3.1 shows, the public expenditure on education (as a percentage of GNP) in Iran has been higher than Turkey and Pakistan from 1980 to 2010 and it has been higher than Egypt from 2005 to 2010. In other words, the relatively high investment in education in Iran appears to have eliminated the gender gap in education, but gender inequality in the labor market is still a major concern in Iran. This inequality is very significant and high compared to other nations. High investment in female education can lead to more economic development when women have equal opportunity to contribute to the economic growth as their male counterparts; otherwise, the country cannot capitalize on the full potential of women's human capital. Farzanegan (2011) found statistically no significant direct effect from education expenditures on labor productivity in Iran. According to the World Development Report 2012, the number of women in secondary school as a percentage of the eligible age group more than doubled from 30 % in 1979 to 81 % in 2009. Statistics also show increasing enrollment of women in Iran in universities. Women made up 23.7 % of university students in Iran in 1966; this ratio increased to 31 % before the 1979 revolution and moved up to 50 % in 2001 (Abbasi-Shavazi 2013).

In the last decade the number of women in college enrollment has surpassed the number of men in Iran. Women accounted for more than half of all Iranian university students in 2009, 68 % enrolled in science and 28 % in engineering. While this has led to an increased supply of highly educated women, the country has not been able to benefit from this well-educated and accomplished section of the population.

The integration of young women in the social and economic development of the country has not been notable. Accelerating the number of high-educated individuals in the country does not necessarily reflect the utilization and optimization of human capital. Education is a means, not an end, and education investment needs to generate productive returns in order to lead to economic development. "The essence of human capital theory is that investments are made in human resources so as to improve their productivity and therefore their earnings. Costs are incurred in the expectation of future benefits" (Benjamin et al. 1998). Hence, the goal of spending significant amounts of the government budget in education is expected to be higher productivity, more value added generation and growth in national welfare. In spite of high spending on education, the economic participation rate of the working age population in Iran is only 37.9 % (Statistical Center of Iran 2014). The unemployment rate for 15–24 year old women has also been reported as 46 % in 2014. In other words, women as a huge source of potential human capital in Iran cannot fully utilize their ability to increase productivity (Mian et al. 2013). "The efficient use of the women's potential human capital is a key driver of competitiveness" (World Development Report 2012). The recent attention of the government to develop entrepreneurship in Iran may provide regulatory environment for the prevalence of entrepreneurship among youth and women.

3.10 Education and Entrepreneurship

Education plays a crucial role in building the individual's capacity of recognizing opportunities and developing an entrepreneurial competencies. Wilklund and Shepherd (2003) suggest that growth increases with growth aspirations at a faster rate for entrepreneurs who have higher levels of education or experience. How is women's entrepreneurship affected by education? A number of recent studies attempt to clarify the impact of education on entrepreneurship; e.g., Strohmeyer (2003) indicates that, in Germany, the growth in women's entrepreneurship during the 1990s can be mainly attributed to the considerable increase in the number of female university graduates entering self-employment. Foss and Ljunggren (2006) argue that in Norway, the impact of education on women's entrepreneurship depends on their education levels. They showed that the impact of primary and lower secondary education as well as the high level education (graduate level) on women's self-employment is negative while the attainment of education at the 4-year undergraduate level has a positive impact on women's self-employment in Norway.

Table 3.3 Percentage of students at the University of Tehran by gender and the areas of entrepreneurship, 2013–2014, The University of Tehran, GEM Iran Office

Areas of entrepreneurship	Male	Female	Sum	Percentile point gap
Organizational entrepreneurship	7.2	6.5	13.7	0.7
Business entrepreneurship	8.6	3.2	11.8	5.4
Entrepreneurship promotion	7.0	5.5	12.4	1.5
Technology entrepreneurship	8.4	4.9	13.3	3.5
development entrepreneurship	6.1	5.3	11.4	0.8
MBA in entrepreneurship	8.9	4.6	13.5	4.3
Entrepreneurship in e-business	8.0	5.1	13.1	2.9
Tourism entrepreneurship	7.2	3.6	10.8	3.6
Sum	61.4	38.6	100.0	

Hassanzadeh (2007) found a strong effect of higher education on Iranian women's academic, sociocultural, and economic achievements.

3.11 Faculty of Entrepreneurship in Iran

As part of fostering entrepreneurship in the country, in the Third Iran National Economic, Social, and Development Plan (2000–2004), the Faculty of Entrepreneurship was established at the University of Tehran (UT) in 2007 as the first of entrepreneurship in the Middle East. The school of Entrepreneurship at UT offers a conventional program, an E-Learning Program, and a Non-Degree Program at the Master and Doctoral levels. Currently, 850 students are studying in different areas of entrepreneurship at UT, among whom 380 (45.2 %) are female and 470 (55.3 %) are male. Table 3.3 shows different areas of entrepreneurship at UT and also displays a higher percentage of male students than female students at all areas of entrepreneurship at UT. However, the difference in the tendency of women, compared to men, to study entrepreneurship is lowest in the area of organizational entrepreneurship and is highest in business entrepreneurship. UT also publishes an open access journal called "Journal of Global Entrepreneurship Research" via Springer.

3.12 Sharif University of Technology Entrepreneurship Center

The Entrepreneurship Center of Sharif University of Technology (ECSUT) was established in 2000 as part of the College of Management and Economics. In 2005, it was decided to broaden the goals of the center to an Entrepreneurship University to create a new linkage between the University and technology. The Center is in

close collaboration with Sharif Advanced Technologies Incubator and Technology Parks. According to the ECSUT website, "the activities of the center include research, human resource development, and organizing various educational and promotional workshops on/off campus in the field of Entrepreneurship" (The Entrepreneurship Center of SUT 2015). Moreover, the Center organizes the annual Entrepreneurship Festival.

3.13 National Organization of Skill

One of the key players in promoting entrepreneurship education in Iran is the National Organization of Skill (NOS). This organization was established by merging the National Technical and Vocational Organization with the Applied and Scientific Comprehensive University. NOS can play a critical role in the prevalence of entrepreneurship in Iran. It has different responsibilities, including providing educational and training services for those who want to become an entrepreneur, issuing permits for entrepreneurs, and running and managing professional centers and some colleges in entrepreneurship. NOS provides technical and vocational training and education in 100 specialized areas of entrepreneurship to prepare the participant for self-employment and entrepreneurial activity (IEA 2013). The organization has the largest network in Iran and collaborates with the International Labor Organization and Entrepreneurship education centers in Turin, Italy (ibid).

3.14 Women's Networks

Networks play a crucial role in encouraging, generating and fostering female entrepreneurship. However, the degree to which entrepreneurs can benefit from a network depends on the quality of the network they can access, its density, diversity (formal and informal), and its usefulness for business intentions. The idea of establishing networks for women in Iran arose as part of Iran's Third National Economic, Social and Cultural Plan (2000–2004) with the support and supervision of the Department of Labor. Firouzeh Saber together with nine women entrepreneurs formed the Association of Women Entrepreneurs (AWE), under the permission of the Department of Labor in 2005, to provide links between entrepreneurs, government, universities, NGOs and media. Subsequently, the National Association of Women Entrepreneurs (NAWE) was created under the permission of the Iran Ministry of Interior for the promotion of women's entrepreneurship. The two organizations, AWE and NAWE, have some members in common and work with consensus to provide close cooperation in exchanging information, and organizing joint annual conferences, seminars, forums, and workshops for women empowerment. They have set up national networks for women entrepreneurs and have made substantial contributions to the prevalence of female entrepreneurship in Iran. They

have a strong voice representing the challenges, experiences, and solutions for female entrepreneurship.

In accordance with the UN Women's Beijing+20 Campaign, the two networks of The Foundation for Development of Cooperation and Entrepreneurship for Women (FDCEW) and the Comprehensive Center for Empowerment, Employment and Entrepreneurship for Women and Families (CCEEEWF) were also established for women's empowerment in Iran ("National Review on Women's Status in the Islamic Republic of Iran [Beijing+20]," 2014). The main goal of FDCEW is to promote entrepreneurship and create jobs for women and families through cooperative sector. The CCEEEWF focuses on capacity building through increasing the managerial and production skills of women and families for entrepreneurial activity, and also aims to build and develop self-confidence in women. The two entities have been established with the goal of strengthening women's economic capacity and their commercial networks. Another active entity for developing entrepreneurship in the area of research, development, education and consulting plans in Iran is the Women and Youth Entrepreneurship Development Foundation (WYEDF), established in 2005. The main objective of WYEDF is to create capacity among women and youth and also to establish a network of new entrepreneurs, adolescents, entrepreneurs and social entrepreneurs (IEA 2013). The website of WYEDF[1] has been created to encourage creative thinking and productive activities among youth and women, and prepare them to be entrepreneurs. The WYEDF website contains 15 e-courses and 20 video clips about the entrepreneurs' experiences and expertise.

Due to the availability of Internet access in some rural areas in Iran, organizing multiple rural and urban networks for women to share their knowledge, discuss and overcome their challenges seems to be necessary and helpful. The increasing number of networks in Iran is promising. Diversity in networks and establishing specialized networks may help female entrepreneurs to find a suitable social network.

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Chapter 4
Women's Entrepreneurship Challenges and Policies in Iran

4.1 Women's Entrepreneurship and Culture

Women have faced gender discrimination throughout history. However, the degree of gender discrimination has been, and still is, different across the globe depending on culture, religion, social status of women, race, and institutions that define women's roles and expectations in a society.

There is no universal pattern for culture. Different historical and environmental factors have shaped numerous cultures across the world. Society's culture is the collective mind of the society, which distinguishes its members from others (Hofstede 1991). Cultures shape personality through socialization patterns (Maccoby 2000). Women and men's cognition of themselves and their position, role, and responsibility in the society, as two different genders, originates from their environment, i.e. family, kindergarten, school, television, and media. "Though located in the heart of Middle East, Iranian culture which is full of paradoxes and mixed values is distinctive from the Arab culture" (Javadian and Austin 2011). This is even truer for women. For instance, women in Iran were granted the right to vote in 1963, long before the advent of women's suffrage in the Arab countries of Jordan (1974), Qatar (1997), Iraq (1980), Bahrain (2002), Kuwait (2005), United Arab Emirates (2006), and Saudi Arabia (2015). There are also some other examples of women's self-confidence's and capability in Iran in the twentieth century. For instance, Shovkatolmolouk Jahanbani, a descendant of Fathahli Shah Qajar, was the first Iranian woman to obtain a driver's license and drive in 1940. Fakhrol-dovleh, the mother of then Prime Minister, Dr. Ali Amini, was a pioneer in establishing the first taxi services in Tehran (the capital of Iran) in 1940. Narjess Marefat Khah was the first female to become a cab driver, at the age of 50 in Tehran in 1968. While the first two women came from an elite family, Narjess came from an ordinary family. Maude (2011) also suggests that "categorizing cultures by alternative classification systems—by religions for instance—is misleading" and

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mentions the example of Iran, Iraq, and Indonesia as the Muslim countries that have distinct, contrasting culture."

"Iranian culture is distinguished by a paradoxical mix of strong family ties and connections and a high degree of individualism" (Javidan and Dastmalchian 2003). The stories of the successful Iranian women entrepreneurs featured in this book, including those who were born and still live in Iran and those who were born and spent their childhood to teenage years in Iran and then moved abroad, also confirm this cultural characteristic. For instance, Khiyal Abdolahzadeh's link to nature arises from her childhood experience of accompanying her father to the greenhouse of Mehabad Bridge, the place he was working. Her father's interest in stone collecting made her also interested in collecting herbs and motivated her to establish her own herb company later on (see Chap. 5). Mahrokh Falahi, the co-owner and CEO of Fallahi Agriculture and Animal Husbandry, mentions her interest in husbandry due to her childhood interest in her grandfather's cattle (see Chap. 5).

Inner nature, personality and characteristics of an individual can sometimes turn cultural constraints into positive drives for success. Haydeh Shirzadi is a good example of a successful entrepreneur who lived in a deprived village where girls could learn up to the fourth grade of elementary school and had to move to a bigger village to continue elementary school. In her pursuit of knowledge, she ended up earning a PhD degree in the field of ecology of environmental protectionism at Kassel University, Germany. She established Kermanshah Recycling and Composting Company to protect the environment in Iran. Her company manages to recycle 100 % of Kermanshah's waste into useful products and is now responsible for consulting and surveying projects for approximately 80 cities and 3000 villages in Iran (see Chap. 5).

The impact of the childhood experiences is observed in the success of Iranian-born women entrepreneurs who have lived abroad as well. For example, Anousheh Ansari, the first female private space explorer and one of only two women featured on a Fortune magazine list of America's 40 most successful business figures under the age of 40, mentions that, "I have a lot of roots in Iran and feel very close to the Iranian people and the culture of the country." Her childhood dream of traveling to space took shape under the night sky of Iran, which was full of stars. According to Anousheh Ansari, her interest in engineering was initiated from her childhood when her grandmother used to tell her, "Anousheh, you are a brilliant child, you should become an engineer or a physician in future" (Ansari and Hickam 2010) (see Chap. 6).

The grandmother's role in the family is also observable in Ashoobgar Cook's case (see Chap. 6), the entrepreneur who left Iran at age of 14. Ashoobgar Cook attributes her success in running the "Happy Child Care and Education Chain Company" in London mainly to her mother and grandmother "who provided the family a loving, happy and a stable home." Ashoobgar Cook considers her mother and grandmother as her role models and applies her childhood experience of enjoying light and space to provide a happy and healthy atmosphere for the children in her care. She also mentions that "the other influencing aspect from my childhood experience was the importance of food in children's diet. My mother and

grandmother like all families in Iran ensured that we had freshly cooked food every day." The good management and reputation of Cook's childcare center converted a small childcare center to over 25 day-care centers with over 10,000 children in London and Germany. In general, these cases also confirm the findings of Scott and Twomey (1988) that "parental role models and experience shape the perceptions of an individual as an entrepreneur".

"Ecologies shape cultures; cultures influence the development of personalities" (Triandis and Suh 2002). The case of two sisters, Mojgan Harriri and Gisu Harriri, who are internationally well-known New York-based architects, shows the influential factors of environment and childhood lifestyle in shaping the entrepreneurial mindset. Mojgan and Gisu Harriri attribute their creativity to their simple childhood lifestyle in southern Iran. Due to their father's job, the family had to move to deprived areas where Mojgan and Gisu used to make toys for themselves since they had limited or no access to toy shops. Nowadays, they combine their creative imagination shaped by the mysterious beauty of the deserts in Iran, where they spent their childhood, with the top digital technology of where they live now, to create unique, modern, and breathtaking designs (see Chap. 6). Hence, it seems that Ansari, Cook, and Hariri sisters who moved to western countries, and have been exposed to a different culture, have enjoyed and applied the traditional values and heritage of Iranian culture as well as the mobility and diversity in the culture of their adopted countries. While all cultures have their strengths and weaknesses, it seems that these successful entrepreneurs have combined the strengths of the two cultures.

Is the Iranian culture characterized by individualism or collectivism? While Triandis et al. (1990) consider the culture in Asia as collectivism; Javidan and Dastmalchian (2003) believe that "the Iranian culture is distinguished by a high degree of individualism." Iranian culture is full of paradoxes and contradictions. The study of successful women entrepreneurs' cases in this book also shows a combination of collectivism and individualism culture in Iran. For instance, the fact that successful Iranian women entrepreneurs were led to be more decisive as the consequence of being rejected appears to be consistent with the findings of Heine et al. (2000) that, "motivation in collectivist cultures increases following failure". Individuals do this in collectivist culture because they focus on how to change the self and improve the fit between self and the demands of the social environment (ibid). Lee and Seligman (1997) show that people in the individualist cultures have more positive self-esteem and are more optimistic than people in collectivist cultures. However, in this aspect, it seems that the women entrepreneurs featured in this book have high self-esteem and are optimistic, which is a characteristic of individualistic cultures.

4.2 Challenges in Women's Entrepreneurship

Regardless of gender, entrepreneurs in Iran face some common challenges at the macro-level in starting a venture. For instance, government domination over the economy, unfavorable business laws, international economic sanctions, restricted trade policy and protectionism, and a rent-seeking environment can be mentioned among the barriers to entrepreneurship in Iran. However, women entrepreneurs in Iran tend to face more challenges than their male counterparts due to gender inequality and social stereotypes in the country. Technically, there is no historical or legal/official barrier to women's employment, economic participation, and doing business in Iran (Khorshidi 2009). A World Bank survey (Chamlou 2010) also confirms that the MENA regions' business and investment laws are largely neutral or gender-blind. While women constitute 49.6 % of the population in Iran, the percentage of economically active women in the country is only 18.4 %. Currently, women outnumber men at the university (undergraduate) level by a factor of 2 (World Bank, country brief, 2010). In spite of high potential human capital, the entrepreneurial activity rate of women in Iran is 9.9 versus the corresponding rate of 16.2 for men. These differences are, to some extent, due to various structural and institutional factors.

Studies show that socio-cultural as well as economic factors are the main gender barriers to entrepreneurial activity of women in Iran. These factors include familial (i.e. immediate family and the family of the spouse), educational (general knowledge, managerial and marketing skills, legal knowledge, know-how, financial and accounting knowledge, and informational skills), social (inconsistency in role expectations, stereotypical beliefs, and labor environment problems), and financial barriers. Financial obstacles also include lack of personal capital, liquidity constraint, high loan interest rates, inability of women to provide collateral to apply for loans, short periods of repayment, women's lack of information about the availability of loans and credits in the financial system, and inadequate access by women to micro-credit funds. Women in Iran have attempted to overcome these barriers through persistency and patience, even by involving their husbands or other family members in their businesses to get more financial security as well as family support.

The use of social connections to exchange ideas and information is another strategy that Iranian women entrepreneurs have adopted to overcome these barriers. Arasti (2006), through a survey of 105 Iranian women entrepreneurs with higher education degrees, finds that the socio-cultural factor is not an important barrier to women entrepreneurs with a college degree in Iran. Rather, she identifies laws and regulations (83 %), administrative bureaucracy (81 %), obtaining licenses (79 %), financing (71 %), gender discrimination (69 %), market inaccessibility (65 %), management and cost control (58 %), human resource recruitment (55 %), searching for suppliers (54 %), role conflicts (52 %), finding an appropriate partner (40 %), and managing the business (31 %) as the main barriers faced by women entrepreneurs in Iran.

Women's motivation to start a new business in Iran varies with respect to different cultural and personal parameters such as age, education, social status, and family background. Galard (2007) identifies financial needs, job satisfaction, and achievement of authority and reputation in society as the major sources of motivation and the driving force for women entrepreneurs to start a business in Iran. Galard considers job satisfaction as the most effective factor in the startup process. On the other hand, Arasti (2006) considers personal satisfaction (90 %), the need to have power (62 %), financial needs (54 %), and security needs (51 %) as the main sources of motivation for women entrepreneurs with higher education.

The competency of successful women entrepreneurs in male-oriented businesses and the setting up of award ceremonies by the government to honor successful women entrepreneurs are among the factors that may positively change the way Iranian society views women's entrepreneurship.

4.3 Balancing Family and Work

A World Values survey investigates the importance of family and work across countries by asking individuals the following two questions: How important is family in your life and how important is work in your life? The answer options are very important, rather important, not very important and not at all important; the results determine the degree of importance. Table 4.1 shows the results of the 1999-2004 survey regarding the importance of family and work for individuals in Iran and the following countries: Egypt, Pakistan, Saudi Arabia and Turkey. These four countries comprise our benchmark group, which shall be referred to as the "comparator economies" in the remainder of this chapter. As Table 4.1 exhibits, 94.1 % of individuals in Iran consider family "very important", whereas 77.3 % believe that work is "very important". People in Iran are more likely to consider work "very important" than the compared countries. A higher percentage of individuals (97.3) in Turkey believe that family is "very important" compared to the comparator economies.

In general, balancing family and work seems to be a challenge for those who work, regardless of gender. However, multiple responsibility makes balancing family and work more challenging for women. "Whereas men generally see

Table 4.1 How important family/work is in your life (1999–2004 WV Survey), World Value Survey 1999–2004[a]

	Family very important	Work very important
Iran	94.1	77.3
Egypt	96.4	71.5
Pakistan	92.6	60.3
Saudi Arabia	94.7	61.5
Turkey	97.3	69.8

[a]Iran was not included in the World Values Survey in 2010–2014, and, Egypt, Pakistan, and Saudi Arabia were not included in the 2005–2009 WV Survey, hence the 1999–2004 WV Survey is used here to include and compare all the five countries

entrepreneurship as a business decision, many women view it as a life choice-a way of integrating family and career needs" (Buttner 1993). Alwin et al. (1992) studied the attitudes toward women's labor-force participation in Germany, Great Britain, and the United States and found that "the attitudes of both men and women reflect a substantial preference for a primary familial role for women, especially when young children are present."

In 2013, the Global Entrepreneurship Monitor studied and measured "balancing family and work" in 54 economies and the following interesting results were found. At all levels of economic development, entrepreneurs gain more work-life balance satisfaction than those not engaged in entrepreneurial activities. Women entrepreneurs express more satisfaction with their work-life balance than men. Entrepreneurs have less satisfaction in factor-driven economies compared to efficiency-driven economies. Necessity-based entrepreneurs also show a lower degree of work-life balance satisfaction, which is very understandable (Amorós and Bosma 2014).

Iranian women entrepreneurs are undertaking a dual responsibility of running a business and caring for family members as cultural heritage for women. In general, family is considered a priority for the majority of women in Iran. While Iranian women entrepreneurs put efforts into pursuing their business goals, they attempt to avoid conflicts with their family and spouse in particular. Less favorable attitudes toward women and work may be considered as a barrier to female entrepreneurship in Iran and the MENA region. In general, the attitudes toward working women are less positive in Iran and in the MENA region than in most other developing regions (Chamlou et al. 2008).

In a survey conducted by the World Bank (Chamlou et al. 2008) to create an index for attitudes toward working women, both women and men were asked the following two questions: Can a working mother have as warm a relationship with her children as a woman who does not work and should a husband and wife both contribute to the household income? The answers were measured to demonstrate the attitudes toward working women by both women and men. As Table 4.2 shows, both Iranian women and men exhibit more positive attitudes toward women working than the individuals in Egypt and Saudi Arabia. However, the attitudes toward women working in Iran are less favorable compared to the averages in Latin America and the Caribbean, and Europe and Central Asia regions. While across the

Table 4.2 Index of attitudes toward working women in Egypt, Iran, MENA and other developing regions, Chamlou et al. (2008)

	Women	Men
Egypt	3.89	3.47
Iran	4.22	4.25
Saudi Arabia	3.88	4.06
Middle East and North Africa	4.11	3.83
Latin America and the Caribbean	4.39	4.32
South Asia	3.86	3.8
Europe and Central Asia	4.34	4.27
East Asia and the Pacific	4.22	4.19

regions and countries displayed in Table 4.2 women have more positive attitudes toward working women than men, women's attitudes are less favorable than men's attitude toward women working in Iran and Saudi Arabia.

It seems that most of the women entrepreneurs in Iran, even those residing in metropolitan areas, have the challenges of convincing their husband that their business will not threaten their family. This is true even for college-educated women entrepreneurs. While none of the entrepreneurs featured in this book suggest family issues are a serious problem preventing them from establishing their businesses, they do mention that they have faced different challenges. As a matter of fact, some of them, like Anousheh Ansari and Fatemeh Daneshvar, involved their spouses in their enterprises from the very beginning. The women entrepreneurs who live in Iran emphasize that they have been very cautious not to let business ruin their family life. While women entrepreneurs are motivated to be successful, they are also concerned with being good mothers. Fatemeh Moghimi says she used to take her infant kids to work to eye on them.

Iran has joined the United Nations' *Beijing Declaration and Platform for Action* to promote women's status. One of the strategic objectives of this declaration is to support harmonization of work and family responsibilities for women and men. Iran has identified the following plans and legal amendments to promote balancing family and work in the country:

- The proposal to resolve the implementing problems regarding the Housewives Insurance Plan (2010);
- The proposal to resolve the ambiguity of paragraph 1 of Article 87 of the Civil Service Management Code regarding decreasing the working hours of women workers (2009);
- The proposal to increase women's maternity leave for 9 months and consider a two-week leave for them (2013);
- The proposal of the step-by-step plan of insurance for all housewives (since 2013);
- The proposal for giving one day off on Thursdays for the employed mothers having two children under 7 years of age and a child that goes to primary school that can be compensated during the week or through working at home;
- The proposal to increase the allowance of childcare within the annual budget law, appropriate with the standard fees announced by the State Welfare Organization by the age of six years. (National Review on Women's Status in the Islamic Republic of Iran (Beijing+20) 2014).

In spite of some proposals for the improvement of the well-being of women in Iran, different studies criticize family law, including articles 1106 of alimony, 1199 of child support and 11177 of the prohibition on women working in the absence of the consent of the husband. Moreover, according to law, husbands are the head of the family and wives need their husband's written permission to leave the country, a deterring factor that can cause serious problems for women, in particular those who are doing business abroad.

4.4 Gender Gap in Iran

While the gender gap is still a worldwide issue, the extent of inequality varies across the world. Gender equality plays an important role as the basis for female entrepreneurship support. "When women do not participate equally in entrepreneurship, society loses out on the value that can be created by half its populace" (Kelley et al. 2011). "Women's tremendously varied and important roles in economic activity did not translate into economic, legal, or political equality with men" (Chamlou et al. 2008).

To provide a comparative understanding of Iranian women's relative position, in terms of gender equality in the region, different dimensions of gender inequality in Iran will be compared to Egypt, Pakistan, Saudi Arabia and Turkey. While the gender gap in having access to resources is significant in favor of men in Iran, throughout the history of the country, Iranian women have had the right to own property and have control over their real and financial assets.

Table 4.3 shows that the percentage of women who have an account at a formal financial institution in Iran is significantly higher than that of comparator economies. Moreover, the difference between the percentage of women and the percentage of men who have an account in a formal institution is considerably lower in Iran compared to Pakistan, Saudi Arabia, and Turkey.

In general, the gender gap is very high in Iran, the neighboring areas and the MENA region compared to other regions of the world. In order to reveal the relative strengths and weaknesses of the gender gap in terms of some gender indicators in Iran, the 2014 Global Gender Gap Report will be studied. The Global Gender Gap Index, first introduced by the World Economic Forum (WEF) in 2006, is "a framework for capturing the magnitude and scope of gender-based disparities and tracking their progress" (WEF 2014b).

The indicators include gender gaps on economic, political, educational and health benchmarks and are calculated based on 0.00 for complete inequality and 1 for complete equality. In this study, the focus is on the economic participation and

Table 4.3 Percentage of women and men who have an account in a formal financial institution in comparator economies, World Economic Forum (2014b)

	Iran	Egypt	Pakistan	Saudi Arabia	Turkey
Women who have an account in a formal financial institution	62	17	3	15	33
Men who have an account in a formal financial institution	85	13	17	73	82

Table 4.4 The rankings and scores of gender gap in Iran, Egypt, Pakistan, Saudi Arabia and Turkey, World Economic Forum (2014b)

	Overall		Economic participation and opportunities		Educational attainment	
	Rank	Score	Rank	Score	Rank	Score
Iran	137	0.5811	139	0.3589	104	0.9574
Egypt	129	0.6064	131	0.4609	109	0.9467
Pakistan	141	0.5252	141	0.3094	132	0.8054
Saudi Arabia	130	0.6059	137	0.3893	86	0.9868
Turkey	125	0.6183	132	0.432	106	0.9527

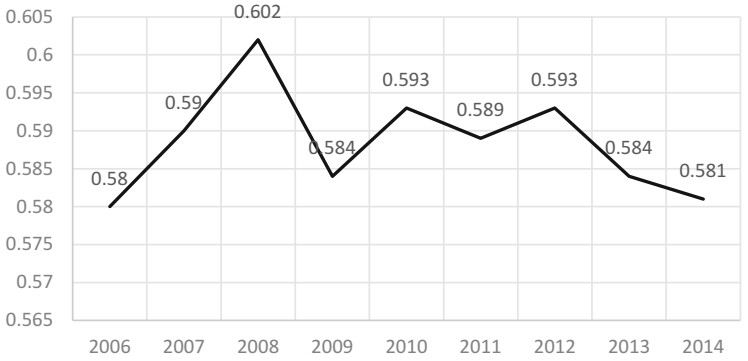


Fig. 4.1 The overall gender gap score fluctuations in Iran during 2006–2014 (World Economic Forum, the Global Gender Gap Report)

opportunities, and the educational attainment aspects of gender gaps. Table 4.4 shows that, in terms of educational attainment, Iran ranks higher than Turkey and Pakistan, and with respect to economic participation and opportunities, Iran ranks higher than Pakistan but lower than the other comparator countries.

Figure 4.1 shows the overall scores of gender-based equality in Iran since 2006 (over the period of 2006–2014). As Fig. 4.1 demonstrates, for the 2006–2014 period, the gender gap scores fluctuated over time and they did not follow a particular trend; indicating the lack of a consistent and solid plan for removing the gender gap in the country. Moreover, gender-based parity decreased significantly in the years of 2013 and 2014 compared to the years of 2008, 2010 and 2012; indicating the recent deterioration of the overall position of women in Iran compared to their male counterparts.

Iran has maintained a relatively high level of education expenditures to close the gender gap in education. However, the attempt of the Iranian government to close the gender gap in educational attainment has not provided women equality in having access to resources and economic participation. "Despite significant achievements in health and education of Iranian women, there are a number of challenges in promoting gender equity, equality and empowerment of women in

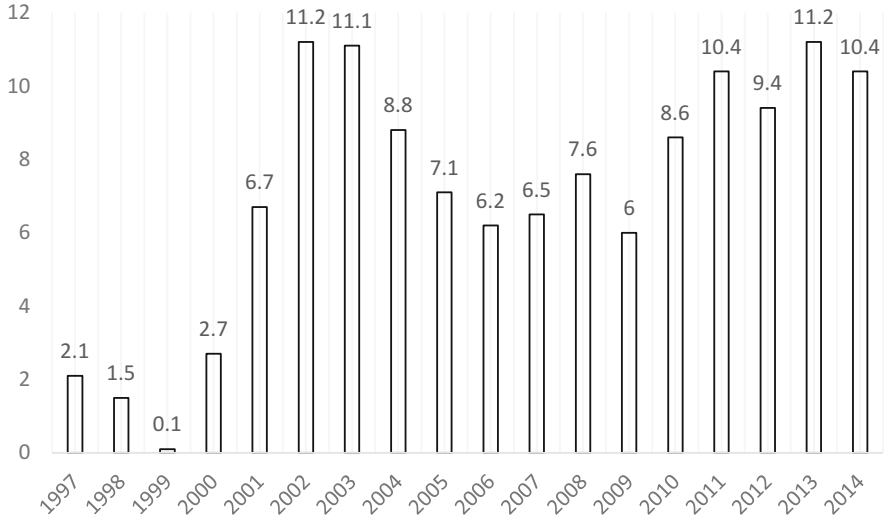


Fig. 4.2 Gender gap in unemployment rate in Iran 1997–2014 (Statistical Center of Iran)

accordance with the United Nations Millennium Development Goal (MDG) 3 and pertinent international conferences and conventions" (Abbasi-Shavazi and Sadeghi 2013).

Table 4.4 shows that, while in terms of educational attainment, Iran ranks higher than all the comparator countries after Saudi Arabia, with respect to the economic participation and opportunities, Iran ranks very low; lower than Turkey and Egypt. Moreover, the gender gap in labor force participation in Iran is very noticeable. The reasons behind the gender gap between women's and men's economic participation could be attributed to the following: institutional weaknesses such as the inability of regulatory agencies to provide equal opportunity for women and men in acquiring resources; visible and invisible socio-economic discrimination against women; environmentally based barriers such as roles and responsibilities defined by culture and society.

Significant differences between male and female participation rates demonstrate the high gender gap in unemployment rates. Figure 4.2 displays a significant increase of the gender gap in the unemployment rate in Iran in recent years. This can, to some extent, be explained by the view that men, as the breadwinners, deserve priority for employment when jobs are scarce.

As mentioned before, the gender gap inequality is not notable in educational attainment. According to the UNESCO's statistics (2007–2011), the youth literacy rate for Iranian females and males was 99 % and, at higher level education, women stand at a higher position than their male counterparts in general. In other words, investment in women's human capital in Iran has been less effective in terms of job creation for women.

4.5 Women's Entrepreneurship and Challenges in Rural Areas

"Women perform 66 percent of the world's work, produce 50 percent of the food, but earn 10 percent of the income and own 1 percent of the property" (World Bank, Women, Business and the Law 2012 Report). At the global level, women are more active in the agricultural sector than men—some 38 % vs. 33 % (SOFA Team and Doss 2011). In Iran, this rate is 38 % for women versus 21 % for men. Since men have more chance than women to migrate to urban areas in search of work, the male unemployment rate (7.6 %) in rural areas is only slightly lower than the female unemployment rate (7.8 %). Women in rural areas are more vulnerable than women in urban areas and have less access to social facilities. The socio-economic structure, lack of education and social networks in the villages usually impede women from having their own ventures. The Iranian government has implemented some plans through cooperatives to encourage entrepreneurial activity in rural areas.

Fostering entrepreneurship among women in villages would generate income, increase women's economic participation, improve resource utilization and reduce poverty. This would probably prevent migration to urban areas and, as a result, make rural areas more sustainable places to live. Women's entrepreneurship in rural areas may also be considered an important factor in changing social attitudes toward women. Cooperatives are the main empowering institutions in rural areas. The performance of cooperatives in rural areas has not been as satisfactory as expected. Jamshidi et al. (2013) found socio-cultural, economic, educational, managerial, psychological, technical, and regulatory factors to be the main barriers to female entrepreneurship in Miankuh village in Iran. The negative view of women's presence in the labor market and in business, lack of belief in the ability of women in entrepreneurship, lack of organizations for supporting women's entrepreneurship and family constraints have also been considered as the main obstacles with respect to socio-cultural factors in Mian Kuh village (ibid).

Mismatched supply of graduates and demand in the workplace has been an issue in Iran for a long time. Oversupply of college graduates in the agricultural sector and the inability of the labor market to generate job for graduates of agriculture has been particularly significant in recent years. The graduates of agriculture, natural resources and veterinary practice are planned to be trained for a period of 11 months at the private, cooperative and government centers to gain skills and technical expertise while receiving financial assistance (Standing Committee on Economic and Commercial Cooperation of the Organization of Islamic Cooperation (COMCEC) 2015). The Ministry of Jihad and Agriculture provided training for 20,000 graduates of agriculture in 2014 to gain practical skills and enter the business market (ibid).

Despite developing "rural infrastructure and social facilities throughout the 1980s and 1990s" (Hooglund 2009) in 70,000 villages of Iran, deprived villages still live in a disadvantaged situation. Cooperatives are the main players in creating and developing job creation and entrepreneurship in villages. Azkia (2011) has pointed out that many women in rural areas have entered entrepreneurship through

cooperatives that have given them access to credit and government assistance. Based on his evidence, Azkia concludes that in the past, the policy of promoting the cooperatives met with some degree of success.

The Organization of Islamic Cooperation has identified the deficiencies of cooperatives in terms of lack of residents' involvement in policy formulation and implication, lack of residents' familiarity with basic functions and the role of a corporate, poor management, lack of resources, lack of autonomy due to government interference, inadequate access to markets, lack of collaboration culture, lack of cooperatives' leaders' knowledge, and dependence of cooperatives on government (COMCEC 2015). It is essential to the success of cooperatives that villagers create them according to their needs and interests. Mashhad Al-Zahra Women's Cooperative is a good example of a successful cooperative that was initiated by Sedigheh Giyahi, a villager in AmirAbad, South Khorasan province, described in Box 4.1.

Box 4.1. Short Bio of Sedigheh Giyahi

Sedigheh Giyahi was born in 1952 to a mother who used to live in Mashhad and a father who was living with his other wife, his son and daughter in the village of AmirAbad. When Sedigheh got married in 1979, she motivated her husband to move to AmirAbad. Sedigheh's father, the chief (Kadkhoda) of the village, was also the CEO of the village cooperative, the position that was followed by Sedigheh's husband. Because of the good reputation of Sedigh's family, she was also asked and accepted to be the CEO of the cooperative in 2003.

Upon being the CEO of the village cooperative, she established the Mashhad Al-Zahra Women Cooperative (MAWC) and started motivating other women to join the organization. They established three stores that sold different consumer goods.

At that time, people in the village did not have access to basic infrastructure, such as adequate water supply and sewage systems, electricity, telephone lines, and paved roads. This was because social facilities were usually provided for villages of more than 100 households while her village had only 75 households. Sedigh decided to improve the living conditions in her deprived village and encouraged 120 women from 14 neighbor villages to join the cooperative. Once, when she was passing through different villages to motivate other women to join the cooperative, a random man told her, "You would rather stay home and do your housekeeping and not bother to change something that men could not change through the cooperative." However, the efforts of the MAWC Board brought electricity and potable water to the city and when a social need was satisfied, they decided to ask for another social facility for the village. Since their cooperative was registered, they were also able to borrow some money from Bank Melat and the Bank of Agriculture to fund some of their projects. When Sedigheh's third son finished primary

(continued)

Box 4.1 (continued)

school, there was no middle school in their village, so her son, together with nine other boys, had to attend the boarding school in kalat Naderi, 80 km from AmirAbad. One day, Sedigheh heard that her son, together with other students, was on hunger strike. Sedigheh went to the boarding school and asked the students about the reason for their strike. They mentioned the very improper and poor conditions of the school as the reason for their strike. Sedigheh called the head of education in Kalat Naderi and made an appointment with him. In a session together with the mothers of the nine children, Sedigheh discussed the problems with the head of the education and he promised to—and did—establish a middle school in their village. This experience showed Sedigheh that will is power. In continuous effort to improve the social condition, the cooperative also opened a bakery shop. Moreover, in a milk project, MAWC built two tanks with the capacity of holding 4 tons, and 6 tons of milk collected from the villages. MAWC has received about US $70 thousand in funds from the government of Japan for the milk project.

Sedigheh Giyahi is a social entrepreneur with little primary education who has inspired and managed other uneducated women to bring social facilities to her village as well as 14 small neighboring villages. Her case shows how a woman has persistently struggled to reach her individual/social goals by motivating other, less capable, female villagers to participate in the economic activities. Through her inspiring leadership, Sedigheh Giyahi has created and promoted a culture of teamwork in 15 villages. For women who live in traditional rural areas with relatively limited network interactions outside of family and friends, this is a great event that brings women self-confidence and social support. This is what makes entrepreneurship interesting!

The collaborative efforts of these women to find solutions for themselves and their society may shed light on how villagers can alleviate poverty in deprived areas. Sedigheh Giyahi, as a role model, has changed the negative stereotype about women in the small villages of Mashhad. Thus, it seems that, even in a hostile environment, there are some extraordinary women like Sedigheh Giyahi who can be the agent of social and economic changes. Sedigheh Giyahi is not the only example here; the two other cases of Khiyal Abdolahzadeh and Haydeh Shirzadi (who have been featured in this book in Chap. 5) also show how successful women entrepreneurs from rural areas can become agents of change in their society.

Box 4.2. Short Bio of Sara Shahverdi

Sara Shahverdi is a young villager who was recognized as a distinguished entrepreneur in 2005. Her story is a unique one, indicating how a teenage

(continued)

Box 4.2 (continued)

female entrepreneur can be an element of social change in a highly institutionalized, gender-discriminating community, uncovering the multidimensional nature of entrepreneurship. Sara was born in 1979, in Bezin, a small village in Zanjan, northwestern Iran. Living in a cultural background, where having a son is a divine grace for a family, Sara was born as the fifth daughter of the family. Traditionally, in villages in Iran, the first son usually accompanies his father in daily life activities while daughters stay at home and take care of their siblings. Deprived of having a son, Sara's father would ride with his 9-year-old daughter, Sara, on his motorcycle to his daily work. This formed a close and friendly relationship between them. Her father, an unskilled laborer, worked multiple jobs as a miller, construction worker, and farmer. When he died, Sara had to drop out of high school, in her sophomore year, to protect her family (four older sisters, three younger brothers and a younger sister). She decided to work hard to support her family, hence she continued her father's multiple jobs. She also wanted to gain skills to earn extra money to be able to spend some money for herself. Thus, she decided to spend 7 months in midwifery training and become a midwife. To date, she has delivered over 400 babies in her village and the neighboring villages.

As a teenager, Sara broke with tradition and made her childhood dreams of riding and owning a motorcycle come true. By borrowing money from the Agricultural Bank, Sara was able to open her own motorcycle shop, where she repairs and sells motorcycles in her village and provides a service to 35 neighboring villages. In 2003, she also established Shamim Spring, a cooperative for recycling plastic waste collected from 12 nearby districts, in her village, Bezin, and has since created jobs for 30 people.

There are other successful award-winning women entrepreneurs from small villages in Iran. Among them, I would like to share with you another interesting case of a young woman entrepreneur in the small village of Bezin who played the role of a son for her family and went on to become a successful entrepreneur. Sarah Shahverdi's case, described in Box 4.2, shows the importance of the role of gender in the family and society in small villages of Iran.

4.6 Disability and Success in Entrepreneurship

Human beings are a complex phenomenon and entrepreneurship is a complicated subject that still needs to be researched. Case studies may help us to understand hidden sides of both entrepreneurs and entrepreneurship.

Box 4.3. Short Bio of Tahereh Javan

Tahereh Javan was born in 1959. At the age of 12, her body suffered severe burns when she was turning the stove gas on, due to a gas leak and fire in the space. Hence, she had to spend 3 years of her teens in hospital and, after undergoing 24 surgeries, she regained a degree of health, though with burn scars on her face and body. It took time for her to heal physically and emotionally. Afterward, she decided to attend the free training classes offered for the disabled by Jahad Sazandegi to learn sewing and painting. Learning how to sew was a turning point in Tahereh's life. Tahereh met her future husband, who also has burn scars on his face, at the Jahad workshop. After marriage, the couple moved to Hasteh, a small village in Isfahan, and started looking for jobs to cover their costs of living. Tahereh had a creative idea of sewing a dress in a short period of time for a reasonable price. Customers could sit and wait to receive their clothes in a short time. As her business developed, she moved to a bigger house where she allocated three of the six rooms to her business. Her business was thriving and Tahereh hired 25 tailors who worked from 8 am to 11 pm to make clothes (average number of 150 per day) ready at the maximum time of 24 h. Tahereh raised two daughters; one has studied clothing design and the other is an expert in herbal medicine. She was hospitalized for diabetes in 2013 and passed away due to cardiac arrest in 2014 at the age of 55 (Tebdarman 2014).

In my research to identify successful women entrepreneurs in Iran, I learned about Tahereh Javan, a distinguished and disabled woman in a deprived village in Iran who had the unique characters of a successful entrepreneur. Since I could not directly contact and interview with her, I did not feature her in a separate chapter of this book and I used her interviews with Medias (in Farsi) to share with you her story in a short bio in Box 4.3.

The intention to start a venture may depend not only on personality but also on people's life situations, changing goals, or contingent preferences (Vesalainen and Pihkala 1999). Tahereh is a typical example of a successful disabled entrepreneur who overcame health adversity through her entrepreneurial mindset. Mindfulness helps individuals achieve a high sense of awareness by consciously recognizing and accepting the present and turning negative thoughts into positive energy (Buang 2012). Tahereh's smart understanding of her circumstance, her ability to manage her status in a positive way, her intrinsic motivation and strong drives are among the reasons for her success. Motivation is the result of a dynamic and self-regulatory mechanism rather than a linear and static state of entrepreneur's motives (Manz 1986). With her resilient disposition and religious convictions, Tahereh could maintain her self-confidence and self-control. In one of her interviews, she shows her self-confidence by saying, "I did not like to cover my face," or, "people were reluctant to talk to me, but I liked to communicate with others and live among them" (Ahmadi 2009).

Tahereh's case provides a better understanding of the application of the resilience theory in the real life. Resilient characteristics lead to "addressing problems rather than avoiding them, a positive, optimistic outlook and a flexible, adaptive disposition" (Buang 2012). It seems that Tahereh had the characteristics of a resilient disposition including "fostering acceptance, finding meaning in life career, developing gratitude, addressing spirituality, retraining one's attention" (Buang 2012). "Resilience is the ability to work with adversity in such a way that one comes through it unharmed or even better for the experience" (Houston Area Women's Center 2009).

Tahereh's hope and optimism helped her to cope with her disability and improve her lifestyle. According to the Hope Theory, "successful entrepreneurs demonstrate higher hope scores than others, and as such can expect many side benefits, from increased social satisfaction to better health outcomes" (Morrow 2006). Existing research indicates that the prefrontal cortex of the brain regulates emotion, thinking and behavior (Buang 2012). Tahereh's professional and personal identity was tied to her sewing business and, as her business was thriving, she was becoming more generous and helpful. Tahereh felt lucky and wanted to share what she had gained with others. She had skills, money and know-how about running a sewing business. Hence, she decided to hold free training classes 3 months a year and taught sewing to the women interested to learn. She pointed out in her interview that, "if the trained women be needy and cannot afford to buy a sewing machine, I buy it for them and send them customers or hire them."

"The best of all leaders is the one who helps people so that, eventually, they don't need him" (Tzu, quoted in Manz and Sims 1987). Tahereh created the culture of team group and team spirit. She pointed out that, "When a new customer comes to our tailor store, she cannot figure out who is the manager and all employee feel that the store belongs to them" (Ahmadi 2009). "A leader is best when people barely know he exists when his work is done, his aim fulfilled, they will say: we did it ourselves" (Tzu, quoted in Manz and Sims 1987). Tahereh's leadership skills at work made it possible to get things done on time.

4.7 Public Policy and Entrepreneurship

In recent years, the issue of high unemployment rates among youth has become more pronounced. 750,000 young individuals are expected to come into Iran's labor market each year and, to reduce the unemployment rate to 7 % by 2015, the government estimated that 8.5 million jobs needed to be created (COMCEC 2015). Entrepreneurship has recently received attention from policymakers in Iran as a useful and necessary solution to overcome the youth unemployment issues (Sarfaraz and Faghih 2011).

It seems that the first initiation by the government to foster entrepreneurial activity in the country was in Iran's Third National Economic, Social and Cultural Plan (2000–2004), by launching the "entrepreneurship development project in the

universities" (called KARAD). This project was launched by the Ministry of Science, Research and Technology in 12 universities in 2001 and was expanded to 44 universities based on the acts of "21" and "48" of Iran's Fourth National Economic, Social and Cultural Plan (2005–2009) (Maleki 2007). The University of Tehran, Sharif University of Technology, Khajeh Nasireddin Toosi University of Technology, Iran University of Science and Technology, Shahid Beheshti University, Esfahan University, Esfahan University of Technology, Shiraz University, Shahid Chamran Ahvaz University, Ferdowsi Mashhad University, Tabriz University and Amir Kabir University of Technology are among the universities that are implementing the KARAD plan. The objective of running an "Entrepreneurial Center" in the universities was to develop entrepreneurship and create positive attitudes in educated youth toward entrepreneurial activity by offering related courses, training seminars, workshops, conferences, and introducing best-selected entrepreneurs at the universities. Following this strategy, the University of Tehran launched its Faculty of Entrepreneurship at graduate level for the first time in Iran and the region in 2007 (the University of Tehran website[1] 2015), which currently also offers both PhD and masters programs in entrepreneurship.

Besides the University of Tehran, in recent times, five other universities in different parts of Iran have offered entrepreneurship as a field of study. Moreover, all the universities have been encouraged to offer an elective course in entrepreneurship for all undergraduate students with different majors. The idea is to establish the spirit of entrepreneurship as a core part of learning in the education system and develop the culture of entrepreneurship among youth.

The National Organization of Skills (NOS) is another key player in educating entrepreneurs. NOS was formed by the merger of the National Technical and Vocational Organization (NTVO) with the Applied and Scientific Comprehensive University (ASCU). NOS offers training and education in about 100 specialized fields of entrepreneurship, technical and vocational fields and "has a close cooperation with entrepreneurship education centers in Turin, Italy and the International Labor Organization" (IEA 2013). NOS has the largest network with the technical and professional centers, colleges, etc. (ibid).

The Ministry of Cooperatives, Labor and Social Welfare (MCLS) is the main governmental agency responsible for the development of an entrepreneurship plan and job creation for the country through various entrepreneurship development programs. The Directorate of Entrepreneurship and Occupation in MCLS is in charge of entrepreneurship and SME-development. The National Occupation Development Plan (NODP) was developed on the Fifth Development Plan to "define the tasks and duties of all the organizations and institutions, active in the area of entrepreneurship, with a strategic, future-based and plan-based approach, and also portrays the future situation of entrepreneurship in the country, in accordance with other legal obligations" (IEA 2013). According to the NODP, it is predicted that "in the year 2025, some 50 % of GDP would be provided through

[1] http://ent.ut.ac.ir/en

entrepreneurial activities" (ibid). NODP has the responsibility of providing a comprehensive plan that covers different aspects of entrepreneurship including education and training, cultural issues, financing, networking and entrepreneurial research in society. Moreover, based on the fifth development plan, the MCLS is supposed to establish the Entrepreneurship System Organization (officially does not exist yet) to lead the management in the area of entrepreneurship development. In accordance with Article 44 of the Constitution, "all the governances, in the area of investment and entrepreneurship will be delegated to the non-government sector" (IEA 2013).

One of KARAD's strategic approaches to promote entrepreneurship in Iran has been to provide basic resources for entrepreneurs to start a business. Hence, Development Centers (Incubators) and Parks of Science and Technology (PST) were established in 2000 to support entrepreneurship and offer numerous services to the companies that are eligible to use them. These facilities include general and office services, administrative and executive services, equipment and facilities services, and technical and specialized services. Presently, there are 99 PST with official permits running in different parts of the country. The majority of PSTs are founded in Tehran (21); "some seven parks of Tehran are affiliated to the University Jihad, and other parks and centers are affiliated to Tarbiat Modares University, Tehran University, Ministry of Energy, Ministry of Health and Medical Education, Amir Kabir University, etc. Fars Province, with 8 parks and Khorassan Razavi Province, with 7 parks, are ranked second and third after Tehran" (IEA 2013). However, it seems that the Development Centers in Iran have not been productive enough to foster entrepreneurship in the country (ibid).

MCLS has also initiated the Business and Sustainable Employment Plan called "Takapoo" (meaning Prowling) to promote local businesses and focus on agricultural and industrial development in the poorer provinces through its branches in all 27 provinces in Iran. Special committees have been assigned in each branch to research local manufacturing opportunities and promote marketing for local production. All affiliated institutions of MCLS are required to participate in the project on a provincial level and are to provide funding for businesses. The project has been implemented in 6 of Iran's 27 provinces. While all the aforementioned projects are promising, the way they are implemented should also be a matter of concern. Some studies criticize and question the quality implementation of these plans and projects. "One of the greatest challenges in policy making system refers to the gap between current and desired situation of institutions at experts' view point in this field. These institutions are far off their desired role" (Rahmati et al. 2011).

4.8 Women's Entrepreneurship in Public Policy

The government's consideration of the importance of women's involvement in business activities in Iran began with the third National Economic, Social, and Development Plan (2000–2004), in a chapter devoted to the topic of "women and

the economy". This trend has continued since then in the fourth (2005–2009) and fifth (2010–2014) National Economic, Social, and Development Plans. In these plans, the Beijing Strategic Objectives of the United Nations (1995), for instance, facilitating women's equal access to resources, employment, markets and trade, access to savings and credit mechanisms and institutions, etc., have been considered for fostering women's empowerment in Iran. However, in reality, evidence shows that these objectives have not yet been fully realized. Different domestic and international surveys and studies related to women's entrepreneurship indicate the marginalization of women in business in Iran.

Important steps have been made since the 2000s to improve the status of women in Iran. For instance, in Article 182 of the Third National Development Plan, Several NGOs have been established in recent years to develop women's entrepreneurship in the country. Among these organizations are the Women Entrepreneurs' Network,[2] the Association of Women Entrepreneurs[3] registered by the Ministry of Labor and Social Affairs in March 2005, and also the Women and Youth Entrepreneurship Development Foundation[4] (WYEDF). The purpose of WYEDF is to encourage entrepreneurship development among young females through providing education, research support, and consulting services. WYEDF has the following two basic roles: 1—creating capacity, and 2—creating a network of new entrepreneurs, adolescent entrepreneurs and social entrepreneurs. "In line with its major objective, namely promotion and development of entrepreneurship, three concepts are included in the objectives of the WYEDF: conceptualization, modeling, and institutionalization." (IEA 2013).

"Some entrepreneurship associations, such as the Association of Women Entrepreneurs, and the Association of Young Merchants and Entrepreneurs do have office at the Chamber of Commerce and are members to this chamber" (IEA 2013). Moreover, the Women's Employment Bureau at the Ministry of Labor and Social Security organizes entrepreneurship seminars, forums, and workshops for women. There are also education centers like the Khorshid Entrepreneurship School[5] and some universities such as Alzahra University[6] that offer free classes and training for women interested in entrepreneurship.

In order to develop the entrepreneurial culture in Iran, Tehran Municipality has created 23 entrepreneurship centers. According to the IEA, each of these municipal community centers contains 13 entrepreneurship houses. The house of empowerment and popular collaborations is one of these houses that is responsible for "education, observation, promotion and development of entrepreneurship and local businesses" (IEA 2013).

[2] www.iranzanan.com

[3] www.ea.wenet.ir

[4] www.foundationed.net

[5] www.khorshidschool.org

[6] http://jahad-alzahra.blogfa.com/post-415.aspx

"Iran is also a member of the Forum for Businesswomen in Islamic Countries, organized by the Islamic Chamber along with the Islamic Development Bank that offers entrepreneurship workshops for women on a regular basis" (Sarfaraz 2011). The award ceremonies held by the government at local and national levels to honor successful women entrepreneurs and the media attention have been a great source of motivation for women to become entrepreneurs in Iran.

4.9 E-Entrepreneurs

Technological diffusion and the growing possibility of greater access to information across the world promise a brighter future for the youth in developing economies. According to Abbasi et al. (2008), "Iran is among the first five countries which have had a growth rate of over 20 percent and the highest level of development in telecommunication and has been awarded the UNESCO special certificate for providing telecommunication services to rural areas". According to the Internet World Stats,[7] the number of internet users in Iran has increased from 250,000 in 2000 to 42,000,000 (46.7 % of population) in 2012. This may provide great opportunities for educated women and men to create value out of their human and social capital. It could enable women to become involved in e-entrepreneurial activities. While women's entrepreneurship in information technology is not a new phenomenon in Iran, it seems that the new generation of entrepreneurs is more likely to move towards technology ventures. One of the pioneers in high-tech venture start-ups in Iran is Azadeh Dannandeh (Box 4.4), the co-founder of the Baharan IT Company.

Box 4.4. Short Bio of Azadeh Dannandeh

Azadeh Dannandeh graduated from Shiraz University with a Bachelor's degree in Computer Sciences in 1983. After graduation, she started working at the Iran Data Processing Company. After gaining 8 years of work experience in the field, Azadeh started to work as an independent consultant for different organizations. One of the turning points of her career was participating in a project in partnership with the Ministry of Industry, the High Council of Informatics, and the Informatics Companies Association in 1999. The objective of this project was the design and implementation of quality management systems in software companies, to enable the companies to offer their products to international markets and promote the export of software. The project was successful and, as a result, six companies in Iran obtained an ISO 9001/Tick IT Certificate for the first time in the country's history

(continued)

[7] www.internetworldstats.com/me/ir.htm

Box 4.4 (continued)

(Dannandeh 2015).[8] Azadeh established Baharan IT Company (BITC) in 2005. The company has three female shareholders and all of its management team are also women. BITC is a company engaged in the development of information technology and focuses on the production of software, hardware, and network infrastructure. Azadeh received the first "Women in Sustainable Economy" award in the field of ICT from the Ministry of Industry, Mine and Trade In 2015.

A more recent distinguished young woman entrepreneur in IT is Nazanin Daneshvar, the owner of a website called Takhfifan in Iran (Box 4.5). Daneshvar, who was featured in the Washington Post[9] as a young IT entrepreneur, also owns an active website similar to Yelp with 20,000 written reviews.

Box 4.5. Short Bio of Nazanin Daneshvar

Nazanin Daneshvar is the founder of the first "Group Discount Buying" website in Iran, named Takhfifan. Daneshvar is 32 years old and has a Bachelor's degree in Software Engineering and IT from the University of Amir Kabir. She gained experience working in the technical department of Groupon in Germany. When Daneshvar was working in Germany, she had the dream of having her own business in Iran. Hence, in 2011, with the help of a German colleague, Daneshvar launched her Group Discount Buying Website business with 500 million rials (~$50,000 in 2011) (Bahramzadeh 2015).[10] She started Takhfifan with her sister and now more than 60 people in Tehran and other cities of Iran are working for her (Bahramzadeh 2015). The website has created a win-win situation for buyers and sellers. Takhfifan has more than one million subscribers with 25 deals a day. More than 10,000 merchants are working with Takhfifan and the company has expanded its services to different big cities of Iran including Shiraz, Mashhad, Esfahan, Tabriz and Karaj. "Daneshvar was chosen as Entrepreneur of the Year in 2011 and was also the winner of the Web Festival Award in the years afterwards" (Nasihatkon 2014).[11]

[8] http://www.ea.wenet.ir/index.php/fa/how/571

[9] https://www.washingtonpost.com/news/innovations/wp/2014/11/10/heres-what-its-like-to-launch-a-start-up-in-iran/

[10] http://mehr-it.ir/author/mehrnaz71/

[11] http://techly.co/2014/08/get-a-bright-discount-online-takhfifan-com/

4.10 Successful Women Entrepreneurs: Role Models

Entrepreneurship does not take place in a vacuum. While there are some similarities among entrepreneurs, the context of socio-economic and cultural backgrounds of different nations determine the quantity, quality and dynamicity of their entrepreneurial activity. These background characteristics also seem to shape the performance of female vs. male entrepreneurship inside a country. In developing countries, women entrepreneurs are more likely to be under the pressure of stereotypes, including culture, tradition and patriarchal perspective that has been formed through formal and informal institutions. Successful women entrepreneurs in these economies seem to break away from tradition, change gender stereotyped roles, build a new rule of the game and demonstrate the diversity of women's capabilities in the professional world.

Successful entrepreneurs are distinguished from others in different ways. It is likely that they do not just look around, they see their environment, search for opportunities, persistently challenge the status quo, explore shortages and attempt to find a solution for a better life. They usually observe something that others may not be able to notice. Successful entrepreneurs are likely to be proactive change agents, creative and not satisfied with the present situation. It seems that they have strong innate talent and energy for entrepreneurship comparable to that of artists, musicians and poets.

The study of successful women entrepreneurs featured in this book shows that, in some cases, no matter where and how these women entrepreneurs have been raised, whether in rural areas or urban areas, in rich or poor families, with educated or illiterate parents, they have an innate ability to be observant, creative and persuasive to reach their goals. They have the ability to guide themselves toward reaching their dreams. They learn from failure and go ahead for further success. Successful entrepreneurs play a major role in economic growth and serve as role models for their generation and future generations. In other words, they contribute not only to economic growth but also to the improvement of quality of life and social development. Successful Iranian women entrepreneurs can create values in their society by changing the community norms and social attitudes toward women. They explore opportunities, combine various resources, and use their social relationships to make their dreams come true. While these competent women are patient, confident and flexible, they find creative solutions at the right time to overcome numerous socio-economic obstacles.

Some researchers have identified the following characteristics of successful entrepreneurs: being creative and having good interpersonal, mental and technical skills (Hodgetts and Kuratko 1992), having clear mission, purpose and values (Thompson 1999), having the ability to identify and explore opportunities both at a firm or individual level (Shane and Venkataraman 2000), and being good leaders (Cutting and Kouzmin 2000). Swiercz and Lydon (2002) discuss the two kinds of leadership that are required for the success of entrepreneurs; functional and self-competencies. While functional competencies consist of operations, finance,

marketing, and human resources, self-competencies include "intellectual integrity, promoting the company rather than the individual leader, utilizing external advisors, and creating a sustainable organization" (Makhbul and Hasun 2010).

Successful entrepreneurs can serve as ambassadors and role models for future entrepreneurs. There is a common agreement that entrepreneurs, regardless of gender, rely on role models and social networks for information and access to resources (Vossenberg 2013). Minniti (2010) found a strong positive and significant correlation between knowing of another entrepreneur and a person's involvement in starting a new business. Shapero and Sokol (1982) also explored a strong connection between the presence of role models and the emergence of entrepreneurs. Moreover, studies show that the influence of role models is gender related. That is, an individual will be more influenced by another individual of the same sex (Deaux et al. 1998), implying that women will be more inspired and influenced by the success of female entrepreneurs. Role models and mentors (in particular, within family) are the major sources of support and motivation for female entrepreneurs through providing them guidelines and advice (Carter and Cannon 1988; Naffziger et al. 1994; Orhan and Scott 2001). According to Khan et al. (2005), the lack of role models in entrepreneurship is one of the many obstacles for women entrepreneurs in Oman. "Recognition and rewards right from the local up to the national level will energize and encourage new entrepreneurs" (Entrepreneurship in India National Knowledge Commission 2008).

The second part of this book is dedicated to the cases of successful Iranian women entrepreneurs who live in Iran and abroad. All 11 entrepreneurs featured in this book were born and raised in Iran; 4 of them moved from Iran to England and the United States of America in their teenage years. Table 4.5 displays a summary sheet of those women entrepreneurs. More detailed information on successful Iranian women entrepreneurs is presented in part II of the book.

The women entrepreneurs featured in this book cover a wide range of age, education, social status and income, representing a diverse array of industries in urban and rural areas of Iran and abroad. The lack of information about these capable women (and women like these) deprives the world community, especially women, from learning about and believing in the competency of women and their value as important resources. Studying the lives of these influential and inspirational entrepreneurs who have overcome numerous barriers would be encouraging for women who are interested in running—or need to run—their own business. This would also give an insight into a better understanding of the entrepreneurship concept, with a special focus on women's issues in developing economies.

4.11 Prognosis for the Future

23.5 % of the population in Iran is under 15 and 31.5 % of the population is aged between 15 and 29 years. It seems that the millennial generation (born after 1980) is open to change and is going through a transition. A survey conducted by Moaddel

Table 4.5 Successful Iranian women entrepreneurs' summary sheet

Name	Age group ATS	Position in the company	Education level ATS/recently	Work experience ATS/years	Industry sector	Jobs created/number of employees	Name of the current/former company	Source of financing ATS
Abdolahzadeh, Khiyal	41	Founder & CEO	High school diploma/B.A. in Economics	None	Extractive/ agriculture	20 permanent employees/ 350 seasonal employees	Anthemia Herb Production & Distribution	Self-financing
Baradaran Ghahfarokhi, Parvin	36	Founder & CEO	B.Sc. in Chemistry/Ph.D. candidate in Entrepreneurship	12	Business services/ knowledge-based technology	10 postgraduate experts	Kuhrang Knowledge-Based Instrumentation Technology	Self-financing, award grants
Daneshvar, Fatemeh	26	Founder & CEO	M.A. in Business Management	2	Extractive/ mineral import-export	100 direct employees/500 indirect employees	Sepehr Asia Mineral Import-Export	Outsourcing
Fallahi, Mahroakh	34	Co-founder & CEO	B.A. in English	14	Transforming/ animal husbandry	12 families permanent employees/35–40 daily employees/450 indirect employees	Fallahi Livestock	Self-financing and outsourcing
Moghimi, Fatemeh	27	Founder & CEO	B.Sc. Civil Engineering/B.A. in Management, DMBA	5	Transforming/ transportation (international)	300	Sadid Bar International Transportation	Self-financing
Sedigh, Banafsheh	24	Founder & CEO	B.Sc. Electrical Engineering/ MBA	2	Manufacturing/ auto parts and gas valves	100	Shigal MFG	Outsourcing
Shirzadi, Hayedeh	38	Founder & CEO	Ph.D. in the Ecology of Environmental Protectionism	None	Transforming/ recycling and composting	245/700	Kermanshah Recycling and Composting	Outsourcing

Ansari, Anousheh	27	Co-founder & CEO	M.S. Electrical Engineering	4	Business services/innovative technology	250	Prodea Systems	Savings, credit cards
Cook, Ashoobgar	34	Founder	B.A. in Psychology	None	Transforming/ daily care	850	Happy Child	Outsourcing
Hariri, Gisue	28	Co-founder	B.A. in Architecture	3	Transforming: residential and commercial	10	Hariri & Hariri Architecture	Self-financing
Hariri, Mojgan	26	Co-founder	B.A. in Architecture and M.A. in Urban Design	3	Transforming: residential and commercial	10	Hariri & Hariri Architecture	Self-financing

ATS at the time of startup

(2010) showed "a shift toward social individualism, liberal democracy, gender equality, and nationalism" among Iranians between 2000 and 2005. Since then, the trend of change may be expected to be even more significant considering the increasing awareness created by the exponential increase in the number of internet users in the country. Iran has the biggest group of Internet users in the Middle East; the number of internet users has increased from 250,000 in 2000 to 7,500,000 in 2005 and to 42,000,000 in 2012. About 60 % of the population aged 18–28 have internet access at home (Wojcieszak et al. 2012).

In spite of the low rate of entrepreneurial activity among Iranian female entrepreneurs, according to the GEM 2010 Women's Report, women who have started businesses in Iran are likely to have high expectations for growth. The most recent GEM survey shows that female early-stage entrepreneurship in Iran has seen more than a twofold increase in 2014 compared to 2008. Moreover, the current generation of women entrepreneurs in the country, compared to the previous generation, is likely to be more educated (due to the increasing number of female college graduates) and less restricted to traditional gender roles. It would appear that young women in Iran are becoming more knowledgeable and interested in entrepreneurship. According to the GEM 2010 Women's Report, "22.4 percent of Iranian entrepreneurs are in the age group of 18–24 and 39.5 percent in the 25–34 age group" (Sarfaraz 2011).

Evidence shows the prevalence of IT in the country. The Iran Web Festival (IWF) was created in 2008 to promote on-line businesses and identify the best websites in the country. According to the Washington Post Report, the number of applicants for IWF increased from 100 in 2008 to 8000 in 2014 and 450 people attended a related conference (McFarland 2014). Raising awareness of the importance of information technology and the role high-tech can play on Iran's sustainable development is not limited to the entrepreneurs in the country; a jumpstart program named "BRIDGE"[12] has also been launched at an international level in fostering entrepreneurship in high-tech in Iran. The first BRIDGE event was held in UC Berkeley, USA on September 6th, 2014, followed by the second gathering of Iranian entrepreneurs from Europe and America, and their counterparts in Iran on June 4–6, 2015 in Berlin, Germany. The participants shared their experiences and knowledge and also discussed the challenges impacting high-tech entrepreneurship in Iran. The aim is to build educational bridges between Silicon Valley (as a symbol of high-tech) and Iran to foster entrepreneurship and innovation, and develop the culture of "Entrepreneurs without borders" in the country.

"BRIDGE attempts to reduce the gap between the 'global innovation model' and the prevailing conditions in Iran. Another aim has been to bring the attention of the world to 'High Tech Entrepreneurship in Iran: Opportunities and Challenges'" (Bridge 2014). Bridge events received international media attention in some

[12] http://www.bridge2014.org/

popular newspapers like the Guardian[13] and the Financial Times.[14] The Guardian (May 31, 2015) quoted one of the non-Iranian attendees of the Bridge event, Dave McClure, the founder of 500 Startups, saying that, "If the country opens up and relations are restored with the US and other parts of the world, I think there is going to be a lot of economic growth. Definitely there are many interesting possibilities for tech startups, too." The Washington Post also featured some of the young women and men high-tech entrepreneurs in Iran; a part of the report stated that, "Shades of Silicon Valley are bubbling up in Tehran, which is home to a growing tech ecosystem with start-ups, accelerators, and a venture capital firm, despite the significant challenges of launching a tech company in Iran" (McFarland 2014).

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[13] http://www.theguardian.com/technology/2015/may/31/amazon-iranian-style-digikala-other-startups-aparat-hamijoo-takhfifan

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Part II
Role Models, Growth-Oriented Iranian Women Entrepreneurs

Chapter 5
Iranian Women Entrepreneurs Living in Iran

5.1 Entrepreneurship and Poverty Reduction

Khiyal Abdolahzadeh, Anthemia Herb Production & Distribution, Ltd.



Khiyal Abdolahzadeh is an anthemia herb producer who started her company from scratch by harvesting wild herbs from free land. Khiyal's case demonstrates the extra-ordinary power of a villager who challenges numerous educational, social, and financial problems in her life, and eventually, successfully uses her emotional intelligence and inborn entrepreneurial talents to overcome all serious obstacles.

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She is a romantic, sensitive, industrious and naturalist social entrepreneur who is well known for her tireless efforts to improve the lives of others. Her creativity, humanitarian activities, hard work, and understanding of the limited opportunities available in a deprived village inspired her to use natural lands as a source of revenue for herself and the needy illiterate women in her village. She has been successful in creating jobs for women who have no income or limited income sources in a deprived village where there are very limited local job opportunities.

Khiyal was born in 1962 in Dashtjure, a small village near Mahabad in the northwest of Iran. When Khiyal was 1 year old, the village was destroyed because of the construction of the Mahabad Bridge. Her father, who was a farmer, was employed in the greenhouse of the Mahabad Bridge. He was a naturalist and had a collection of stones. As a child, Khiyal used to accompany her father to the greenhouse and was exposed to nature. This is how her passion for plants started. She was curious to know about the uses and the benefits of plants and had a passions for the secrets of nature. Khiyal's interaction with nature and her father's hobby of collecting stones motivated her to collect herbs. Her interest in collecting different kinds of herbs and pasting them into an album became the basis for her future entrepreneurial activity.

The family of Khiyal lived in a poor district where the majority of people were almost illiterate and children were not motivated to learn and study. While attending elementary school in early 1970s, she built up good relationships with her classmates, teachers, and the personnel of the school. Khiyal was concerned about the people around her since childhood. At school she took various leadership roles, such as taking the younger students to their homes after school.

Khiyal has been a self-starter, interested in learning and motivating others to learn. Since there was no library in the area, Khiyal asked her father to allocate two rooms in their house for a local public library. In middle school, she finally managed to found a library with the help of her parents, teachers, and the Department of Education in her town Mahabad. Cleverly, she organized contests for the readers in the library and motivated them with small prizes that she provided.

Tragically, Khiyal's father died when she was a high school freshman. As a result, she became responsible for the care of her three younger brothers, because her two older sisters were married. The family was in a desperate financial situation and the war between Iran and Iraq made their situation even worse. Their house was bombed in the war and she had to work one summer at a brick kiln to support her family and continue her education up to the high school diploma. However, she never lost her high spirit and motivation, even finding time and energy to assist those with war injuries.

Marrying an agriculture propagator who lived in a village, moved Khiyal from town to village Hamzeabad, where she had to live in her husband's extended family of 40 relatives. Quickly she became friendly with people in the village and made connections with the school where no educational facilities, such as library, science laboratories, and other complementary services, were available. Khiyal dedicated 45 books to the school to promote reading; she also wrote stories for the students, directed a few plays, and encouraged poetry reading. Her efforts made a great

change in the lives of the children, some of whom could not even speak Farsi,[1] the official language of the country. She shared her knowledge and arts with the people of the village, by administering injections and first aid, and teaching sewing, embroidery and crochet. Khiyal points out that helping people gives her more energy, and believes that she has learned a lot from the numerous challenges of her life.

After the death of her father-in-law, Khiyal had to live with her brothers-in-law and their families, and her mother-in-law, a total of 18 people. Because the inheritance was divided among many heirs, not much heritage was left for Khiyal, her husband, and her two sons and two daughters. Khiyal's family were placed in a difficult financial situation, and therefore, she started looking for a paying job but it was not easy to find a job.

Living close to nature and Khiyal's interest in plants motivated her to learn more about the herbs and their applications in treating different diseases. Her husband, who was quite supportive, provided her with agricultural magazines to increase her knowledge of herbs. This was an essential step toward entrepreneurship. With the help of her husband, Khiyal participated in a training program conducted by Jahad on reproduction of herbs at Mahabad. Khiyal says that her husband has played a key role in her life by supporting her and providing her with agricultural magazines to increase her knowledge of herbs, which has been her passion since childhood. Moreover, Khiyal acquired more knowledge about herbs from Euromiyeh Medical School Research Center.

Khiyal established her own herb business in 2003 and invited the experts in the Euromiyeh Medical School Research Center to evaluate her activities and express their opinions. It was very encouraging to have her work confirmed by this expert group. This was the beginning of a new life. Khiyal started recognizing and providing different species of herbs. She now owns and manages Anthemia Herb Production & Distribution Company, Ltd., whose products include various components of herbs e.g., flowering tip, fruit, root and rhizome. Khiyal could cure her sciatica pain by using herbs.

In the first year, her company produced only 90 kg of herbs. As part of a plan to expand service across the country, the company has representatives in the main cities of Iran such as Tehran, Kermanshah, Isfahan, Yazd, etc. According to Khiyal, there is a strong market for her products; each year she sells all her inventory before the beginning of Persian New Year (marked by the beginning of spring). Her company has employed 20 full time (permanent) women and 350 seasonal unskilled women producing a daily average of 900 kilos of 69 herb species. The daily amount of the product collected varies depending on the weather, and the species. In 2008, the production was 16 metric tons, of which 10 tons was exported to Iraq (neighboring her native town Mahabad). She plans to install plant and equipment for extraction of essential oils, distillation of plant material, and produce herb facial cream as well.

[1] People in Mahabad speak Kurdish.

Khiyal was so excited when she was selected as the Most Active Woman in Mahabad by the Agricultural Jahad and could participate in the Mahabad Great Exhibition of Development. She compares her feelings to the Wright Brothers, when they first took off with their plane. Nobody in the village could believe that a woman from a small village could be honored for her efforts and capabilities. She mentions that her husband always accompanied and motivated her. In 2003, Khiyal was chosen as the best-selected social assistant in Mahabad and received a prize from the Mahabad governor and the Agricultural Jahad top manager. In the same year, the Deputy of the Agricultural Ministry visited her workshop and she was awarded a trip to India. In 2004, she was recognized as a distinguished herbal instructor, and started teaching classes on herbs in the West Azerbaijan Province. From 2004 to 2009, more than 600 people participated in her herb training program and obtained certificates in these courses. In 2004, 2005, 2006, 2007, and 2008 she became the best selected herbal producer in the West Azerbaijan Province. Participating in different national and local conferences and exhibitions have helped her improve and extend her activities. In 2007, she was selected as the distinguished entrepreneur of the year. She was also selected by the Faculty of Entrepreneurship at Tehran University (representing Global Entrepreneurship Monitor in Iran) as a distinguished entrepreneur in 2008–2009.

A thesis called "Khiyal" was defended by Mohsen Ahmadi at the Faculty of Entrepreneurship at Tehran in 2013. The thesis was a documentary film about the life of Khiyal Abdolahzadeh. This documentary explores the life of Kyial and her challenges in the three stages of: childhood to adolescence, starting a business, and becoming a distinguished national award winner entrepreneur. The film also describes the personality characteristics, personal experiences and family background of Khiyal. Khial's granddaughter played the role of Khiyal's childhood in that documentary.

Being a self-directed learner and a self-starter since childhood have been the key success factors for Khiyal. She also has other qualities and characteristics that distinguishes her from others. She enjoys working in groups and appreciates the significance of teamwork. Khiyal is realistic and listens to her internal thoughts and voice, believes in change, and welcomes criticism. She highly respects ethics and morality, and delightedly passes on her skills, knowledge, and experiences to others. She thinks that motivation produces creativity and innovation as the key to success. She is rational and decisive, believes in consulting and in diversity of opinion and has strong self-confidence. In her opinion, high achievements require dreams, plans, tolerance, and a strong belief in what one is doing. She counts work, competitiveness, love, family, thought, arts, excitement, and challenge as the effective factors for her development and growth. Moreover, radio and Television broadcasts of her activities, interviews and participation in exhibitions and conferences are considered by her as personal encouragement and helpful in marketing her products.

5.2 Entrepreneurship and Technology

Parvin Baradaran Ghahfarokhi, Kuhrang Knowledge-Based Instrumentation Technology, Ltd.



Parvin Baradaran Ghahfarokhi is an influential entrepreneur, who is passionate about using technology to protect the environment from pollution. She has established a company with the motto of improving quality of life through environmental technology.

Parvin was born in 1971 in Farokhshahr, Chaharmahal and Bakhtiari province in central west of Iran. She was born to a family of 6, as the second child and the first daughter. Pravin's father and her uncle (father's brother) cofounded a grocery wholesale store. Parvin's family and her uncle's family, including five children, lived in the same building. The kids used to play together and learned how to interact in groups and develop relationships. When Parvin, her siblings and cousins, reached the age of 12 they worked as salespeople in their father's store. They marketed the products of the wholesale store to retailors and received tips for their work. In the other words, Parvin learned teamwork, communication skills and marketing skills through her childhood experiences.

Parvin earned Bachelor of Science in Chemistry from Shahid Chamran Ahvaz University in 1995. Studying chemistry provided Parvin with a solid foundation for her future research and business growth. In 1996, she was hired as the first woman to work at Chaharmahal and Bakhtiari Pak Dairy Industry (PDI), a well-known company in Iran, to serve as a lab manager. Parvin was very hard-working,

knowledgeable, and caring, and soon became director of R&D and the head of the establishment of the ISO system and the food safety management system (HACCP).[2] Parvin was inspired to learn more; hence during her 13 years of work in PDI, she attended a variety of workshops including accounting, computer and management that were held by PDI. Parvin was selected as distinguished employee of Chaharmahal Bakhtiari Province three times.

Learning about the policy of the Iran's Industrial Development & Renovation Organization regarding encouraging the industries to establish systems management by subsidizing 80 % of the costs in 2006, Parvin decided to represent a reputable company and provide management consulting to the industries in the region. Therefore, she requested time off from work and visited factories, became acquainted with the production lines and gained valuable experiences.

Parvin also learned that the Labor and Social Security Institute (LSSI) had planned to hold 72 hours of entrepreneurial training workshops in different parts of Iran. Parvin contacted LSSI and when she was informed that a committee from LSSI had a visit to Chaharmahal Bakhtiari, Parvin arranged a meeting with them and requested for setting up entrepreneurial training workshops in the area. This is how the first company of Parvin named Pishgame Pouyesh Sanat (Pioneer Dynamic in industry) was established in early 2007. In a successful program, the Company offered 72 hours training on entrepreneurship development for the local industries. Parvin was chosen as the best-selected entrepreneur of the province of Chaharmahal Bakhtiari in Amir Kabir Festival and her company was selected as Top Private Company in the field of entrepreneurship education and development in 2007.

The environment has always been a concern to Parvin and in the process of holding workshops, she also found entrepreneurship very interesting. Hence, she decided to expand her knowledge in both areas. Parvin participated in the university entrance exam in 2007 and was admitted to the graduate program of the University of Tehran to study entrepreneurship (at the Faculty of Entrepreneurship), and environmental economics (at the Faculty of Economics) at the same time. From the beginning, Parvin had a plan to acquire the knowledge she needed for starting her own knowledge-based business and adjusted her thesis topics accordingly. For this purpose he established a new company, Kuhrang Knowledge-Based Instrumentation Technology (KKIT), in 2008.

In 2009, Parvin started networking with post-graduate relatives and friends who had a diverse range of skills, knowledge and experience to join her company. They formed a group of ten experts and Parvin served as the CEO of KKIT company. The team conducted a needs assessment survey for recognizing local, national and international needs and resources. They planned to identify new

[2]"HACCP is a management system in which food safety is addressed through the analysis and control of biological, chemical, and physical hazards from raw material production, procurement and handling, to manufacturing, distribution and consumption of the finished product" (U.S. Food and Drug Administration).

business opportunities in technology and then patent, produce and commercialize them. The group has considered the uniqueness of the ideas in the domestic/international markets as the criteria for their selection. The members of the team (including university faculties and Kharazmi Festival award-winners) attempt to explore and create new ideas. In 6 months, they could patent five inventions. The company's patents won the first prize of Innovation and Development of the Iran's National Elites Foundation (NEF) Festival among 3000 participants in 2009/2010. In the same year, the KKIT group was selected as the top winner of the Shahid Chamran's Prize, at the first Festival of Technological Innovation held by Faculty of Entrepreneurship at the University of Tehran. Furthermore, Parvin and her colleagues submitted papers to the conferences that were relevant to their patents and to their surprise, they received best paper awards. These accomplishments motivated the group to work harder.

The company undertakes different projects in the areas of environmental health and natural resources, nanotechnology, biotechnology, polymers, materials and metallurgy, and environmental chemistry. In their Project, KKIT group's concern has been to improve human health and quality of life by using advanced technology in the area of environmental chemistry. One of the eight patented projects of KKIT is the production of mulch ground-cover material by using Nano-polymer technology to prevent environmental hazards and desertification in particular. Traditionally, desertification was combated primarily at the local level and involved dune stabilization measures, especially the use of oil mulch, re-vegetation and windbreaks (Amiraslani and Dragovich 2011). The KKIT new product has been designed as an alternative material to replace oil mulch spraying.

Combating desertification in Iran has been a priority area of support by the government. One fourth of Iran (43 million ha) suffers from desertification. In recent years, the desertification hazards due to the influx of dust flows into the air of west and south of Iran have caused environmental hazards on public health. It has also disrupted people's lives, industrial activities and agricultural production. The problem is not limited to Iran, dust storms have had considerable impacts on various regions of the world and neighboring regions of Iran.

In recent years, this environmental health hazard has taken on new dimensions in Iran and has changed from a local problem to a national issue (Boloorani et al. 2014). In 2004, a national plan to combat desertification was approved with the emphasis on community participation (Amiraslani and Dragovich 2011). Furthermore, protection against desertification and dust storms is indispensable and needs broad and considerable cooperation between the government of Iran and the neighboring countries (Sharafi et al. 2015). In 2010, the KKIT proposal of using organic mulch ground-cover material to combat desertification was financially supported by the Vice Presidency for Science and Technology of Iran and Malek Ashtar University was chosen as project mentor. This helped the project progress, an anti-desertification project that was designed to control wind erosion, fix sand dune and suppress dust.

Parvin and her colleagues have not been interested in borrowing from banks or investors, they have been looking for quick funding sources and grants. One of their

policies is that they design a specific product for a company for free provided that the customer buys raw material from KKIT. This method has been very successful. Furthermore, they may win cash prizes and grants for their design creativity in different festivals. In 2012, KKIT was awarded as a specially selected knowledge-based company in the third National Festival of "Science to Practice" and received 1 billion Rials (~US$33,380). The "Science to Practice" Festival rewards the commercialization of inventions in knowledge-based companies.

Up to now, four patented inventions of KKIT have been financially supported by the NEF of Iran. The amount of the financial support depends on different factors (levels) defined by NEF. For instance, KKIT received 20 million Rials (~US$6676) financial support and in addition NEF covered 80 % of the cost of international patenting for one of their inventions.

The anti-desertification project of KKIT was successfully tested in Khoro-Biabanak, and Na'in deserts in Isfahan and was patented in 2009. Then the project was implemented in the following places: Dasht Sagzi in 2011, Abu Ziad, Kashan province, in 2013, Talab (pond) Mighan, Central Province, in 2014, and the salt marsh of Urmia Lake, Azerbaijan Province, in 2015. KKIT trains local people to be involved in seeding and mulching erodible soils. The KKIT group have identified the dust crisis centers in 18 provinces of Iran and estimate to have the potential of producing 120 tons of organic mulch per day to combat desertification and dust storms. Some Gulf region countries have also shown interest in the anti-desertification project of KKIT and the company has started negotiation with them.

KKIT has taken big steps in environmental protection. Among other KKIT patented inventions in the area of environmental protection is the design, synthesis and manufacture of a product called fire sulfide. This product, patented in 2009, is a water-based solution composition for extinguishing huge forest and grassland fires.

Parvin has received various awards and has been recognized as the best selected entrepreneur several times. Her most recent awards include: best selected growth-oriented tech entrepreneur in Shekh Bahaei Festival in 2015, top selected as Lady of Water in the National Conference of Environmental Protection in 2015, first placed patent-owner of Scientific Award of Alborz Institution in 2014, first placed National Skill-Elite of the Technical and Professional Organization in 2014.

Parvin shares the knowledge of KKIT group by presenting their ideas, experiences and research in seminars, conferences and media interviews. She also has a live radio program in the National Radio of Iran every other Tuesday night. In spite of her busy work schedule, Parvin has successfully balanced work and family life with the help of her husband (married in 1998) and her mother. Parvin says that without her mother's help, she could not have raised her son, Mehdi, while also studying and founding a company. Since his early childhood, Mahdi has been exposed to entrepreneurship. At the age of 11, he is now learning game design at Iran Computer Game Institute, and aims to follow in his mother's footsteps and become an entrepreneur.

5.3 Entrepreneurship and Philanthropy

Fatemeh Daneshvar, Sepehr Asia Mineral Import-Export, Ltd.



Fatemeh Daneshvar is a successful, distinguished entrepreneur and philanthropist, who is the first woman in Iran to serve as co-founder and CEO of a mineral company. Her perseverance with research to explore business opportunities, as a young educated woman, distinguishes her case from other cases. Moreover, she is credited for allocating 25 % of the income earned from her mineral company, Sepehr Asia, to fund Mehafarin, a charity organization that supports 4500 children with inadequate adult supervision.

Fatemeh was born in a traditional family in Tehran in 1974. Since childhood, she was exposed to a variety of rich Iranian literature introduced to her mainly by her grandmother. This motivated her to read different books and develop a broad perspective with big dreams. She was impressed by and learned from writers such as Joseph Murphy, Dale Carnegie, and Anthony Robbins to believe in the power of thoughts with which mankind has been blessed. Since her early teenage years, Fatemeh liked to be financially independent and she was paid for painting decorative ceramics for a relative who owned a ceramic making plant when she was in middle school. She was praised for her accuracy and doing a good job.

After receiving her master's degree in business management from the University of Tehran (had already earned a bachelor degree in healthcare management), Fatemeh was hired by a car-manufacturing company. Her supervisor was very strict and watched her while she worked about 8 hours a day. At that time, she did not like to be under such strict control; however, now she believes working hard in that company made her ready for the future. Two years later, she quit the job to pursue her major goals and established her own company. Fatemeh established a trading business and started by importing toys from China. Because of the large amount of Chinese toys in the Iranian market, the project failed and the result was

indebtedness after several attempts at importing toys. Fatemeh had borrowed from the informal financial market, charging interest much higher than formal financial markets. In the process of borrowing from another lender, Fatemeh was introduced to her future husband, Mehrdad Akbarian, who had experienced a similar unsuccessful trade experience. In their interaction with each other and sharing similar business failure experiences, they decided to get married.

Fatemeh and Mehrdad established Sepehr Asia Company (SAC), in collaboration with two Chinese businessmen. Fatemeh and her husband complemented each other and shared the same objective. She was very persuasive and her husband had a good talent in negotiation that made them a perfect business team. Fatemeh hired three of her college friends, who were hardworking and motivated, to continuously study the markets of Iran and China. They tried different unsuccessful trades with China in 8 months. Importing and exporting different commodities with China had no positive outcome. During the years 2004 and 2005, Fatemeh and her husband worked very hard from 8 am until 11 pm and had meetings with experts on importing different commodities such as mineral machinery, yarn and fabrics, cloths, tools, and stripes. In attending foreign fairs and different foreign production lines, they found good Chinese business connections.

Sepehr Asia decided to put more efforts into exporting. Studying different opportunities, they also found out that exporting minerals from Iran to China was profitable. Research on China's market showed that iron ore, manganese ore, and chromite ore had good markets there while iron ore price was more stable in China and iron ore was more accessible in Iran. Their study showed that exporting Iranian minerals was an untapped area for business. According to Fatemeh, Iranian investors were not interested in the mining business since it is a difficult job with high risk and high costs; moreover, there would be no guarantee of return on investment in mining. Fatemeh kept in touch with the Industries and Mines Organization and obtained different information and advice from mine experts. Meanwhile, she met someone in Iran who had plenty of unsold iron ore and was desperately looking for a buyer. After negotiating in several meetings, Fatemeh and her husband convinced the seller to accept a check due in 2 months. They only had 6 million Rials (~US $660 in 2005) cash and needed 60 billion Rials (~US$6.6 million) to export the merchandise to China. The owner of iron ore accepted the deal, and that was the beginning of a new era in Fatemeh's career. They could open a letter of credit and export iron ore safely. Fatemeh gives credit for their success to the talent of her husband in negotiation with the iron ore seller and convincing him to accept a check in the deal.

When they came back from China, Fatemeh and her husband paid all their liabilities and established good credit in the market, which enabled them to buy solely on their credit. They decided to allocate 25 % of their earnings to establishing Mehr Afarin Charity. They stayed in the mineral industry and negotiated with several private mine owners. At that time, mine owners had basic technical problems in their production schedule as well as the desired quality and quantity of the ordered products. Fatemeh could not count on the promise of the mine owners to receive the ordered product on time. Mine owners, who were mainly uneducated,

had no clue about international agreements that the tonnage of goods with contract specifications needs to be delivered according to the agreed schedule. So, they decided to make a contract with a mine owner in Kerman (located in central Iran), and started the production and transportation of minerals in 6 months. They were able to produce 40,000 tons per a customer's order which was transported to the harbor in Bandar Abbas and shipped to the customer. This was another huge step toward future success.

According to Fatemeh, Sepehr Asia has had a significant positive impact on the areas in which the mines are located. Sepehr Asia has created both direct and indirect jobs in its mining areas. The project has generated new demand mainly for land and sea transportation companies. Sepehr Asia has a permanent contract with the Islamic Republic of Iran Shipping Lines to carry freight containers and charters vessels on a regular basis. They have provided 100 direct jobs and 500 indirect jobs in mining, manufacturing, and mineral processing, in addition to hiring 16 experts to work in their commerce and exporting department. Moreover, they contribute to the national economic development plan for export promotion, and earn foreign exchange in the non-oil sector of the economy.

The next step after mineral exploration, extraction, and processing was marketing in the consumers' foreign markets. However, they soon understood that despite having rich mineral resources in Iran, no systematic and coherent management system was available for marketing minerals. Fatemeh put great effort into marketing, and sales promotions were put in place to attract the customer's attention. The company became an active member of related commercial websites. After 4 years, Sepehr Asia became one of the most specialized mineral exporter companies in Iran. 300 trucks for transportation of minerals carry ores to Bandar Abbas harbor, where the company has built a warehouse, provided equipment for loading and unloading, and also has trained specialists. All procedures including extraction, exploitation, processing, and ground transportation to Bandar Abbas, warehouse, deposit, ship rentals and CFR (CFR stands for cost and freight, which requires the seller to clear the goods for export) sales operations to the final customers are now done by Sepehr Asia Co. They are one of the few companies that control all phases of their business to the end user. As of now, the company owns seven mineral companies; for three of them (Kerman, Zanjan, and Semnan Provinces), they have mining exploitation licenses, and Sepehr Asia is in the process of obtaining mining exploitation licenses for the rest.

Family is very important to Fatemeh. She has a special respect for her husband who is also her business partner. She has a 5-year-old daughter and wishes she had more time to spend time with her. Fatemeh tries to compromise when her husband disagrees with her opinions, or otherwise, she makes sure that her husband has been convinced to approve of what she wants. Fatemeh believes that God has empowered mankind with valuable gifts that need to be explored, recognized, and unleashed. This is how one can have high spirits, great objectives, and strong self-confidence. As a result, Fatemeh says, "you take big steps in your life and even a failure can make you stronger to recover and start again." This mindset helped Fatemeh to overcome several business failures in her 20s. She thinks a lifelong goal can seem

far away and too big, but it can be split into small pieces to be achievable. According to her, many people are satisfied with having a secure job with a fixed salary, but those who like to be entrepreneurs and hire many employees need to be dynamic and move forward. It is essential to work hard and spend time to find the secrets of success in any areas; nobody can teach you the whole thing. Moreover, you need to understand crisis management. Fatemeh disagrees with those claiming that her success was coincidental.

MehrAfarin established by Fatemeh, is a charity organization that provides living services such as food and clothes to street children and children with poor family support. Although it offers a variety of services such as treatment, psychological consulting, and legal, its main mission is to promote education. The document the households must provide to receive these services is the official report of the progress in the education of the children who use these services. When the transcripts of the children are not satisfactory, MehrAfarin offers additional remedial classes (private lessons).

As of now, MehrAfarin has offices in Tehran and the following provinces: Sistan and Balouchestan, Kerman, Hormozgan, Central and South Khorasan, and East Azerbaijan. These are the places where Sepehr Asia has mine plants and operations. MehrAfarin Group videotape the training courses they hold in educational workshops in Tehran and send their CDs to their charity centers in deprived villages to deliver to needy families. They also provide a collection of free CDs teaching family planning and healthcare as well as life and work skills. Watching these films are mandatory for those receiving help and families will be asked some questions regarding the related subjects. A magazine named Hamraz is also published by MehrAfarin to offer more information in health and education. The charity's motto is "reaching a day that no one feels helpless and alone". MehrAfarin attempts to find jobs for single mothers and make them financially independent after 2 years of using the charity's services. Fatemeh cleverly uses her social capital to offer a better life for the charity customers; there are 200 physicians, experts, and lawyers who voluntarily provide free service to MehrAfarin. There are also 52 paid employees working for MehrAfarin whose salaries are funded by Sepehr Asia Company. From the beginning, MehrAfarin was a charity organization funded completely by Sepehr Asia; however, in 2009, due to the economic crisis, Fatemeh decided to change it to an NGO to get more support. She believes social problems need social solutions. She asked Fatemeh Motamed Ariya, a well-known Iranian actress who has a friendly face and motherly figure to become their ambassador to get more support for their charity organization.

Fatemeh is involved in Iran Chamber of Commerce. The Chamber of Commerce consists of 18 committees. Fatemeh is holding the chair of one of these committees as the only woman there. She says that the chairmen of the other committees do not confront the obstacles that confront Fatemeh. Therefore, to achieve the same goal, Fatemeh needs to put much more effort than the other 17 chairpersons. Fatemeh believes that women reach their goals later than men do, because of gender discrimination. Men are welcomed for various opportunities in society, while women have to confront and overcome different obstacles. Women get to the finish

line later than men because of the unequal starting point. Thus, for the same level of success and social status, women need to spend more effort than men.

For the consecutive years of 2010 and 2011, Fatemeh was recognized as the best selected entrepreneur of the Industry Hero Festival. In addition, she is a member of the following organizations: Women Entrepreneurs Association, National Association of Women Entrepreneurs, Executive Council of Tehran Chamber of Businesswomen, Representatives of the Iran Chamber of Commerce, Industries and Mines, as well as Tehran Chamber of Commerce, Industries and Mines. She is also a board member of Iranian Women's Business Council. Recently, Fatemeh cofounded with her husband, Madan Kav, a mineral consulting company.

5.4 Entrepreneurship and Retirement

Mahrokh Fallahi, Fallahi Livestock, Ltd.



Mahrokh Fallahi is a true example of lifelong learning. Mahrokh was born into an educated family in Tehran in 1943. Her father, a senior judge who was very interested in her daughter's education, sent Mahrokh to elementary school 1 year earlier than usual for her age. Upon graduating from high school, she successfully passed the university entrance exam and started studying English Language at the University of Tehran in 1960. Early in her freshman year of college, Mahrokh married her husband, who was also studying at the University of Tehran and gave birth to her first son at the end of that academic year.

Upon completing her B.A. with honors at the age of 20, Mahrokh started teaching at Tehran High Schools. After 4 years of teaching, she was recognized as a national distinguished teacher. Subsequently, after passing an English

proficiency test, together with two of her colleagues, Mahrokh was sent to the USA and Spain to learn "special education for blinds". After completing the training courses in English, she came back to Tehran and was later appointed by the Ministry of Education as the president of the Institute for Blind Children's Education and held this position for 16 years. This appointment coincided with the birth of her second son.

In 1968, Mahrokh, together with her husband, also a nationally best-selected teacher, and her father who was retired by that time, obtained the license to establish the Avicenna Educational Institute. The Institute was a co-educational complex that provided academic services to students from kindergarten *to* grade 12 (it started with 250 students in the first year and its enrollment had grown to 800 students at close-down). However, after 12 years of hard work, due to the Ministry of Education's decision to dissolve private educational institutions, the Avicenna Educational Institute was closed down in 1980. That incidence led Mahrokh to think of establishing her own business in retirement. About 8 years prior to her retirement, Mahrokh, together with her husband, decided to sell their big family house in Tehran, and allocate the money from its sale along with their lifetime savings to an apartment and a 1-ha ranch in the suburbs of Karaj. Now, after 34 years, the ranch has been changed to the Fallahi Institute of Agriculture and Animal Husbandry (FIAAH) that includes Fallahi Livestock Company and Fallahi agriculture Company.

Mahroakh's interest in livestock comes from her childhood memories of her grandfather's ranch in the northern part of Iran. Now, the Fallahi Institute has 250 pure Holstein cows, a 150-ha farm producing wheat, maize, fodder, and livestock, and also several hectares of apple and pear orchards. Twelve families live in the ranch and 3–4 members of each family work permanently in the ranch and live there for free. Furthermore, between 35 and 40 daily workers including experts and unskilled labors work in the husbandry. In general, Fallahi Institute has created 450 direct and indirect jobs.

The success of Fallahi Institute, to a large extent, is owed to its board of directors. The board includes Mahrokh, her husband, and their two sons, Farid (who followed in her mother's footsteps and became an English major) and Omid (who is a medical doctor). Since all four members of the management board are educated, they participate in researching the principles and methods of agriculture and livestock production, both in the farm and garden sector, and they are directly involved in the operation of their business. The garden produces fruits and is managed and run by Esfandiyar Fallahi, Mahroakh's husband.

The purebred cattle breeders of Fallahi Institute was recognized as the Best Cattle in the Iranian Holstein Dairy Cattle Show organized by the Iran's Livestock Breeding Center for three successive festivals of 2000, 2003 and 2006. The festival is held every 3 years and is judged by professionals and experts from America, Canada and Germany. Mahroakh's prize was a trip to Canada and Germany. Mahrokh was also the winner of the Best Entrepreneur of the year 2012 Iran National Awards. She also received the Distinguished Entrepreneur of the year award 2005 at Sheikh Baha'i Festival, Isfahan University of Technology.

Mahroakh was interested to know more about industrial husbandry and decided to educate herself in that area. Thus, she participated in university training workshops about industrial husbandry and attended the related national and international seminars. At the *start* of her animal husbandry business, she bought seven calves for which she borrowed money from Agricultural Bank (Bank Keshavarzi). The number of calves increased from 7 in the first year to 14 in the second year and reached 28 in the third year. After 4 years, the revenue obtained from selling milk was enough to pay all debt from Agricultural Bank. As the calves increased exponentially, Mahroakh decided to breed the cows and bought imported frozen semen and inseminated cows. As a result of breeding, the daily milk production of a cow increased from 18–20 L (39.7–44 lbs.) to 65 L (143 lbs.). Mahroakh did not stop at this point and started embryo transfer project. Mahroakh does all the procedures herself, including cattle hormone therapy during the estrous cycle in cattle. As a result, a cow releases 30 eggs instead of an egg and gives birth to 30 instead of 1. She used vet for the delivery of the calves in the first year but soon she learned how to deliver a calf. Now she only uses vet for cesarean section of the calves. These innovations are key reasons for her success. Male calves are sold to the Breeding Center at reasonable prices and according to Mahroakh, this to some extent compensates the economic damage resulting from selling cheap milk.

Mahroakh believes that entrepreneurship is like a ball when it falls down it moves higher. She has faced many challenges in her entrepreneurial venture. She considers price fluctuations of livestock products (meats, milk) and agricultural products in the internal market, uncontrolled import of agricultural products and livestock, and more than these epidemics livestock disease, and the financial worries such as on time payments on loans as her main challenges in her adventure. She believes that leaving the luxurious life of Tehran, time management and living on her family's dairy farm with her husband and balancing family and work life are the secrets to her success.

Due to her good reputation, Mahroakh was elected as the president of the Islamic council of Mohammad Shahr (Karaj) for the period 1999–2003. She was also elected for the consecutive term of 2003–2007 as the city council president, but due to her busy schedule, she served as the vice president of the council. She thinks that if women show their competencies, the gender discrimination disappears. Gender discrimination will continue as long as women underestimate themselves.

Mahroakh is satisfied with her life; she says that "compared with many of my peers I feel stronger, healthier, more dynamic and enthusiastic". She has two sons and four grandsons and more than that she enjoys her work. Her house is located in a tower at the middle of her livestock where she can keep an eye on and control her livestock by the hunting camera. Mahroakh starts her day with milking cows at five in the morning and ends her day with milking cows at 12 o'clock at night. When a cow is sick or is about to give birth, Mahroakh carefully watches and takes care of her.

FIAAH is an appropriate and conducive place for academic and scientific research. With a mutual agreement reached between FIAAH and some universities, FIAAH holds internship program for college students who are studying animal

husbandry and veterinary. This provides the students, especially female students, the opportunity to integrate their theoretical knowledge with the practical application and experience under the supervision of the FIAAH Board of Directors. Certificates of the completion of practical training will be issued by the *FIAAH* Board and sent to the participants through the university of origin.

Mahroakh participates in seminars, translates articles (from English to Farsi) in animal husbandry journals, and is an active member of National Association of Women Entrepreneurs, Association of Women Entrepreneurs, and Association of Women and Youth Entrepreneurship Development Foundation.

5.5 Turning Unemployment into Growth-Oriented Entrepreneurship

Seyedeh Fatemeh Moghimi, Sadid Bar International Shipping and Transportation, Ltd.



Seyedeh Fatemeh Moghimi is a civil engineer with multidimensional talents and high leadership quality. Her case demonstrates how an accident can successfully change the life path of a person, who has managerial talent with entrepreneurial thought and action.

Fatemeh was born in 1958 in Tehran. After earning her high school diploma in mathematics, she went to England to pursue her studies in Civil Engineering. Returning back to Iran in 1979, Fatemeh as a young female engineer, was full of energy and excitement to participate in the socio-economic development of her country. At that time, the atmosphere did not provide appropriate environment for a female civil engineer to work in the field of construction. Fatemeh was desperately looking for a job in Tehran. One day, as she usually did, she bought a newspaper looking for job opportunity advertisements. Since Fatemeh couldn't wait to go home and respond to newspaper employment advertisements, she decided to use public phone. Hence, she joined a public phone queue, and started talking to a

young woman standing next to her. When Fatemeh told her that she is fluent in English and had studied abroad, the woman told Fatemeh that her brother in law was a stockholder of a company that needed an interpreter to work with their international affairs. This conversation was the beginning of a new chapter in Fatemeh's life.

Having no other option, Fatemeh contacted the company the next day. She learned that the company was an international transportation company with a crowded and noisy environment, a male-dominated workplace and very different from what Fatemeh had expected. Fatemeh had no experience in the field; moreover, in her family's culture, nobody had any experience in the area. However, in the absence of a better alternative, she started working. She began by translating and understanding the international transport documentations and soon she was in charge of English translation, interpretation, and conversation, and also, taking care of telexes. After 5 months of working hard in the company, she became responsible for the truck sector and became the traffic manager. Many of the truck drivers employed by the company were from Greece and Turkey. After working for a month in her new position, when the board of directors was abroad, the company attorney came to the office and announced that the company was closed.

Becoming very interested in forwarding services, Fatemeh decided to work in another similar company. At that time, almost 35 years ago, there were only 40–50 international transportation companies in Iran. Through the Transportation Ministry, she found the Transportation syndicate address, applied for a job, and was hired by a similar company in 1980. In a short period of time, she became the internal manager of the company. By 1983, Fatemeh managed 150 trucks, which were transferring commodities from Iran to abroad and vice versa. During this period, she was married and her first son was born. She worked conscientiously during her pregnancy and when she wanted to deliver her baby, she went directly from her work to the hospital. She came back to work after 2 weeks. Her extreme interest and persistence, made her husband interested in transportation as well, and he encouraged her to establish her own business.

In 1984, Fatemeh applied to the Transportation Ministry for permission to establish an International Forwarding Company. As she expected, her application was rejected. Her condition, being a woman, educated and young was not quite compatible with the traditional characteristics of the male-dominated job of international transportation. However, these constraints never stopped her from pursuing her goals. Once she was referred to the Department of Commerce to get commercial license, her request was rejected, but one of the managers proposed her to work with his wife who had an education center and needed someone to teach professional techniques. She believes that these apparently unsuccessful experiences have encouraged her to keep moving forward. Fatemeh says that each negative response made her stronger and more persistent in attaining her objectives. With strong motivation and self-confidence, Fatemeh continued on her way

patiently, and believed in what she was doing. She believes that God likes and helps patience and rewards the activities and efforts of human beings. She did her best to achieve her rights. Fatemeh tried to use and rely on all the positive cultural understandings that she had learned from her religion and from Iranian history. She remembered some parts of the will of king Darius the Great to his son, Khashayar Shah, over 2500 years ago; "Your mother Atosa has a right on me and people, and you are responsible to satisfy not only her wishes, but also the rights of all women in the country". She says how wonderful it is demanding one's rights and trying to achieve them.

In her continuing attempts, Fatemeh could finally obtain the license to establish her own company, Sadid Bar International Transport, from the Transportation Ministry in 1984. Hence, Fatemeh's new life started as the first female CEO of an International Transportation Company in Iran. This was not the end; rather, it was a beautiful fresh start that was achieved through perseverance. Through all these processes, Fatimah's husband was protecting her and had a great role in her success. She started her company with a secretary in a small room (12 m^2) in her husband's newly established engineering company, where she could use his office facilities. However, her main asset was her experience, knowledge of the job and good reputation. The experienced drivers who already knew Fatemeh joined her; she increased the number of the employees very soon and moved to a bigger office (120 ft^2). At that time, Fatemeh started traveling within the country for internal forwarding activities and established representative agents in the main arrival and departure borders and large cities that had customs. She felt responsible to promote the capabilities of her personnel by providing them more education and training to the extent that some of her former staffs have their own International Transportation Companies now.

Accompanied by her husband, Fatemeh started traveling to some of the European countries to obtain representation. In order to develop the leadership skills demanded by her position, she studied at Azad University, Iran, and received a degree in Transportation Management. In 1990, she, as the first Iranian woman, received the truck and bus driver's license, which is quite unusual for a woman in the Iranian culture. This could help her to become more familiar with the technical parts of her job and also to create a tight link between the management and the industrious workers who played an important role in the success of the company.

Fatemeh's second son was born in 1991 and kept her away from her work for 2 weeks. She kept her baby next to her in the office so that she could give him enough love and affections. To start and run her business, Fatemeh had to face many challenges. She was not taken seriously when she applied for establishing an international transportation company. Her application was rejected because transportation company was considered mainly as a male oriented business. Moreover, when Fatemeh, as the only woman among the applicants, wanted to take Commercial Driver License (CDL) road tests, she was faced with a stereotype reaction of the

crowd. However, as a young educated woman with high self-esteem she could overcome serious barriers. Fatemeh says every new positive movement brought her more enthusiasm and power to move forward faster with more energy.

Besides the opposition of traditional government management to women engaging in the so called "men's jobs" and the anti-women attitude of companies existing in the area of transportation, Fatemeh had to face external problems as well. Facing the sanctions by U.S. and allies during the war with Iraq and the existing sanctions against Iran, facing the danger of theft of the shipments by smugglers especially at Iraqi and Afghani borders have been among the main external constraints for Fatemeh's business. Furthermore, Fatemeh had to handle some issues related to the nature of the transportation business. For instance, she had a hard time when her company's trucks were overturned, were stolen, or caught fire, and the cargo was lost. Sometimes their insurance did not cover all the damages. Worse than that was the time when the accidents led to the death of the driver. Fatemeh says that she was suffering even more when the driver behind the wheel had a stroke, died or fell into the valley. In some occasions, she had to transfer the body from the scene to the driver's family and manage to bury it. Besides the emotional involvement, she was frequently being summoned to court for various lawsuits related to these accidents; however, she was vindicated every time.

What makes Fatemeh unique is the combination of innate leadership skills, practical experience, management skills, and high achievement motivation. With these competencies, she could expand her accomplishments, share her knowledge and explore new ideas. After experiencing success in her business, Fatemeh's next challenge was serving the transportation industry and the women community. She participated in social syndicates and became active in related seminars, conferences and TV interviews. In 1998, Fatemeh nominated herself in the International Transportation Syndicate election, now called International Transportation Companies Association (ITCA), and was elected as one of the nine members of the Board of Directors. During the period of 1993–2006, she became the head of ITCA's Arbitration Committee which convenes weekly to resolve various problems and claims in the syndicate. In her new responsibility she expended great effort to promote the social and economic rights of the transportation industry both domestically and internationally.

Since 1993, Fatemeh, as the representative of Iran's ITCA has participated in many dual and multiple joint committees between the Islamic Republic of Iran and other countries to formulate and draft transportation protocols and agreements. In 1994, she became a member of Iran-Russia and Iran-Italy International Transport Advisory Board. At that time, Iranian truck drivers were faced with many problems obtaining transit permits in Italy, and also the ambiguity of regulations in the newly independent Soviet Union countries which resulted in extensive delays and eventually transportation stoppages. She accompanied a government board, to provide expertise, on a trip to Russia to negotiate and

solve the problems. Finally a successful protocol was signed in 1998 between the two countries. Fatemeh was also appointed as a member of Functional Group of North–South Corridor (1999–2008) and as a member of Establishing Board of Iran & Canada Friendship Association (2001–2003).

Fatemeh made use of both internal and external media to present and introduce the real face of an educated, successful Iranian woman entrepreneur. Since 1998 Fatemeh has given talks at numerous national and international conferences such as the 3rd Businesswomen Forum in Islamic Countries held in Doha, Qatar (Jan. 2008), and Woman Global Economic Forum, Abu Dhabi, UAE (Nov. 2007). Making women's voice heard, Fatemeh nominated herself in the board of representative election at Tehran Department of Commerce (TDC) and was elected as the only female of that board (2006), a membership that was always Fatemeh's dream. She says that this board plays a key role in industry, mining, agriculture and commerce in the country. Fatemeh is also chair of the Women Trader's Council of Tehran Department of Commerce. According to her, 2500 trade business cards had been issued for women in Tehran by 2009. Fatemeh has also been Executive Manager of National Association of Women Entrepreneurs (2006–2008). Since 2008, she has been appointed as a consultant to the President of Iran Chamber of Commerce, Industries and Mines (ICCIM) in Women Entrepreneurship, and she also serves as a member of the Tehran Women's Council at Chamber of Commerce, Industries and Mines.

During 1998–2008, Fatemeh taught in the fields of International Commerce, Technical Correspondence, Shipping, Transit & Road Transportation, Packing, Commercial & Transport Conventions, Customs Regulations, Incoterms, and Entrepreneurship. Fatemeh became the director of the Office of Entrepreneurial Development at Sharif University of Technology from 2012 to 2013. She is also the author of Economic Strength.

Fatemeh is successful not only in business but also in holding numerous management/professional positions and responsibilities at the country level. Some of her more achievements are as follows: Vice President of Board of Directors in ITCA Terminal Investment Co. (2000–2008), Vice Chairman of Customs & Transportation Commissions of ICCIM & TCCIM (2008–2009), participation and cooperation in the specialized commissions of the Iranian Parliament for the discussion and approval of the National Fifth Economic, Social, and Development Plan (2010–2014), adviser to the National Iranian Tanker Company since 2013.

As an entrepreneur, Fatemeh has received different awards, among which she was selected as the Top Entrepreneur of Muslim countries in 2012 for creating jobs and empowering women in the transport sector by the Islamic Development Bank.

5.6 Entrepreneurship and Technology Transfer

Banafsheh Sedigh, Auto Parts and Gas Valves Shigal MFG, Ltd.



Banafsheh Sedigh is a young, successful and dynamic entrepreneur who is involved in other community activities such as women entrepreneurs' networks, serving as a board of trustees, and co-founding a charity organization that supports orphans and vulnerable children in rural areas. Travelling internationally and attending workshops and seminars, Banafsheh attempts to introduce Iranian women entrepreneurs outside Iran.

Banafsheh was born in 1977 in Tehran. Her father was the founder and manager of an industrial business. Since childhood, Banafsheh was interested in her father's business activity and listened to his conversations with other business owners and customers. Moreover, his father had a workshop in the basement of their house where he used to work at night. Hence, Banafsheh was exposed to business environment early in her life and wanted to follow in her father's footsteps. Her mother who worked at Iran Shipping Lines before marriage, decided to become a housewife and take care of her children and family after she got married. She encouraged Banafsheh and her older sister to become educated and work. Banafsheh earned a bachelor's degree in electrical engineering in 2001.

During her studies, Banafsheh worked part-time at her father's company. She got married in 2000 and had her daughter Golnar in 2002. With the help of Banafsheh's mother in taking care of her 9-month-old daughter, Banafsheh was able to obtain a full-time job at Pars Shir Industrial Company (PSIC). Her 2 years of experience at PSIC, particularly her managerial roles in information technology, production, and quality control, prepared her for her future venture.

Banafsheh was interested in challenging herself, becoming independent, and proving her competence. Hence, she started thinking about founding her own company and asked her father for advice. Although her father told her that she was choosing a rough and difficult path, Banafsheh was determined to have her startup. She established Shigal Company (SC) that produces auto parts and gas valves in Shahrake Sanati Kaveh (Kaveh Industry Town), Saveh, in 2001. At the beginning, Banafsheh raised funding for SC by borrowing from banks and the support of her family. As the company grew and Banafsheh received down payments from customers, there was no need to take out a loan.

The main automobile industries in Iran including, Iran Khodro Company, Saipa Company, Mazda, and Iran Pars Khodro Company are the customers of SC. Since its establishment, SC has designed and made different products. Banafsheh has developed the line of production from synchronizer rings in 2001 to bus heater valves in 2007. The Company has also designed and developed break connection, with the aim of problem solving, which was patented in 2008 in Iran. Participating in Green environment National Project, SC made a joint-venture contract with VTI, a well-known company in valve technology in Germany, for transferring the technology of producing Compressed Natural Gas (CNG) valves to Iran in 2010. By localizing VOSS connection in 2013 and making the product in the country, Banafsheh could transform the threat of sanction against Iran to an opportunity. As part of the development and innovation policy of SC, in 2014, the Company improved the production process of piston bar of shock absorber. SC produces and sells one million auto parts and gas valves per year.

As a young entrepreneur, Banafsheh's activity has not been limited to founding and managing her company. Banafsheh has joined the board of directors of PSIC since 2010, the company where Banafsheh started her first full time job. She has recently served as chair of the board of directors of that company, a company with 400 employees. Furthermore, Banafsheh has been passionate about encouraging women to be involved in entrepreneurship. She has been a member of the board of directors and the board of trustees of the Entrepreneurship Development Foundation for Women & Youth since 2009. She was also the managing director and a member of the board of directors of the National Association of Women Entrepreneur (NAWE), the first and the most important women's entrepreneurship network in Iran, from 2007 to 2014. Banafsheh serves as a steering committee member of The Women's Alliance for Virtual Exchange (WAVE), an alliance of women civil society organizations from Middle East, North Africa and Central Asia with the aim of improving women's well-being through the use of Information and

Communication Technology. She is also a member of the committee of Young Entrepreneurs and Business-owners in Iran's Chamber of Commerce.

As a high-achieving woman, Banafsheh has a diverse set of personal characteristics and attitudes including self-confidence, leadership and teamwork skills, accountability and objectivity, that allow her to be successful in any endeavors she undertakes. She also has a strong desire to help others and to make a difference in families that have been affected by loss of a breadwinner in rural areas of Iran. Hence, in order to support orphans and vulnerable children in rural areas, Banafsheh co-founded Morsal Charity Foundation in 2009, and since then she has served as a member of board of trustees and chair of the board of directors of Morsal organization. Qualified children in rural areas who can use the facilities of Morsal Charity will financially be supported by the organization from primary school to college. Morsal charity also holds seminar for parents and students to increase the awareness and knowledge of orphan families in rural areas.

Believing in their ability and in the proverbs that "Where there's a will, there's a way" and "will is power" are among the motivations that Banafsheh and most of the women entrepreneurs featured in this book mention for their success. Furthermore, Banafsheh credits her family business background and support, good networking, and learning about role models as the reasons for her success. In her path to success, Banafsheh faced different challenges including balancing work and family and the difficulty of working in a male dominated work environment. Her passion for challenge and belief in lifelong learning, enriches Banafsheh's life and gives her life purpose. In order to expand her knowledge and skills in business, Banafsheh received an MBA from the Julius Maximilian University of Würzburg, Germany, in 2015.

At the national level, Banafsheh received Certificate of Appreciation for "Young Industry Women" in 2009. She was also the "Top National Producer of Iran" in 2011 and was selected as one of the "Leaders of Entrepreneurs" in the second national festival of entrepreneurial leaders that was held by the Faculty of entrepreneurship at the University of Tehran in 2014. At the regional level, she was one of the five finalist for Young Entrepreneur of the 5th Asia Pacific Young Entrepreneur Award, presented by Confederation of Asia-Pacific Chambers of Commerce and Industry (CACCI), 2014 in Kuala Lumpur, Malaysia.

Banafsheh considers exhibitions, fairs, and workshops held by the government very helpful for fostering women entrepreneurship in Iran. She has expanded her skills and knowledge in entrepreneurship through workshops/courses offered by universities/organizations in and outside the country. She also writes articles in journals and interviews with media about women entrepreneurship ecosystem, challenges and opportunities faced by women in Iran. Furthermore, she speaks at conferences and events on women entrepreneurship, e.g. she had a workshop on Iranian Women's Leadership at the University of Massachusetts, USA in 2010.

5.7 Entrepreneurship and Environmental Protection

Haydeh Shirzadi, Recycling and Composting Kermanshah, Ltd.



Individuals have different dreams. Many people have their personal dreams, but some people attempt to fulfill society's dreams. Haydeh is one of those rare people who has persistently pursued her dream of an environmental cleanup project. Her life story tells the tireless efforts of a woman who has faced different challenges and overcome many barriers since childhood. She is an extraordinary social entrepreneur who has devoted her time and energy to accomplishing her big dream of making a healthier environment for everyone.

Hayedeh Shirzadi is an environmental ecologist. She was born in 1960 in Gilangharb, a deprived small town in Kermanshah Province, northeastern Iran. The residents of the town were mostly nomads and farmers. Since her childhood, she dreamed of being educated; however, this was not feasible because there were no high schools for girls in her hometown. Girls used to work at home after finishing their primary school. She decided to study high school courses at home and successfully graduated with a high school diploma. She then went on to earn an associate's degree in Arts and Letters, and became a middle school teacher. After 2 years of teaching, her position was not renewed. Hence, she decided to go to Germany to continue her education in 1986. Since her Persian high school Diploma was not accepted in Germany, Haydeh had to learn German and received the needed diploma in Germany.

In 1992, Haydeh earned a bachelor's degrees in International Agriculture at Kassel University, Hessen, one of the pioneers in organic agriculture and new energy sources in Germany. Being aware of the projects undertaken by the researchers at Kassel University in organic residual separation, she decided to work at applying her knowledge in her homeland. She returned to Iran In 1994. Upon her return to Iran, she proposed a composite recycling project for the city of Kermanshah. However, her persistent efforts to convince Kermanshah Provincial Authorities of the vitality of her proposed project for the city were unsuccessful. When Haydeh received no positive response despite her frequent follow-ups, she became curious about the possible reasons behind the negative reaction, and thought the problem could be either because the administrators were not sufficiently knowledgeable to help her or she had not gathered enough logical evidence to convince them to approve the project. Her deep concern for protecting the environment motivated her to return to Germany to pursue a doctoral degree in the field of ecology of environmental protectionism at Kassel University. She completed her doctoral thesis on "Environmental Protectionism and Composite Recycling in Iran" in 1997 and obtained exclusive knowledge in this field. Haydeh met her future husband in Germany and accepted his marriage proposal on the condition that they move back and live in Iran. Haydeh then came back to Iran, now equipped with more knowledge and understanding of how she could achieve her dream to protect the environment and provide a better life for everybody. She says, "I challenged and succeeded. We should think globally and perform locally."

Haydeh addressed her proposal to the Kermanshah administrators again, but did not get any positive response. So she presented her proposal to administrators in bigger cities like Tehran and northern cities of Iran which had more waste disposal problems. However, again explaining the vital importance of environmental ecology protection and its urgent necessity application in Iran did not work and she could not convince the administrators to implement her project.

Following her unsuccessful experiences with local and central official decision makers, in 1998, Haydeh finally decided to start her own business and applied for a Bio-Composite License from the Kermanshah Provincial Industries Center. She says "my asset was my knowledge and I could not change my human capital into money capital in a short period of time". Thus, she applied for a loan to establish her company. After following up many times, she received the loan by convincing the bank officers that her collateral was her knowledge. In 1998, the Kermanshah Municipality decided to participate in implementing the project, and the Recycling and Composting Kermanshah (RCK) Company was established. The company began operating in 2001. This was a new era in the life of Hayedeh and also an opportunity for people in different parts of Iran to gradually dispose of waste in a hygienic way, protect the environment, and also change the waste into useful composite fertilizers.

In her broad perspective on the environmental issues, Haydeh's consideration on a healthy environment is not limited to establishing a waste recycling company. In her research and interviews, Haydeh attempts to increase awareness about environmental issues in her city and the country. She says, "We can import numerous

products from outside, but it is not possible to import environment, soil, water, sanitation and health to our country". Haydeh draws the public's attention to the high rank of Iran (10th) in terms of the world's environmental degradation indexes. She also adds that "Iran is the 14th producer of greenhouse gases in the industrial sector, 13th producer of greenhouse gas emissions in the transportation sector and the 6th producer of greenhouse gases in the agricultural sector". According to her, "the culture of protecting the environment has originated from Iran. We have had a special day, April 22, for preserving the four elements of water, wind, earth, and fire in our history. The same day, now universally called "Earth Day", is celebrated globally to demonstrate support for environmental protection."

Haydeh thinks managing waste while protecting the environment is a vital issue. She turned an environmental disaster into an opportunity. She says every human being is born with waste and is buried as waste as well. For daily activities, one uses different products and food which need to be disposed, so it is imperative to have special plans and use the appropriate technology to manage waste and protect the environment. Haydeh refers to the closeness of women and nature, and their role in environmental sanitation and waste management education. The consumption pattern of women as housewives and their use of chemicals, sewage and detergents play important roles in environmental protection. This is the concern of social ecology as well. If women are educated properly, they can protect their economy and society more effectively. This is especially true in the less developed areas where majority of women are housewives. She also endeavored to alert the authorities and make them understand the fundamental importance of environmental problems.

Over many years, the most common way to dispose of waste in Iran has been landfill; a method causing environmental destruction. According to Haydeh, food wastes are sources of pollution and cause greenhouse gas, while their conversion into fertilizers can prevent environmental destruction. She adds, "the daily waste in Iran is 50 thousand metric tons, of which only 8 percent is recycled". At the moment 100 % of Kermanshah's waste is recycled and converted to useful products by RCK.

The project started with the participation of Kermanshah City Municipality to turn food waste into quality compost fertilizer in a 3-month process of aerobic activity useful for the soil recovery. Upon the completion of the project at two phases in 2004 and 2007, all the city's residuals (500 metric tons) were transferred to the project place and 80 % of them were completely recycled. Moreover, a machine was purchased to convert the remaining 20 % of waste materials to alternative energy. The company converts waste to energy and useful products. The recycled wastes are used as agricultural fertilizer and also as the fuel for Kermanshah cement factory. There are 70 people working directly in RCK, and an additional 700 related jobs resulted in the city including education and training in the residual management. After 10 years of RCK activities, it is a significant model for other cities of Iran. Haydeh has trained an engineering team of 35 Kermanshahi youth to study and monitor the recycling composite phase of other cities in Iran.

Since its establishment, RCK has conducted many positive projects largely in Kermanshah and some other parts of Iran as well. The company is the exclusive representative of Komptech (waste recycling machinery manufacturer from Germany and Austria), and Austropersen (the Manufacturer of press machines). RCK[3] has been responsible for consulting, and surveying projects for approximately 80 cities and 3000 villages in Iran since 2001, and has completed the following plans with the Zana Engineering Consultation Office:

1. Waste management projects in the city of Kermanshah including running the project of bio-composite in the city of Kermanshah for the first time in Iran, implementation of training in organic waste source separation for the first time in Iran, Implementation of waste processing methods to reduce their physical-biology size, monitoring the first project of mechanized waste collection in Iran, monitoring the implementation of dry waste recycling collection (paper, glass, metal, and plastic), and developing "waste-to-energy" plants and "refuse-derived fuel (RDF)" systems.
2. Consultation phase; study and administrative waste management projects in other regions of Iran such as Zahedan, Lanjan (eight towns), Sanandaj, Khomeyn, Boroujerd, Marand, Anzali Port, Saveh, Najaf Abad (six towns), Barkhordar and Meymeh (nine towns), Bam, Ilam, Sistan, Chahbahar, Iranshahr, and Bojnourd.
3. Monitoring the Waste Management Projects in the cities of Zahedan, Lenjan, Najaf Abad and Khomeyn.

Since 1996, Haydeh has been actively involved in research and development, publishing articles and presenting papers related to environmental protection including composite technology, waste management, and compost versus chemical fertilizers. She has presented numerous research plans and projects. Her more recent research is plastic pelletizing production. She has received multiple awards at the national as well as provincial level. Among numerous awards she has received, some of her national awards are as follows: National Environmental Award in 2003, 3 years (2003–2005) of Waste Management Award-winning in Iran, the Nation's Distinguished Entrepreneur in 2006 for creating 700 jobs, Iran's Elite Lady elected in 2007, winner of the first rank Industrial Design Excellence in the Second National Conference on Fuel, Energy and Environment, 2010, best selected Regional Technology Entrepreneur in Technology Entrepreneurship Zagros Festival, May 2010.

Haydeh has received multiple honors at the Kermanshah province including: Distinguished Employer of Kermanshah Province in 2005, 2009, and 2010, Amir Kabir Festival's top ranking Provincial Entrepreneurs in 2006, the province's Top Researcher in 2009, Distinguished environmental activist and eco-friendly in 2010, and Top Entrepreneur of Kermanshah Province in 2011 for employee protection.

[3]Related internet resources, www.rck.co.ir and https://www.youtube.com/watch?v=LxY5yj9J_RM

Haydeh serves as a member of the committee for selecting distinguished entrepreneurs in Kermanshah. RCK was also recognized as the Top Company in 2011 and the Top Industrial Unit in Research and Development Centers in 2012.

Hayedeh warns people of the harmful effects of environmental pollution. According to her, only 8 % of about 50,000 tons of waste, produced by the 75 million inhabitants of Iran per day, is currently recycled. These data show that 600 ha of mountains of garbage is created each year. She educates and asks people to shift to a less disposable lifestyle and helps government to manage waste disposal, handle and convert waste materials to increase resource productivity in different parts of the country.

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Chapter 6
Iranian Women Entrepreneurs Living Abroad

6.1 The Power of Dream and Technology Entrepreneurship

Anousheh Ansari, Prodea Systems Ltd.



Anousheh Ansari is a universally well-known brilliant Iranian-American immigrant entrepreneur and international role model who became the first female private space explorer in 2006. She is a Co-Founder and Chief Executive Officer of Prodea Systems, a market transforming technology, simplifying people's complex digital lifestyle.

Being an extraordinary entrepreneur, Anousheh, at the age of 35, was featured as one of the two women among America's 40 most successful business figures under the age of 40 by the Fortune magazine where she was named as the boldest newcomer. Her case is exceptional, starting from her childhood dreams of going into space, overcoming significant challenges to make her dreams a reality and co-founding a telecommunications firm in 1993, selling it for $750 million in 2000,

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and achieving her life-time goal of soaring among stars in 2006. Anousheh, as the first Muslim woman in space, stimulates and encourages women and children to dream big, believe in their dreams, work hard and study hard to reach their dreams.

Anousheh, the first daughter of her family, was born in 1966 in Mashhad, Iran. Her family moved to Tehran when she was 4 years old. Feeling close to nature and stars shaped her childhood dreams. She was raised in a family that valued education. Anousheh was enrolled at Jeanne D'Arc, a well-known French Catholic school, staffed mostly by tough nuns, where she studied French, Farsi, and Arabic. Anousheh and her family faced adversity during Iran's 1979 Revolution, followed by the Iran-Iraq war which started in 1980. They finally immigrated to the United States in 1984.

Arriving in the United States as a young immigrant who could not speak English, Anousheh had to face many challenges. Her goal was to study hard and hence considered obstacles as merely problems to be solved. Anousheh earned a bachelor's degree with honors in electronics and computer engineering from George Mason University in 3½ years. In her book, "My Dream of Stars", Anousheh divides her life into two parts: "everything before May 18, 1988, and everything that happened afterward. That was the date I met Hamid Ansari, my future husband" (Ansari and Hickam 2010). In 1991, she married Hamid Ansari, her supervisor at MCI, a company where she was working even before her graduation. When Anousheh learned that MCI paid for college courses, she continued her studies and received her Master of Electrical Engineering from George Washington University.

Anousheh and Hamid began their marriage believing they could do anything together. Hamid wanted to become a millionaire before age 30 and Anousheh intended to help him reach his goal. Shortly after their marriage, MCI announced they were closing their Virginia offices, where Anousheh and Hamid were working, and moving to Texas. They lost their jobs. Looking for a new job, they decided to recondition and sell used cars. Amir, Hamid's younger brother, also joined them sometimes as the salesman. After a while, they found the job frustrating and started their consulting practice and working with MCI.

Eventually, in 1993, Anousheh, Hamid, and Amir decided to start their own family business and set up Telecom Technologies, Inc. (TTI) which brought a turning point in their lives. They made a good family team. Telecom Technologies created a product called a Fastest, Fast, and Test, which automated and saved the testing cycle in telecom service. In 1995, the first customer was MCI followed by other companies. This innovation brought a turning point in their family adventure. By then, Atousa, Anousheh's only sibling moved to Texas to work with the family team as an electrical engineer. Now they had enough confidence and motivation to investigate more innovative product ideas. Having 200 creative, competent, and qualified employees, along with two large labs equipped with the latest technology, made the innovation of three more products possible over time. The most significant of all three was Amir's design called, "softswitch" that allowed voice communications over the Internet and positioned them as an innovative, cutting-edge company.

In 2001, Anousheh served as Co-Founder, Chief Executive Officer and Chairperson of the board for Telecom Technologies, Inc. The company successfully merged with Sonus Networks, a provider of IP-based voice infrastructure products, in a deal worth approximately $750 million, after receiving three key US patents and growing the company to 250 employees with 100 % sequential growth year over year since inception.

On September 18, 2006, Anousheh accomplished her lifelong dream when she boarded Russian Soyuz TMA-9 spacecraft blasting off from the Baikonur Cosmodrome in Kazakhstan, en route to the International Space Station (ISS). Achieving her dreams has been a strong motivation for Anousheh to accomplish her extraordinary entrepreneurial success. The power of her dreams, not only changed her personal life, but also had significant impact on the perspective of many people around the globe. Having an emotional soul, Anousheh had a childhood dream of going to the stars. When this dream came true, Anousheh realized the power and strength of her dreams, as she mentions in one of her speeches, "my dream never died in my heart and it was like a little glowing coal under the ashes, waiting to come to life with the first wind that would wipe off the ashes" (Anousheh's Experience 2009).[1] She, as a role model, inspires youth, and especially girls, around the world to pursue their dreams.

What makes Anousheh a distinguished cosmonaut is her pleasant personality and concern to share every moment of her space observations, feelings, knowledge and happiness with people all over the world. During her 9-day stay onboard the ISS, Anousheh wrote a blog that invited readers to share her experience. She described the Russian Soyuz capsule during liftoff, talked about the sights and smells of the space station, and explained the intricacies of everyday activities such as eating and washing up in zero gravity. The blog captured international attention. Her personal web site and her blog garnered more than 25 million hits from readers around the world. In spite of her very short time in space, she participated in some scientific experiments, including filling out a lower-back pain questionnaire for the European Space Agency and studying muscle atrophy with a NASA astronaut.

Anousheh's social and emotional aspirations are prominently highlighted in her statement that her ultimate goal is "to bring her experience and the ability to fly to space to more and more people and to inspire young women and men to go into the fields related to STEM" (BBC News 2006).[2] She considers herself a "space entrepreneur" for commercial space travel. She is the first space cosmonaut who has put much effort into advocating private space flights. In 2002, Anousheh, along with her family, became title sponsors of 10M XPRIZE, which was set up to develop "the first non-governmental organization to launch a reusable manned space craft into space twice within two weeks" (Ansari XPRIZE).[3] The Ansari X Prize was awarded to the aircraft designer Burt Rutan, and financier Paul Allen,

[1] www.anoushehansari.com/experience/jerusalem.php

[2] http://news.bbc.co.uk/2/hi/science/nature/5345872.stm

[3] http://space.xprize.org/ansari-x-prize

who led the winning team to build and launch a "Spaceship One" craft capable of carrying three people to 100 km above the earth's surface, twice within 2 weeks in 2004.

When Anousheh started her historic flight as the first female private space explorer on September 18, 2006, Prodea Systems was founded by members of the Ansari family, including Anousheh, her husband, Hamid, and her brother in law, Amir. Anousheh is a Co-Founder and Chief Executive Officer of Prodea Systems, transforming the Internet of Things market with their powerful software platform that makes it technically possible and economically feasible for service providers to deliver creative services using smart devices and sensors in a user friendly and fully integrated, seamless way. She is also an active proponent of world-changing technologies like Prodea's Residential Operating System and social entrepreneurship. Prodea Systems' vision is to "simplify people's complex digital lifestyles and to enrich and facilitate easy interactions within their communities." Prodea Systems is headquartered in Richardson, Texas, with additional sales offices in Los Angeles, London, and Dubai.

Anousheh is determined to help NGOs and social entrepreneurship organizations to create a better life for people in different parts of the world. Two examples of these organizations include ASHOKA, A global organization that identifies and invests in leading social entrepreneurs—individuals with innovative and practical ideas for solving social problems. (Including the Middle East and Central Asia) and Peace First, a national nonprofit organization that exists to create the next generation of peacemakers. Peace First views children as natural problem solvers and creative thinkers, and invests in their ability to see themselves as leaders. Furthermore, to nurture young talent, Anousheh runs the "What If?" competition for middle school students to encourage free thinking and Science, Technology, Engineering, and Mathematics (STEM) education.

Anousheh is on the advisory board of the Teacher's in Space project, and also a life member in the Association of Space Explorers. She serves on the board of directors/trustees of several organizations, Challenger Center, Peace First, Astronomers without Borders, and X Prize Foundation's Vision Circle (Trustees). She has received numerous honors, including, the Working Woman's National Entrepreneurial Excellence Award, DFW International Community Alliance Hall of Fame Award, the World Economic Forum Young Global Leader 2007, Ellis Island Medal of Honor, the Ernst & Young Entrepreneur of the Year Award for Southwest Region, George Mason University's Entrepreneurial Excellence Award, and George Washington University's Distinguished Alumni Achievement Award. She also was awarded an honorary doctorate by the International Space University when she was in space as well as George Mason University and Utah State University. In celebration of her accomplishments, the "Anousheh Ansari Day" was announced in the City of Plano on November 2 and in the City of Dallas on November 7, 2006.

Anousheh is a multi-dimensional entrepreneur; techno-entrepreneur, serial entrepreneur, immigrant entrepreneur, family entrepreneur, astronaut entrepreneur, and social entrepreneur. She considers herself a "full-time entrepreneur". She is currently working toward a master's degree in astronomy from Swinburne

University and working on a project to use Prodea's Internet of Things technology platform to bring rural India online.

6.2 The Pervasive Effects of Family on Entrepreneurship

Ashoobgar Cook, Child Care and Education Company Ltd., Acquarossa Spa Resort Project



Ashoobgar Cook is an Iranian/British entrepreneur whose passion for education inspired her to pursue an early years education and training business. She set up a multi-million dollar company from 1990 to 2007 consisting of over 25 day-care centers, 5 elementary schools, vocational training college for adults, and a recruitment agency employing more than 1000 staff with over 10,000 children in London and Germany.

Ashoobgar Cook nee Hamidi, the second daughter in a family of five children, was born in 1957 and was brought up in a beautiful orchard in Shiraz, Iran. Her childhood spent in a warm and a loving close knit environment with the love and affection of her family and particularly of her maternal grandmother, Esmat, who used to look after her grandchildren whilst their mum was at work. One of the many influencing aspects from Ashoobgar's childhood experience was the importance of love, security and support.

From early age her role models were her mother and her grandmother, two strong women who provided the family a loving, happy and a stable home. "Iranian women are by nature very caring and maternal and though they are normally the primary careers in the homes they are not by any means subservient." Ashoobgar's grandmother was a pioneer in having home education including learning English in the early 1900s and she went on to encourage her own two daughters to have higher education. Ashoobgar's father Abdul Samad was a university professor who travelled extensively and was a formidable educator whose great wish and ambition

was to install good values in work and family and in ensuring that his children obtained the most out of education, and in life and living with great sense of pride and joy.

At the age of 14, Ashoobgar left Iran to attend an English Boarding school in 1970. She graduated in Humanities Honors degree with Psychology as her major in 1981. At the age of 19, Ashoobgar got married to an English scientist. Her husband's university research took them to Florida, USA, and Saudi Arabia. Ashoobgar says, "I wanted to continue with my studies but like many Iranian women from my generation I decided to stay at home and instead to raise my son and stay at home with him in his formative years".

Ashoobgar was introduced to a *daycare* for the first time in US, a business that enabled working mothers to find a place to leave their children in day-care centers whilst going out to work. She observed that these day-care centers varied from very cooperative highly organized and structured chains to single nursery owners. Ashoobgar was particularly interested in the educational aspects and was convinced that children thrive in smaller settings with careful and structured program with great emphasis on learning through play taught by early years educators who understand, respect and are able to love the children in a purpose built settings. She says, "one thing I never thought I could do at that stage was to ever set up my own chain back in UK as I always assumed that I needed to have a business qualification and therefore lacked the confidence."

When Ashoobgar's family returned to London In 1988, she found living in London very hectic and extremely expensive with a young family of three boys under the age of 6. Ashoobgar was determined to help to contribute financially with family expenses and like her mother in Iran, who also worked part time, she decided to look for a place to leave her sons whilst finding an educated-related job, which she thought would be suitable to her young family's needs. At the same time her husband wanted to change career and had decided to study for an MBA. Ashoobgar thought that they both could not study full time at the same time so again she held back and started to search instead for a part time job. To her surprise, she could not find anywhere suitable for her children where they would be educationally stimulated whilst in a loving and caring environment. Ashoobgar realized that the only way was to set up her own place.

After spending 2 years of searching for a site to set up a daycare center, Ashoobgar found a detached church hall with a lovely enclosed garden in a quite alley near her home in 1990. From the very beginning, her goal was to provide a safe and secure place where working parents would be happy and reassured to leave their children. Ashoobgar had many challenges to face including resistance from her family and in particular her husband. She was determined to introduce the US business model in childcare, though on a small scale and to attempt to set up a day-care center primarily for the needs of her own children. In spite of its risk, Ashoobgar's plan was to put all the life family savings into the project. She says, "it was a huge undertaking as we had very little money and the risk of losing our home was very frightening indeed." But she was determined to make a go of it as she was hoping to provide a more comfortable life and better schooling for her children.

Securing the finance was quite tough as Ashoobgar had no commercial experience to proof and was reliant only on her "one page business plan" to demonstrate the feasibility of the nursery. The business loan was finally secured against the family's tiny apartment. Although the financial terms were quite stringent, Ashoobgar was determined to work all hours and in all capacities to make this venture work for her family. Ashoobgar used £10,000 of life savings for repairs and renovations. The work was completed and the daycare was ready for opening in October 1990. In the early months, Ashoobgar was carrying on many duties alone both for home and the nursery including; cleaning, shopping, cooking, teaching, marketing, banking and accounts, collecting children, supervising homework and carrying on with her own family housekeeping whilst her husband was absent and was on frequent business trips abroad.

While for many months the nursery numbers were worryingly low, Ashoobgar began to receive good recommendations from the existing parents and with the 'word of mouth' the nursery began to fill up slowly. At that stage, Ashoobgar's main concern was not profit but sustainability. Finally at the end of the fiscal year the nursery showed a small profit which was a relief for Ashoobgar and her family.

Ashoobgar's passion to provide a safe and stimulating place where children learn through fun whilst parents feel safe and secure leaving their children in trust of loving and professional careers was the key to her success. She wanted to provide the families with "home from home" experience offering healthy foods in a warm and a loving environment with plenty of natural light and space. Ashoobgar knew some of the children in the nursery would not have cooked home meals on return to their respective homes. She specifically recruited many Iranian ladies who were looking for part time jobs and who were also keen and able to cook fresh meals. Moreover, from her childhood experience, Ashoobgar knew children love the feeling of freedom through space and natural light. Hence, with the help of her Iranian architect, Ashoobgar planned to design specific spaces for different activities and areas of particular interest. The design elements were crucial for children to have opportunities to experience and to explore with different textures, lights, shapes and colors both indoors and outdoors. Landscaping the gardens for each of the nurseries was designed to be child friendly with designated areas for growing organic vegetables and fruits, playground and quiet corners to retreat.

Ashoobgar owes her success to her parents who never discriminated their daughters from their sons and encouraged her to have ambition and above all kindness and respect for people. She says, "the success of my business in childcare was partly due to my strong organizational and people skills, but perhaps also it was due to my highly developed social responsibilities, respect and kindness for people." Ashoobgar owes her success to a simple belief of love and respect for people around her regardless of any discrimination against race, creed, religion, and class differences. She adds, "in my family similar to many at the time we were taught to have great respect especially for the elders and to be kind and honest." As children we were given certain responsibilities to care for others especially to supervise and care for our younger siblings." Ashoobgar was grateful to her parents who gave her great confidence and value as a person regardless of her gender.

Family and education are highly valued in Iran. In spite of her eagerness, Ashoobgar could never managed to return to school and pursue graduate studies. But in return, she gained a wealth of insights on numerous courses and workshops as well as learning "on the job" in running a business, working with people dealing with very sensitive social and family issues, learning about legal aspects of the business, managing the finance particularly the importance of cash flow, acquiring and managing real estate opportunities. Finally Ashoobgar was able to grow a multi-million dollars business from very humble beginnings.

Following the success of her first day-care, Ashoobgar realized the potential of growing her business and therefore started to look for other suitable sites and developed a chain called Happy Child with more than 1000 staff and over 3000 children. During the first 7 years Ashoobgar was managing and running the business by herself and later when her husband was made redundant, he then joined the company and became the finance director. Their roles complemented each other; Ashoobgar was the managing director responsible for the strategic decisions for acquisitions, mergers expansions, recruitment and all issues relating to people and the business, whilst her husband was involved in all aspects of finance. Ashoobgar owned 100 % share of the company and had a "hands-on" involvement in all aspects of the operation.

By 2007, Happy Child was one of the largest Childcare and Education companies in the United Kingdom with groups of daycares in London and Germany. The company owned a substantial portfolio of properties with over 25 day-care centers, 5 elementary schools, a vocational training college for adults, and a recruitment agency. Happy Child achieved many milestones in early year's education and was awarded on many occasions for "best business practice", advisory roles for various educational charities and organizations overseas, and close involvements with local community projects and various children's charities. Happy Child Company was among the first few to win the "Investors in People" award in the United Kingdom.

Ashoobgar successfully sold her multi-million dollar company, at the peak of the market, to a private equity in 2007. The most successful projects are the ones that reach their goals in time, are terminated before the deadline, and stay within budget (Schwab and Miner 2008). Ashoobgar says, "After 17 years of long and hard journey I achieved something more than money. I learned that my dedication, persistence, hard work and above all my passion helped me to strive for a vision which started small and then went on to grow with pride and joy for my family." Ashoobgar adds that she is "proud to be an Iranian woman abroad, who was able to represent the spirit of the Iranian women who are inherently strong and organized with great social and emotional skills with natural instincts to care for others." She has given many lectures on women entrepreneurship and works as a consultant in early years education and in the care sector.

Ashoobgar has been involved in various real estate projects. She is also currently co-developing Switzerland's first group of an all-inclusive thermal spa resorts especially designed for families with age appropriate facilities for both indoors as well as the outdoors whilst maximizing the best of the Alpine nature pursuits. The

total project size for a group of family sap resorts is estimated for 300 million Swiss Francs.

The first Swiss family resort will be situated in Acquaroosa, in Ticino. It is a beautiful Alpine site within the natural park which has a medicinal thermal 'red water' due to its high iron content. The family spa resort has planning for 35,000 m^2 of hotel, apartment developments which will include many sports facilities with various children's clubs from birth to 14 years of age. The project aims to provide an eco-family-friendly resort in a beautiful valley with activities including skiing, climbing, hiking, horse-riding, small farms for children, and a spa therapy center.

As part of her strong dedication to training and staff development, Ashoobgar is planning to build Switzerland's first hotel management school which will offer great training for staff working with children with great understanding and delivering for the developmental and recreational needs of children staying in such family resorts.

Ashoobgar has always worked closely in partnerships with the communities she has worked with and as such has managed to gain trust, and confidence of all who support her projects. Her one dream still remains, which is to 'payback 'to her hometown of shiraz by setting up a school for girls and hope to be working closely with one of the children's charities in Shiraz, Iran.

6.3 Family Entrepreneurship

Gisue Hariri and Mojgan Hariri, Hariri & Hariri Architecture



MOJGAN HARIRI GISUE HARIRI

Two Iranian sisters, Gisue and Mojgan Hariri, are New York based, outstanding architects who have recently won the historical project bid of Sternbrauerei in Salzburg, Austria, beating out strong competitors from Europe, Japan, and the United States. They combine their creative imagination shaped by the mysterious beauty of the deserts in Iran, where they spent their childhood, with the top digital technology of where they live now (USA), to create unique, modern, and breath-taking designs.

Gisue Hariri and Mojgan Hariri, a year and a half apart in age, were born in a small town in the southern part of Iran. Their parents are from Isfahan. Their father, who was an engineer, worked for the Iran National Oil Company and his job required the family to live near oil fields, far away from major cities and market-places. During their childhood, Gisue and Mojgan did not have access to toys for entertainment and used their creativity to make toys with primitive items such as paper, cinderblocks, wire, and wood pieces. Their imaginations flourished under the beauty, stillness and mystery of the desert. Gisue says, "The desert became both deeply isolating and nurturing". They created their own games, toys, and world. When Gisue told her family that she wanted to be an architect, at the age of 12, she had a youthful understanding of what that meant. Now she realizes that architecture is more than just a carrier; "it is a way of life, a way of seeing the world, a way of harnessing the past", Gisue explains.

Like many families in Iran, the Hariri family highly valued education and upon graduating from high school in Tehran, Gisue and Mojgan moved to the US for college in the mid 1970s. Pursuing their passion for architecture and arts, they studied architecture at Cornell University. To their surprise, there were very few girls in the architecture program. Moreover, there were no female professors among the architectural design faculty members. Soon they realized they had chosen a challenging career in a male-dominated field. Gisue and Mojgan had the feeling that their freedom and imagination were constrained in terms of design because most of their professors at Cornell did not like to see any curvilinear elements in their drawings. They needed to release their imagination to create unique designs. This was a mystery to them as the two most admired architects whose focus were on human scale and connection to nature, Frank Lloyd Wright and Le Corbusier, had human figures and natural elements in their drawings. Their work encouraged Gisue and Mojgan to articulate what their inner voice already knew. Gisue says, "Like the desert, the simplest things in life are the most complicated, magical and extraordinary." Gisue and Mojgan earned the B.A. degree in Architecture from Cornell University in 1980 and 1981, respectively. Mojgan also received a master's degree in Urban Design in 1983.

Gisue and Mojgan established Hariri & Hariri (H&H), a New York based architecture and multi-disciplinary design firm, in 1986 to find their inner voice, as they believe the outer world has little to offer. In fact, they made a perfect team in architecture; Gisue has worked on residential and smaller projects while Mojgan has studied cities and worked on large-scale urban projects. Gisue is very concept-oriented, highly artistic and vision-oriented, while Mojgan is extremely focused, artistic and detail-oriented. Their strong belief is that design is fundamental to

improving the quality of life. The hallmark of their work is their continuous mindset, holistic creativity, and innovation. With a continuum approach, H&H is interested in designing a variety of goods from clothing to jewelry, accessories, cars, boats, etc. "Its work spans from master-planning to architecture, interior design, lighting, product design, and fashion-accessories/jewelry design," says Gisue.

Gisue and Mojgan have not cut ties with their country of birth. They are still under the influence of Iran's rich culture. When they visit their grandparents in Isfahan,[4] the hometown of their father, they get ideas for their work. "The tactility, materiality and use of art and craft throughout our projects, regardless of scale and program, are what we carry within us from the visits to Isfahan," says Gisue (Clara Lambert 2013).[5]

The creative imagination of Gisue and Mojgan, shaped by the mysterious beauty of the deserts in their childhood, joined with the top digital technology of the country where they live now, has formed their holistic philosophy and their ability to create unique, modern, and breathtaking designs. They have successfully combined positive elements from two different nations (Iran's rich cultural heritage in arts, literature, poetry, and music since ancient times, and USA's modern arts, knowledge, and technology) to create innovative, extraordinary designs. In particular, they have been influenced and inspired by Omar Khayyam, an internationally well-known twelfth century Iranian scientist, philosopher, astronomer, and poet.

In their nature-inspired design, Gisue and Mojgan articulate geometric forms observed in Persian architecture as well as modern Western architecture. Furthermore, they have recently expanded their work by studying geological formations and crystals. One of their well-known crystal works is the crystal tub and sink designed for AF New York showroom in 2010.

Their work has been displayed at major art institutions including the Museum of Modern Art (MOMA), the National Building Museum, and the Barcelona Museum of Contemporary Art. One of the famous works of H&H is the Digital House, a breathtaking work of art and technology that was designed in 1998 and exhibited in MOMA, New York in 1999. In the Digital House design, H&H exhibited what a house would look like in the next millennium.

[4]Located in the center of Iran, at the intersection of the main north-south and east-west routes crossing Iran, Isfahan has had a significant strategic position in the history of Iran. The city grew by becoming the capital of Persia (for the second time) from 1050 to 1722, particularly in the sixteenth century under the Safavid dynasty (Press TV-Isfahan 2013). The city is acclaimed by generations of art historians, architects, artists, archaeologists, photographers, and tourists for the magnificence and splendor of its architecture, triumphant remainders of Saljuq and Safavid empires (Walcher 1997). Isfahan has been named sister city to Florence in Italy, Freiburg in Germany, Yerevan in Armenia, Xian in China, Kyoto in Japan, Saint Petersburg in Russia, Havana in Cuba and several other cities in the world (Press TV-Isfahan 2013).

[5]http://asiasociety.org/blog/asia/interview-gisue-hariri-her-persian-influences-and-women-architecture

Hariris consider the organic relationship between man, nature and architecture as an important element in their work. They believe in simplicity, efficiency and sustainability in their projects. Before offering a design, Gisue and Mojgan study and learn about the culture, community interests, historical and geographical characteristics of the intended location, and design accordingly. They also try to recognize the essence and the spirit at the heart of each project. This is how they have successfully designed international projects across the world. Besides designing numerous projects for art galleries, museums, residential buildings, offices, beauty salons, etc. in the USA, they have also designed outstanding projects in other parts of the world including Tehran, Vienna, Dubai, Salzburg, and Morocco. Furthermore, as a multi-faceted design firm, H&H works with leading international corporations, developers, and fashion labels (more information in their website).[6]

Gisue and Mojgan Hariri have recently won a master-plan for Sternbrauerei Salzburg, a residential and mixed-use development near the historic center of Salzburg, Austria, beating out internationally recognized competitors from Europe, Japan, and the United States. This project, now under construction, includes 80 luxury apartments, exhibition space, a gallery and lecture hall in the old brewery's underground vaults, a spa and wellness center, a café and restaurant facilities and an underground parking garage.

Their holistic philosophy, technological attitude, curiosity, creative and innovative mind, research dedication, and childhood background have won them many awards including the "Academy Award" for Architecture from the American Academy of Arts and Letters, induction to the Design "Hall of Fame" and Best of Year Award from the Interior Design Magazine, and inclusion by Architectural Digest in their NEW AD100, showcasing today's greatest talents in Architecture & Design.

By the invitation of the luxury lifestyle brand Swarovski to design and "re-invent" a chandelier for their Crystal Palace Collection and to their limited edition Couture-Jewelry collection, Hariris have been interested in designing clothing and jewelry. Gisue says, "Each new project is an opportunity to mine the imagination and to give our clients not only a new spatial experience but also to help them to a new way of living, where furniture, lighting, and the art are all integral part of the environment. To them, designing different products from clothing to jewelry, accessories, cars, boats, etc., is all part of the same continuum. In their book "Hariri and Hariri Houses", published in 2006, Gisue and Mojgan have presented a collection of their work.

[6] http://www.haririandhariri.com/

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Appendix: Questionnaire

- Name and Family
- Your life/family history
- Your educational, training and working background
- Your personal characteristics/skills
- Your motivations for starting the business
- The name, type, sector, size, history and location of your business
- Specific obstacles you encountered/overcame as well as specific enablers
- Key influences (positive or negative) on your entrepreneurial journey
- Kind of products/services you produce
- Total quantity produced per day/month/year
- Number of your employee, do they have work insurance?
- Your internal and external markets
- Your marketing arrangements
- How you finance your costs
- Your future plans for growth and development
- Any other details which you feel would be relevant to the story

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